

CHILE*

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Abstract

We introduce CHILE, an asset-pricing framework with asymmetric information, general utility functions, and arbitrary payoffs. It features a large economy (LE) with continuous-and-heterogeneous information (CHI). The defining feature of this information structure is that the total amount of information stays finite as the economy grows large. We use the framework to study how concentration—the presence of market participants of different sizes—affects market quality, namely liquidity and information efficiency. Greater concentration proves a double-edged sword: it raises liquidity but harms information efficiency.

Keywords: inefficient markets, information aggregation, rational expectations with non-CARA preferences, wealth effects, competition

JEL Codes: D01, D53, D82, E19, G12, G14.

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1 Introduction

How do market prices aggregate the information dispersed among investors? The canonical framework for studying this question assumes that investors have constant absolute risk aversion (CARA) and that asset payoffs are normally distributed.¹ This CARA–normal specification delivers great tractability, but at a price. Normally distributed payoffs place positive probability on arbitrarily negative prices and, combined with unrestricted positions, on negative wealth. More fundamentally, CARA preferences remove wealth from the demand for risky assets, so that investors of different sizes behave identically: the framework leaves no room for wealth effects, and hence no role for the size distribution of investors.

We introduce a framework that accommodates both general, heterogeneous preferences and general asset-payoff distributions, addressing the limitations of the standard approach. Despite the generality of our primitives, the end product is tractable, with all equilibrium objects available in closed form. We illustrate the framework by asking how the size distribution of market participants interacts with market quality—its informational efficiency and liquidity: is a more concentrated market better or worse for efficiency and liquidity?

The framework’s tractability rests on a new way of modeling the information dispersed across a large market. In contrast to the traditional methodology (Hellwig, 1980; Admati, 1985, and consequent literature), we do not assume that traders have signals of finite precision. Instead, following Section 9 of Kyle (1989), we distribute a finite amount of information among all traders, so the total amount of information remains finite regardless of their number. As we show, these information structures can be formally treated as diffusion processes running through a heterogeneous continuum of traders, giving structures we call continuous-and-heterogeneous information (CHI); recognizing that our model also features a large economy (LE), we adopt the name CHILE for the class of models described in this paper.

A novel feature of our large economy is that the degree to which traders internalize their influence on prices has nontrivial implications for equilibrium. Such influence operates through two channels. Through the price level, a trader’s demand may be a non-negligible part of the

¹Classic examples include Grossman and Stiglitz (1980), Hellwig (1980), and Admati (1985).

market total, so that changing it must be met by a price change for the market to clear. Through the informational content of prices, her signal may be a non-negligible part of the information the price conveys, so that different signal realizations change what the price tells other traders. The first channel manifests itself as a trader’s price impact; the second, as a correlation between her signal and the price. Correspondingly, a trader who disregards the first channel makes an assumption we call *price-impact neglect*—the standard price-taking assumption—and one who disregards the second, an assumption we call *signal-price correlation neglect*. Signal-price correlation neglect is a trader’s belief that the price tells the same thing about the fundamental whatever her own signal happens to be; equivalently, that her signal and the price are conditionally independent given the fundamental. It is a behavioural departure from rational updating—a rational trader would recognize that the price partly embeds her own signal and account for the resulting correlation.²

Unlike in traditional large economies—where influence through both channels washes out in the limit, so that the choice of finite-economy equilibrium concept is largely immaterial—our setting permits a granular analysis of the two, with each combination yielding a distinct equilibrium outcome. As we show in Appendix H, the Bayesian Nash equilibrium (BNE), in which traders internalize both channels, differs from the competitive rational expectations equilibrium (REE), in which traders neglect price impact but not the signal-price correlation; and both differ, in turn, from the equilibrium in which traders neglect *both*—which we refer to as the *correlation-neglect equilibrium (CNE)*.³

We adopt the correlation-neglect equilibrium—in which traders neglect both their price impact and the signal-price correlation—as our main equilibrium concept. This choice provides a clean, tractable competitive benchmark, allowing us to isolate the central channel of interest—the effects of changing the size distribution—from additional effects operating via

²This is an instance of *correlation neglect*, documented experimentally by [Enke and Zimmermann \(2019\)](#): people treat correlated information sources as independent and thereby double-count information. In our setting the correlated source is the endogenous market price rather than an exogenous report, and the correlation arises because the price partly embeds the trader’s own signal.

³A related analysis of market power in large economies is conducted by [Kovalenkov and Vives \(2014\)](#), who start from the finite-agent Bayesian Nash equilibrium—where agents fully internalize both dimensions of their price impact—and then take the large-economy limit. We follow the same approach in Appendices G and H, where we derive the limiting BNE.

market power.⁴ Importantly, the key economic mechanisms we identify are not artifacts of this equilibrium concept: as summarized in Section 9, our robustness analysis shows that the direct effects of market concentration on market quality point in the same direction across all three equilibrium concepts.

Beyond providing a clean competitive benchmark, signal–price correlation neglect plays a second, conceptual role: it is what sustains a well-defined equilibrium in our economy without external noise. In a finite economy, the assumption rests on an incorrect conjecture: a trader treats her signal as independent of the information contained in the price, whereas in fact the two are correlated. This correlation—and with it the error in each trader’s inference—vanishes as the economy grows large, so that neglecting it is not a mistake in the limit. Nevertheless, these individually vanishing errors aggregate across infinitely many traders into a nontrivial random component that prevents full revelation—an *endogenous* source of noise in place of the exogenous noise usually assumed in this literature.

We then put the framework to work on the following question: how does the size distribution of market participants shape market quality? The question is economically salient because participants—be they institutions or individual investors—come in very different sizes, and when they are *granular*, shocks to the larger ones move aggregate outcomes rather than averaging out. Whether a more concentrated market is better or worse for informational efficiency and liquidity is thus a genuinely open question, one we motivate more fully in Section 5. Using the model, we study the effect of making the market more concentrated—enlarging the largest participants and shrinking the smallest, where size is measured by wealth for individual investors and by assets under management for institutions. Focusing on decreasing-absolute-risk-aversion (DARA) preferences, a plausibly realistic assumption, we show that increasing concentration makes prices less informative; reducing it has the opposite effect.

The key intuition is that prices reveal a weighted average of all private signals, with weights

⁴The correlation-neglect equilibrium is, in fact, the most competitive of the three concepts, in the sense of Aumann (1964). As Aumann puts it, the essential idea of perfect competition is that the economy “has a ‘very large’ number of participants, and that the influence of each individual participant is ‘negligible.’” In our economy each trader can influence the price through two channels—its level and its informational content—and the correlation-neglect equilibrium is the only one of the three in which each trader treats her influence through *both* channels as negligible; REE and BNE retain a non-negligible influence through one or both.

proportional to agents' trading intensities.⁵ Information would be aggregated best if signals were weighted in proportion to their precision alone. The equilibrium weighting scheme distorts this ideal: accounting for wealth effects, it scales signals by risk tolerance on top of precision, placing more weight on the signals of more risk-tolerant traders. As risk tolerance increases in wealth under DARA, the signals of larger traders receive excessive weight, so that the information content of prices is distorted by the concentration of the market. Increasing concentration—enlarging the largest traders and shrinking the smallest—aggravates this distortion, pushing the equilibrium weighting scheme further from the ideal and making prices less informative.

The same aggregation logic applies to the distribution of *information*. Raising the signal precision of larger traders and lowering that of smaller ones likewise pushes the weighting scheme further from the ideal and makes prices less informative. This yields a surprising corollary: prices can become *less* informative even when every trader is weakly better informed, because raising the precision of sufficiently large traders pushes the weighting scheme further from the ideal. This information-aggregation paradox is a hallmark of imperfect information aggregation under wealth effects—more can be less, as more badly-aggregated information is less information.

Finally, our results are reinforced when information is endogenous. When traders acquire costly information in the spirit of [Verrecchia \(1982\)](#), a wealthier trader optimally acquires a more precise signal, as he trades more aggressively and thus values information more. The response to an increase in concentration then combines two effects of the same sign: a direct one from wealth alone, and an indirect one from the induced change in precision that amplifies it—both reducing efficiency.

All of these efficiency findings carry over to liquidity, but with the opposite sign. Any change that raises efficiency affects liquidity through two conflicting channels: as prices grow more efficient, each trader leans more on the commonly observed price and less on her own view, aligning expectations and dampening the willingness to trade; at the same time, more efficient prices are less volatile, reducing the risk traders must absorb. Because these channels pull in opposite directions, the effect on raw liquidity is non-monotonic. Scaling liquidity by return

⁵As we discuss below, the trading intensity of an agent is defined as the sensitivity of his demand to changes in private information.

volatility nets out the risk component, however, and restores a monotone comparative static—one running opposite to efficiency: increasing concentration, for instance, raises risk-adjusted liquidity.

The mechanisms outlined above are empirically relevant at different economic timescales. For long timescales, our results can help interpret the long-term trend towards higher institutionalization, widening wealth inequality, and growing concentration of the asset management industry.⁶ At shorter timescales, our model provides novel insights into recent evidence connecting changes in fund size distribution to variations in informational efficiency. We develop these and other empirical implications in Section 8.

Our paper relates to several branches of literature. First, a literature extends REE models beyond the CARA-Normal framework.⁷ Breon-Drish (2015) extends it beyond normality with a single asset, and Chabakauri, Yuan, and Zachariadis (2022) further allows for multiple assets. Albagli, Hellwig, and Tsyvinski (2024) study noisy information aggregation in a general setting, though their sharpest, fully tractable characterizations obtain for risk-neutral traders facing position limits. All of these papers rely on CARA or risk-neutral preferences and so abstract away from the wealth effects central to our paper.

Malamud (2015) considers an REE model with a continuum of assets, whose tractability hinges on market completeness.⁸ In contrast, we have one asset and a continuum of states, so our market is incomplete. A central result in Malamud (2015) is that with non-CARA utility the equilibrium is fully revealing;⁹ in our incomplete market, aggregate price noise rules out full revelation for any utility function.

Peress (2004) was the first, to our knowledge, to study wealth effects in noisy REE, in a

⁶The empirical literature offers recent evidence on these trends. See Ben-David, Franzoni, Moussawi, and Sedunov (2021) on the increasing concentration of the asset-management industry and institutionalization, and Saez and Zucman (2016) on widening wealth inequality.

⁷The CARA-Normal framework is also used to model markets where traders have market power (see Rostek and Yoon (2025) for a review). Glebkin, Malamud, and Teguia (2023a) and Glebkin, Malamud, and Teguia (2023b) allow for, respectively, non-normal payoffs with CARA traders and non-normal payoffs with non-CARA utilities. These papers abstract from information frictions that are central to this paper.

⁸Relatedly, DeMarzo and Skiadas (1998) and DeMarzo and Skiadas (1999) analyze REE models where the market is *quasi-complete*.

⁹See Theorem 2.1 in Malamud (2015). See also Chabakauri (2025), who shows that full revelation with non-CARA utility is more nuanced.

model with log-normal payoffs and non-CARA utilities.¹⁰ The key difference is that [Peress \(2004\)](#) relies on a “small risk” approximation in which the asset’s return variance vanishes in the limit, making it unsuitable for quantitative work; our “small information” approximation keeps the asset risky in the limit. Beyond this, in our model equilibrium quantities depend on absolute risk aversion, absolute prudence, and conditional skewness—whereas only risk aversion matters in [Peress \(2004\)](#)—and we allow for generally-distributed payoffs.

Our paper also relates to the asset pricing literature on the implications of preference and wealth heterogeneity for asset prices; examples include [Dumas \(1989\)](#), [Gârleanu and Panageas \(2015\)](#), and [Gomez \(2025\)](#)—see [Panageas \(2020\)](#) for a review. This literature focuses on risk premia and risk-free rates but abstracts away from informational frictions and does not derive the implications for market quality that are central to our paper.

Our paper is also related to the literature on mean-field games; see [Achdou, Han, Lasry, Lions, and Moll \(2022\)](#).¹¹ The analogy holds in our model: the effect of other traders on a given trader is summarized by a few statistics of the cross-sectional distribution of traders’ characteristics, which act as a “mean field” to which each trader contributes only infinitesimally.

On the technical side, our paper relates to work using stochastic calculus outside continuous-time finance: [Malamud \(2015\)](#) models noise in a continuum of assets as a cross-sectional stochastic process; [Gârleanu, Panageas, and Yu \(2015\)](#) use a Brownian bridge to represent dividends for firms on a circle; and [Glebkin et al. \(2021\)](#) derive a marginal value of information in a static model.¹² Most closely related is [Avdis \(2018\)](#), which introduces continuous heterogeneous information but with CARA preferences and hence no wealth effects.

2 Discrete economies

We introduce a sequence of economies indexed by the number of agents n , or equivalently by $\mu = 1/n$, the mass of each agent. All equilibrium objects—demands, posterior beliefs, and

¹⁰The wealth effect can arise even in a CARA model, because wealth may affect the tightness of financial constraints, as in [Glebkin, Gondhi, and Kuong \(2021\)](#).

¹¹See also [Lasry and Lions \(2007\)](#) and [Guéant, Lasry, and Lions \(2011\)](#) for a review.

¹²A related econometric literature on unit roots includes [Phillips \(1987\)](#).

prices—depend on μ , which identifies the economy within the sequence. Our primary interest is the continuous economy obtained in the limit $\mu \rightarrow 0$ ($n \rightarrow \infty$), which we characterize in Section 3.

We now describe a generic economy in this sequence; that is, we fix μ (and hence $n = 1/\mu$). The economy is defined on a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with trading at $t = 1$ and consumption at $t = 2$.¹³ There are two assets, one risky and one risk-free. As we use the risk-free asset as a numeraire, we normalize its gross return to 1. The risky asset pays off $V(v)$ in the second period, where $v \sim N(0, \tau_v^{-1})$ is the *fundamental*, and $V(v)$, a weakly increasing and non-constant function of v , is the *payoff function*.¹⁴

The risky asset trades at price P , determined in the first period. Allowing for both price-inelastic and price-elastic supply components, we write the total supply of the risky asset as

$$\Theta(P) = \bar{\theta} + \theta(P), \quad (1)$$

where $\bar{\theta}$ is a constant and $\theta(P)$ is a continuously differentiable function.

The economy is populated by n agents. Agent i , $i = 1, \dots, n$, is associated with point $a_i = (i-1)\mu$ on the unit interval $[0, 1]$; informally, agent i lives on the interval $[a_i, a_i + \mu)$. The population of traders is characterized by three profiles: initial wealth $W_0(a)$, preferences over terminal wealth $u(W, a)$, and signal precisions $t(a)$ (signals are characterized below). Agent i 's primitives are $W_0(a_i)$, $u(W, a_i)$, and $t(a_i)$; thus, discrete primitives are obtained by “sampling” the profiles at the leftmost points of agent subintervals.

The signal of agent i , who lives in segment $[a_i, a_i + \mu)$, is

$$\Delta s_i = v \cdot \Delta a_i + \frac{1}{\sqrt{t(a_i)}} \Delta B(a_i). \quad (2)$$

Here, $B(a)$ is a standard Brownian motion on $[0, 1]$, independent of v , $\Delta a_i = a_{i+1} - a_i = \mu$, and $\Delta B(a_i) = B(a_i + \mu) - B(a_i) \sim \mathcal{N}(0, \mu)$. The scaled signal $\Delta s_i / \mu$ has a standard “fundamental

¹³In Appendix D, we study information acquisition, for which we add a period at $t = 0$.

¹⁴The normalization $\mathbb{E}[v] = 0$ is made without loss of generality, as any mean can be incorporated into the general form of $V(\cdot)$.

plus noise” structure with precision $t(a_i)\mu$.¹⁵ We refer to $t(a)$ as the *precision* of agent a .

Agent i ’s optimal demand in the economy with size parameter μ , given a signal realization s , solves

$$x^*(s, P, a_i, \mu) = \arg \max_x \int_{\mathbb{R}} u\left(W_0(a_i) + x(V(v) - P), a_i\right) f(v \mid s, P, \mu) dv, \quad (3)$$

where $f(v \mid s, P, \mu)$ denotes the posterior density of v given the signal realization $\Delta s_i = s$ and the price P . The explicit dependence on μ reflects the fact that beliefs—and hence demands—vary across economies in the sequence.

Remark 1 (CHILE vs. traditional large economies). The setup above differs from traditional large rational expectations economies, such as Hellwig (1980) and Admati (1985), in two important respects. *First*, by (2), the precision of agent i ’s signal is $t(a_i)\mu = t(a_i)/n$, scaling as $1/n$ with the size of the economy. As a result, the total (aggregate) amount of information—the sum of individual signal precisions—has a well-defined finite limit as $n \rightarrow \infty$, while per-capita information vanishes. By contrast, conventional models endow each trader with a signal of finite (non-vanishing) precision, so that aggregate information grows linearly in n and diverges in the limit, while per-capita information has a well-defined finite limit. *Second*, the aggregate supply $\Theta(P) = \bar{\theta} + \theta(P)$ in (1) does not scale with n and thus has a well-defined finite limit, while per-capita supply vanishes. In conventional models, by contrast, aggregate supply is proportional to n and diverges in the limit, while per-capita supply has a well-defined finite limit. Together, these two features imply that in CHILE the *aggregate* counterparts of information, demand, and supply are well-defined and finite, while their *per-capita* counterparts vanish; in traditional large economies the roles are reversed—per-capita quantities are well-defined, while aggregates diverge. See also Remark 5. A simple CARA-normal model that isolates the economic consequences of these two scaling assumptions—and contrasts CHILE with traditional large economies under the CNE, REE, and BNE concepts—is developed in Appendix G.

¹⁵Dividing both sides of (2) by μ gives $\Delta s_i/\mu = v + \Delta B(a_i)/(\mu\sqrt{t(a_i)})$. Since $\Delta B(a_i) \sim \mathcal{N}(0, \mu)$, the noise term has variance $\mu/(\mu^2 t(a_i)) = 1/(t(a_i)\mu)$, i.e., precision $t(a_i)\mu$.

2.1 Optimal demands under price-impact neglect and signal-price correlation neglect

Our goal is to study a correlation-neglect equilibrium—a natural competitive benchmark. In such an equilibrium, traders make two assumptions, *price-impact neglect* and *signal-price correlation neglect*: they disregard both their influence on the price level and the correlation between their signal and the price.

These two assumptions have distinct implications for the optimization problem (3). Disregarding one's influence on the price level—*price-impact neglect*—means that the agent treats the densities in (3) as unaffected by his choice of x . Neglecting the correlation between one's signal and the price—*signal-price correlation neglect*—means that the agent assumes that, conditional on v , the price tells the same thing about the fundamental whatever his own signal turned out to be, i.e., that the price and his signal are independent.¹⁶

Under the conditional independence assumption, the posterior density factors as

$$f(v \mid s, P, \mu) \propto \underbrace{\exp \left(t(a_i) \left(sv - \frac{\mu}{2} v^2 \right) \right)}_{\propto \text{density of } \Delta s_i \mid v} \underbrace{g(v, P, \mu)}_{\text{density of } (v, P)}, \quad (4)$$

where $g(v, P, \mu)$ is the joint density of v and P as conjectured by the agent, which we call the *price belief*. In equilibrium, the price belief is required to be consistent with the actual joint distribution of v and P in the limiting economy ($\mu \rightarrow 0$).¹⁷

Substituting (4) into (3) and dropping terms unaffected by the maximization over x , the

¹⁶The price and the signal cannot be independent unconditionally, as they both reflect the fundamental v . Conditioning on v isolates the noise in each, and independence amounts to saying that the agent's idiosyncratic noise is unrelated to the noise in the price.

¹⁷We emphasize that, for $\mu > 0$, the price belief $g(v, P, \mu)$ is an auxiliary object that supports the sequence of demands rather than the exact distribution of v and the market-clearing price P in a finite economy with $n = 1/\mu$ traders. Indeed, correlation neglect is a misspecification for any $\mu > 0$. We therefore do not require $g(v, P, \mu)$ to coincide with the actual joint distribution of v and P at each finite μ ; we impose consistency only in the limit $\mu \rightarrow 0$, where the neglected correlation vanishes.

optimal demand solves

$$x^*(s, P, a_i, \mu) = \arg \max_x \int_{\mathbb{R}} u\left(W_0(a_i) + x(V(v) - P), a_i\right) \exp\left(t(a_i)\left(sv - \frac{\mu}{2}v^2\right)\right) g(v, P, \mu) dv. \quad (5)$$

To summarize, our agents are heterogeneous along three dimensions: initial wealth $W_0(a)$, preferences $u(W, a)$, and precisions $t(a)$. The profiles $W_0(a), u^{(l)}(W_0(a), a), l = 0, 1, 2, 3$, and $t(a)$ are arbitrary functions of a , continuous on $[0, 1)$. We assume that $u(W, a)$ is increasing, strictly concave, and thrice continuously differentiable in W on an open neighborhood of $W_0(a)$. In addition, we assume that the technical conditions introduced and discussed in Section C.1 hold.

Remark 2. Our agents can be interpreted either as individual traders or as fund managers. In the latter case, $W(a)$ represents the value of the assets managed by fund a , with a portion of $W(a)$ serving as the fund manager’s compensation. To lighten notation, we absorb all proportionality coefficients into the parameter a of the utility profile.

Remark 3 (Neglect of price impact and signal–price correlation). In a large economy, each trader’s influence on the price level, and the correlation between her signal and the price, are both vanishingly small, so one might expect price-impact and signal–price correlation neglect to be innocuous. We show that this is not the case: by making both assumptions, traders commit errors that, while individually small, aggregate to a non-trivial effect. As a result, the correlation-neglect equilibrium differs from both REE, in which traders account for the correlation between their signal and the price, and BNE, in which they account for both channels; see Appendices G and H. On the other hand, it is precisely these “errors”—specifically, traders’ neglect of the correlation between their signal and the price—that sustain a well-defined equilibrium without exogenous noise under correlation neglect (see Remark 7).

3 CHILE

We now pass to the limit $\mu \rightarrow 0$ ($n \rightarrow \infty$). In this limit, the population of agents becomes a continuum indexed by $a \in [0, 1)$ —a *large economy* (LE). The discrete signal (2) converges to

the continuous stochastic process

$$ds(a) = v da + \frac{1}{\sqrt{t(a)}} dB(a). \quad (6)$$

We call the increment $ds(a)$ the *signal of agent a* . Because signals vary across agents, the limiting information structure features *continuous-and-heterogeneous information* (CHI). Combined with the large economy, this defines the class of models we call CHILE.

To represent the CHI structure formally, we define the information available to agents in segment $[b, c)$ as the σ -algebra $\mathcal{F}_{b,c} = \sigma(\{s(z) - s(b)\}_{b \leq z < c})$, and assume that $\mathcal{F}_{b,c}$ is precisely the information set obtained by combining the information of all agents within $[b, c)$. The information available in the entire economy is $\mathcal{F}_{0,1}$, denoted hereafter by $\mathcal{F}_1$.

There are two equilibrium objects of primary interest, from which all others are derived: a cumulative demand function, $X_*(a, P)$, and a market-clearing price function, $\mathbf{P}_*$. We denote their out-of-equilibrium counterparts by dropping the asterisk. We discuss the former in Section 3.1 and the latter in Section 3.2; the equilibrium concept is introduced in Section 3.3.

3.1 Cumulative demand

We define cumulative demand in the continuous economy as the limit of its discrete counterpart. In the economy with size parameter μ , the optimal cumulative demand up to agent a is $\sum_{i: a_i < a} x^*(\Delta s_i, P, a_i, \mu)$, where x^* solves (5). The *optimal cumulative demand* in CHILE is then

$$X_*(a, P) = \lim_{\mu \rightarrow 0} \sum_{i: a_i < a} x^*(\Delta s_i, P, a_i, \mu). \quad (7)$$

We first establish when this limit exists.

Lemma 1 (Efficiency condition). *The limit in (7) exists if, and only if*

$$\int_{\mathbb{R}} (V(v) - P) g(v, P, 0) dv = 0. \quad (8)$$

3.1.1 Efficiency condition

The condition (8) can be rewritten as

$$\mathbb{E}[V(v) - P \mid P] = 0, \quad (9)$$

where the expectation is taken under the conditional distribution of $v \mid P$ implied by the limiting price belief, i.e., the density proportional to $g(v, P, 0)$. Therefore, the optimal cumulative demand is well-defined only if the market is weak-form efficient in the sense of (9). The intuition is that a large market with finite asset supply cannot accommodate non-infinitesimal trades from infinitely many agents: if, for example, $\mathbb{E}[V(v) \mid P] > P$, every agent would buy a finite amount, and the cumulative demand would explode. Consequently, price functions that violate (9) are not viable candidates for equilibrium.¹⁸ Hereafter, we restrict attention to price beliefs $g(v, P, \mu)$ for which the efficiency condition holds in the limit.

3.1.2 Itô representation

A key source of tractability in CHILE is that cumulative demands admit the following representation.

Lemma 2 (Representation lemma). *There exist deterministic functions $\beta(a, P) : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$ and $\delta(a, P) : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$X_*(a, P) = \int_0^a \beta(b, P) ds(b) + \int_0^a \delta(b, P) db, \quad (10)$$

Two properties of (10) merit emphasis. First, cumulative demand is an Itô integral, mirroring the structure of the cumulative signal (6). The noise in individual demands does not wash out in aggregation; instead, it accumulates into a finite random variable. Second, the demand of each agent is linear in his signal: the increment $dX(a, P) = \beta(a, P) ds(a) + \delta(a, P) da$ depends on $ds(a)$ only through the linear term $\beta(a, P) ds(a)$. This linearity, which makes learning from

¹⁸If the noisy supply grows proportionally to the number of agents, as in Hellwig (1980), supply and demand grow at the same rate, so their ratio remains finite and an equilibrium that is not weak-form efficient can be sustained.

prices and market clearing tractable, holds for general preferences and payoff functions. Prior literature obtains linearity in signals only under specific preference–payoff combinations, typically CARA–Normal. See Section B.2 for an intuitive discussion of why this property emerges in our framework.

3.2 Market clearing and price inference

Turning to market clearing, we require that the equilibrium price $\mathbf{P}_*$ be $\mathcal{F}_1$ -measurable, as the price cannot reflect more information than what is available in the entire economy. Given (10), when agents observe the price realization $\mathbf{P}_* = P$, they infer a sufficient statistic:

$$s_p \equiv \frac{\int_0^1 \beta(a, P) ds(a)}{\int_0^1 \beta(a, P) da}.$$

This statistic is derived from the market-clearing condition as follows:

$$\begin{aligned} \int_0^1 dX_*(a, P) &= \int_0^1 \beta(a, P) ds(a) + \int_0^1 \delta(a, P) da = \Theta(P) \Rightarrow \\ s_p &= \frac{\Theta(P) - \int_0^1 \delta(a, P) da}{\int_0^1 \beta(a, P) da} \equiv h(P). \end{aligned} \quad (11a)$$

Moreover, s_p can be written as

$$s_p = \int_0^1 \omega(a, P) ds(a) = v + \int_0^1 \omega(a, P) \frac{dB(a)}{\sqrt{t(a)}} \quad (11b)$$

where

$$\omega(a, P) = \frac{\beta(a, P)}{\int_0^1 \beta(a, P) da}. \quad (11c)$$

That is, the price reveals a weighted-average signal, formed by weighting the signal of each agent a by $\omega(a, P)$, which is proportional to a 's trading intensity, $\beta(a, P)$.¹⁹ Although (11c) writes the weight as $\omega(a, P)$, in equilibrium it does not depend on the price. Anticipating the multiplicative separability of the trading intensity established in Lemma 7, $\beta(a, P) =$

¹⁹Thus, our economy features generalized linear equilibria, as in Breon-Drish (2015), Glebkin et al. (2021), and others.

$\beta_a(a) \beta_P(P)$, the common price-dependent factor $\beta_P(P)$ cancels in the normalization (11c), leaving the deterministic weight

$$\omega(a) = \frac{t(a)/\rho(a)}{\int_0^1 t(b)/\rho(b) db}.$$

We therefore write $\omega(a)$ in place of $\omega(a, P)$ hereafter.

We impose the following restriction on $h(P)$ in equilibrium.

Assumption 1. *The function $h(P)$ defined in (11a) is strictly monotone.*

We maintain Assumption 1 throughout, including in our analysis of the competitive REE and the BNE in Appendices G and H. Under this assumption, the equilibrium price $\mathbf{P}_*$ and the sufficient statistic s_p are informationally equivalent; the information conveyed by the price thus *originates in*, and coincides with, the information contained in aggregate demand through the market-clearing relation (11a).²⁰ The sufficient statistic s_p in (11b) has the familiar “truth plus noise” form, and both components—the truth, v , and the noise, represented by the Itô integral in (11b)—are normally distributed. Standard Bayesian updating with normal random variables then yields the following characterization.²¹

Lemma 3 (Price Inference). *The conditional distribution of v given $\mathbf{P}_* = P$ is*

$$\mathcal{N}\left(\frac{\tau_p}{\tau} s_p, \frac{1}{\tau}\right).$$

The sufficient statistic s_p can be computed as $s_p = h(P)$, with $h(P)$ as in (11a). Here $\tau = \text{Var}[v|P]^{-1} = \tau_p + \tau_v$, and τ_p is the precision of s_p , given as

$$\tau_p = \text{Var}[v|s_p]^{-1} - \text{Var}[v]^{-1} = \left(\int_0^1 \frac{\omega(a)^2}{t(a)} da \right)^{-1}. \quad (12)$$

²⁰In particular, Assumption 1 rules out fully revealing equilibria of the kind constructed by Grossman (1976). In such an equilibrium, each trader conditions on a price that already reveals a sufficient statistic for all signals and therefore optimally disregards her own signal, so that her demand—and hence aggregate demand—is insensitive to information and reduces to a pure risk-sharing trade. Aggregate demand is then non-stochastic, yet the price is fully informative: the information in the price does not originate in aggregate demand. Requiring that the price’s information originate in aggregate demand thus excludes equilibria in which prices aggregate information that no individual trade reflects.

²¹See Section B.3 for a heuristic derivation and an intuitive discussion of Lemma 3.

Remark 4. Our model does not require noise traders. Traditional large economies require noise traders to prevent prices from being fully revealing: the noise in private signals washes out in aggregation due to the exact Law of Large Numbers, so without noise traders the price would be fully informative. In our economy, however, the aggregate signal does not reveal the fundamental, despite aggregating a large number of private signals. As in [Avdis \(2018\)](#), the aggregate noise entering prices takes the form of an Itô integral—a random variable with positive and finite variance.²²

3.3 Notion of correlation-neglect equilibrium

We now formalize the equilibrium concept for our economy.

Definition 1. A *correlation-neglect equilibrium* is a cumulative demand function $X_*(a, P)$ and an $\mathcal{F}_1$ -measurable price function $\mathbf{P}_*$ such that

- (i) $\mathbf{P}_*$ clears the market, i.e. $X_*(1, \mathbf{P}_*) = \Theta(\mathbf{P}_*)$, and
- (ii) $X_*(a, P)$ is optimal given $\mathbf{P}_* = P$.

Here optimality means that the cumulative demand $X_*(a, P)$ is given by (7) with the underlying individual demands given by (5). Moreover, we require that the sequence of price beliefs $g(v, P, \mu)$ entering (5) be such that the limit in (7) exists and equals $X_*(a, P)$, and that the limiting price belief $g(v, P, 0)$ coincides with the density of the joint distribution of v and $\mathbf{P}_*$.

Definition 1 combines market clearing with demand optimality, following standard conventions.

Remark 5 (Aggregates vs. averages in large economies). For any quantity Y defined at the agent level, one can consider both the *aggregate*, $\sum_{i=1}^n Y_i$, and the *average*, $\frac{1}{n} \sum_{i=1}^n Y_i$. In a finite economy, both are well-defined and generically nonzero. In a large-economy limit, however, the two cannot both be finite and nonzero: if the average has a finite, nonzero limit

²²Our price noise also ensures there is trade in our large economy in a way that, unlike [Avdis \(2018\)](#), is consistent with the notion of perfect competition in [Hellwig \(1980\)](#). See [Remark 7](#) and our literature review for details.

as $n \rightarrow \infty$, the aggregate diverges; conversely, if the aggregate has a finite, nonzero limit, the average vanishes. Any large-economy model will therefore fail to match real-world behavior for at least one of these two versions of each quantity. This is not a deficiency of the modeling approach—it simply determines which version is the appropriate one to compare with data.

To see this concretely, consider the comparison between CHILE and the economies in [Hellwig \(1980\)](#) and [Admati \(1985\)](#) (hereinafter, “traditional large economies”):

	CHILE		Traditional	
	Average	Aggregate	Average	Aggregate
Wealth	finite	∞	finite	∞
Trade	0	finite	finite	∞
Risk compensation	0	finite	finite	∞

In both frameworks, individual wealth does not scale with the number of agents, so average wealth is finite and nonzero while aggregate wealth diverges. The two approaches differ in trade and the risk premium. In traditional large economies, each agent’s trade and the per-unit risk premium $\mathbb{E}[V \mid P] - P$ are finite and nonzero, so the corresponding economy-wide totals—aggregate trade $\sum_i x_i$ and the *scaled risk premium* $\psi(P) = \lim_{n \rightarrow \infty} n (\mathbb{E}[V \mid P] - P)$ (defined formally in Section 4 below)—diverge.²³ In CHILE, individual trade and the per-unit risk premium vanish in the limit, reflecting the fact that each agent’s risk exposure is infinitesimal. In contrast, aggregate trade $\Theta(P)$ and the scaled risk premium $\psi(P) = \lim_{n \rightarrow \infty} n (\mathbb{E}[V \mid P] - P)$ are finite and economically meaningful objects in our framework, as we discuss below. Appendix J provides a self-contained illustration of this phenomenon in the simplest possible setting: a standard competitive economy with identical uninformed agents.

Remark 6 (Bringing aggregate predictions to the data). We make two observations about taking the aggregate predictions of CHILE to the data. First, in any time-series setting in which the number of market participants is fixed or slowly varying, the aggregate and per-capita versions of a given quantity differ only by a positive multiplicative constant (the number of traders, n). Time-series regressions on either object therefore yield coefficients of the same sign, and our

²³See [Hellwig \(1980\)](#), Section 5, eq. (B.2).

comparative-statics results for aggregate quantities carry the same empirical content as the corresponding statements for their per-capita counterparts. Second, the two scaling assumptions can be distinguished empirically by examining how outcomes depend on market size: finite-aggregate models such as CHILE predict that aggregate quantities (e.g., aggregate trade and the scaled risk premium) are insensitive to n in large markets, whereas finite-average models such as Hellwig (1980) and Admati (1985) predict that per-capita quantities are insensitive to n . Cross-sectional or panel comparisons across markets of different size can therefore inform which scaling system provides the more appropriate empirical benchmark.

4 Correlation-neglect equilibrium in CHILE

We present the main theorem of the paper. Appendix B provides a heuristic derivation of its key statements, while a rigorous proof is given in Appendix C.5.

Theorem 1. *There exists a unique equilibrium. The equilibrium price function has the representation $\mathbf{P}_* = \mathcal{P}(s_p)$, where*

$$s_p = \int_0^1 \omega(a) ds(a) = v + \int_0^1 \frac{\omega(a)}{\sqrt{t(a)}} dB(a) \quad (13)$$

is the equilibrium sufficient statistic and $\omega(a)$ is a weighting function given by

$$\omega(a) = \frac{t(a)}{\rho(a)} \left(\int_0^1 \frac{t(b)}{\rho(b)} db \right)^{-1}. \quad (14)$$

The function $\mathcal{P}(x)$ is given by

$$\mathcal{P}(x) = \int V \left(\frac{\tau_p}{\tau} x + \frac{z}{\sqrt{\tau}} \right) d\Phi(z). \quad (15)$$

Here $\Phi(z)$ denotes the standard normal cumulative distribution function (cdf). Consequently, the price function is completely determined by $V(\cdot)$ and two other quantities, the precision of

s_p , given by

$$\tau_p = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a)} da \right)^2}{\int_0^1 \frac{t(a)}{\rho(a)^2} da}, \quad (16)$$

and the posterior precision of v , given by $\tau = \mathbb{V}ar(v|P)^{-1} = \tau_v + \tau_p$.

The equilibrium cumulative demand function has the representation $dX(a) = \beta(a, P)ds(a) + \delta(a, P)da$, where

$$\begin{aligned} \beta(a, P) &= \frac{t(a)}{\rho(a)} \frac{\tau^{-1} \mathbb{E}[V'(v)|s_p]}{\text{Var}[V(v)|s_p]}, \text{ and} \\ \delta(a, P) &= \frac{\beta(a, P)^2}{2t(a)} \pi(a) \frac{\text{Sk}[V(v)|s_p]}{\text{Var}[V(v)|s_p]} - \beta(a, P) \frac{\mathbb{E}[v(V(v) - P)^2|s_p]}{\text{Var}[V(v)|s_p]} + \frac{\psi(P)}{\rho(a) \text{Var}[V(v)|s_p]}, \end{aligned} \quad (17)$$

where the function $\psi(P)$ is given by

$$\psi(P) \equiv \lim_{\mu \rightarrow 0} \frac{1}{\mu} \frac{\int_{\mathbb{R}} (V(v) - P) g(v, P, \mu) dv}{\int_{\mathbb{R}} g(v, P, \mu) dv}. \quad (18)$$

Here, $\rho(a)$ denotes the absolute risk aversion and $\pi(a)$ denotes the absolute prudence coefficient, defined as

$$\rho(a) = -\frac{u''(W_0(a))}{u'(W_0(a))}, \quad \pi(a) = -\frac{u'''(W_0(a))}{u''(W_0(a))}. \quad (19)$$

The sufficient statistic s_p is related to the price P as follows:

$$s_p = \mathcal{P}^{-1}(P),$$

where $\mathcal{P}^{-1}(\cdot)$ is the inverse of the function $\mathcal{P}(\cdot)$ defined in (15). The conditional moments of $V(v)$ and the function $\psi(P)$ are given in closed form in Appendix C.5.

The theorem above highlights the notable tractability of CHILE. All equilibrium objects are available in closed form despite rich heterogeneity and the generality of preferences. We will make use of this tractability in Section 7, where we examine the effects of changes in concentration on market quality. We now discuss the main features of our equilibrium.

Trading intensity $\beta(a, P)$. The trading intensity has an intuitive structure. First, $\beta(a, P)$ is proportional to $t(a)/\rho(a)$: More informed traders (those with higher $t(a)$) and more risk-

tolerant ones (those with higher $1/\rho(a)$) trade more aggressively. Second, $\beta(a, P)$ is inversely proportional to $\text{Var}(V(v)|P)$: Higher uncertainty makes all traders scale down their trading intensities. Third, $\beta(a, P)$ is proportional to $\mathbb{E}[V'(v)|P]$: Trading intensities are higher for assets with payoffs that are more sensitive to changes in fundamentals.²⁴

The key results of our paper operate through *wealth effects*. With non-CARA utility, the risk tolerance $1/\rho(a)$ depends on the initial wealth $W_0(a)$. In a plausible case of decreasing absolute risk aversion, risk tolerance increases with wealth. In this scenario, Theorem 1 suggests that wealthier investors trade more aggressively, and their signals receive greater weight in the price.

Price function. The price reflects the weighted average of traders' signals $ds(a)$. Moreover, the weights are proportional to $t(a)/\rho(a)$: the signals of more informed and more risk-tolerant traders have greater weights. This is intuitive as such traders trade more aggressively.

We also note that the closed-form expression (15) for the price function is a restatement of the efficiency condition (9). Indeed, Lemma 3 implies that conditional distribution of v given P is Normal with a mean of $\tau_p/\tau \cdot s_p$ and a variance of $1/\tau$. Thus, a random variable $z = \sqrt{\tau}(v - \tau_p/\tau \cdot s_p)$ has standard normal distribution. Therefore, one can write $v = \tau_p/\tau \cdot s_p + z/\sqrt{\tau}$. Substituting this into the efficiency condition yields $\mathbb{E}[V(v) | P] = \mathbb{E}[V(\tau_p/\tau \cdot s_p + z/\sqrt{\tau})] = \int V(\tau_p/\tau \cdot s_p + z/\sqrt{\tau}) d\Phi(z)$. The same change of variable is used below to obtain closed-form expressions for other conditional moments of $V(v)$.

The coefficients $\delta(a, P)$. The first two terms in (17) indicate that our equilibrium is influenced by the higher moments of the payoff (third central moment (“skewness”), $\text{Sk}(V(v)|P)$), as well as by the higher derivatives of the utility function (prudence, $\pi(a)$). This contrasts with the approach of Peress (2004), where these effects do not play a role. Peress (2004) employs a “small risk” approximation (assuming the variance of the fundamental is small), rendering higher-order risk negligible in his model. In contrast, we use a “small information” approximation, where the riskiness of the asset remains substantial even in the limit.²⁵

²⁴The last property is intuitive because higher $\mathbb{E}[V'(v)|P]$ implies that news about fundamentals is more payoff-relevant: signals about fundamental v tell more about payoffs $V(\cdot)$.

²⁵Other papers that use a local second-order approximation to the utility function, such as Samuelson (1970), Campbell and Viceira (2002), Farboodi, Singal, Veldkamp, and Venkateswaran (2025), and Mihet (2025), also

The last term in (17) represents the demand of a trader with no private signal, referred to as “uninformed demand” hereafter. To see this, note that if we set $t(a) = 0$, the term $\psi(P)/(\rho(a)\text{Var}[V(v)|P])$ becomes the only non-zero component in the coefficients $\beta(a, P)$ and $\delta(a, P)$. To better understand this term, let us first discuss the risk premium in our economy.

Risk premium. At first glance, the efficiency condition (9) appears to imply the absence of a risk premium in our economy. However, a more accurate characterization is that the risk premium in CHILE is infinitesimal rather than exactly zero:

$$\mathbb{E}[V(v) - P \mid P] = \psi(P) da.$$

That is, $\psi(P)$ —defined in (18)—is the rate at which the conditional risk premium vanishes as the economy grows large. This mirrors how individual demands in our model are infinitesimal rather than strictly zero.

The expression above indicates that the risk premium in a discrete economy with trader mass $\mu = da$ is given by $\psi(P) da$. We refer to $\psi(P)$ as the *scaled risk premium*: in a discrete economy with n traders ($\mu = 1/n$), $\psi(P)$ equals the risk premium $\mathbb{E}[V(v) - P \mid P]$ rescaled by n , so that $\psi(P) = \lim_{n \rightarrow \infty} n \mathbb{E}[V(v) - P \mid P]$ has a finite limit as the economy grows. By Remark 6, our comparative statics for $\psi(P)$ in Appendix I have the same empirical content as the corresponding statements for the risk premium $\mathbb{E}[V(v) - P \mid P]$, which is the object most commonly measured in the empirical literature.

In CHILE, $\psi(P)$ is finite because a finite amount of aggregate risk is shared among an infinite number of traders: each trader earns an infinitesimal premium, and these premiums accumulate, after rescaling by n , to a finite scaled premium.

With this understanding, the demand of an uninformed trader (i.e., one with $t(a) = 0$) takes

do not account for higher-order effects. However, our results demonstrate that a second-order approximation is not always sufficient. Under small information asymptotics employed here, a third-order approximation to utility is necessary. Even under small-risk asymptotics, the second-order approximation may not be valid. For example, in the setting of Peress (2004), it is valid when traders learn about the mean of the fundamental, but it fails when they learn about the payoff itself (see Peress (2011)).

the form:

$$dX^u = \frac{\psi(P)}{\rho(a)\text{Var}[V(v) | P]} da = \frac{\mathbb{E}[V(v) - P | P]}{\rho(a)\text{Var}[V(v) | P]}.$$

This expression coincides with the standard mean-variance demand function.

Remark 7 (Well-defined equilibrium without noise). Our equilibrium exists and features trade despite lacking external noise—a property that may seem surprising given well-known results (Milgrom and Stokey, 1982; Tirole, 1982). By conjecturing that the noise in their signals is independent of that in the price, our agents treat their signals as incrementally informative relative to the price, in contrast to Grossman (1976)’s agents who treat their signals as dominated by the price. As a result, our agents employ their signals to formulate their demands, and trade ensues. It is this belief structure—the conditional independence conjecture—rather than the continuous heterogeneous information technology per se that sustains trade and prevents full revelation. Indeed, Appendices G and H show that under the REE and BNE equilibrium concepts, where agents correctly account for the informational content of prices, trade breaks down without exogenous noise.

Nevertheless, we stress that our agents make the right price conjectures in the large limit. Although it may thus appear that trade would disappear in the limit, within the logic of our model the no-trade intuition works differently. As the economy grows, the size of individual demands shrinks to zero, but the number of agents grows, with cumulative demand counterbalancing their shrinking size by summing up more demand functions. This balance is maintained all the way to the large limit, guaranteeing that aggregate demand converges to a well-defined and non-trivial quantity. Restated colloquially, and perhaps paradoxically, our model features a no-trade theorem at the individual level—since each individual’s demand converges to zero—but not at the aggregate level, as cumulative demands remain nonzero.

Remark 8 (CNE vs REE). As highlighted in Hellwig (1980), another concern with traditional models of competition under asymmetric information lies within their version of noiseless equilibrium: the information conveyed by prices is unaffected by preferences (see also Grossman, 1976). As Hellwig points out, this is somewhat counter-intuitive: “One would expect that the weight with which agent i ’s information I_i affects the equilibrium price should depend on the strength of agent i ’s reaction to this information, which in turn should depend on his pref-

erences. Presumably, it should make a difference whether the news of an increase in a firm’s profits is passed to somebody who is almost risk-neutral and responds by buying a large number of shares or whether this piece of news is passed to a risk averter who hardly responds at all.” As we discuss below, the informational content of prices *does* depend on preferences in our equilibrium, and in a way that aligns with the intuition in the quote.

5 Market quality

Market participants come in very different sizes: financial institutions differ in the assets they manage, and households and individuals in the wealth they invest. A growing literature establishes that these differences—which it calls “wealth inequality” for individual investors and “concentration” for institutional ones—affect asset prices, and thereby returns, risk premia, and risk-free rates (see [Panageas, 2020](#), for a recent survey). We adopt *concentration* as our umbrella term. Since prices are but one outcome of financial trade, concentration plausibly shapes market quality—liquidity and informational efficiency—as well. This is especially so when participants are *granular*, so that idiosyncratic shocks to them move aggregate outcomes rather than averaging out ([Gabaix, 2011](#)): because shocks to larger investors have a bigger footprint, the size distribution governs how individual shocks aggregate and, in turn, how they shape market quality. Whether concentrating the market improves or degrades efficiency and liquidity is thus a genuinely open question.

Answering it calls for an asset-pricing framework in which (i) wealth influences how investors trade and (ii) individual shocks do not wash out in aggregation—and, since the object of study is market quality, one with asymmetric information. Such frameworks are hard to come by: CARA preferences exclude wealth effects and so fail requirement (i), while models that do admit them, notably [Peress \(2004\)](#), rely on a continuum of investors whose idiosyncratic shocks cancel by the law of large numbers, as in [Hellwig \(1980\)](#), and so fail requirement (ii). CHILE meets both requirements—its general, heterogeneous preferences accommodate wealth effects, while the noise in investors’ private signals aggregates into a finite, non-vanishing random variable rather than washing out—and, together with its tractability, this lets us pose and answer the

question sharply. We now introduce our measures of market quality; Section 7 then examines how they respond to changes in the size distribution.

Our measure of information efficiency is based on how much prices reduce uncertainty about the fundamental and is defined as

$$\mathcal{I} = 1 - \frac{\text{Var}(v|P)}{\text{Var}(v)}.$$

This measure is common in both theoretical (e.g., [Rostek and Weretka, 2012](#)) and empirical work (e.g., [Dessaint, Foucault, and Frésard, 2024](#); [Dávila and Parlatore, 2025](#)). In practice, $\mathcal{I}$ corresponds to the R^2 of predicting fundamentals by prices, where the fundamental v is typically proxied by earnings in empirical work.

Our measure of liquidity is based on how much supply shocks can move prices

$$\mathcal{L} = - \left(\frac{\partial \mathcal{P}(s_p, \bar{\theta})}{\partial \bar{\theta}} \bigg|_{\mathcal{P}=P} \frac{1}{\text{Var}(V|P)} \right)^{-1}. \quad (20)$$

The first term in (20) captures price sensitivity to unexpected price-inelastic supply shocks, a standard measure of liquidity in the theoretical literature (e.g. [Vayanos and Wang, 2013](#)).²⁶ As our equilibrium is generally non-linear, this term depends on the realized price signal s_p , a stochastic quantity with few (if any) empirical counterparts. As we show next, scaling by $\text{Var}(V|P)$ in (20) allows us to obtain the risk-adjusted liquidity measure that does not depend on the aggregate signal, and we thus adopt it as our primary notion of liquidity.²⁷

Proposition 1. *The equilibrium expressions for information efficiency $\mathcal{I}$ and liquidity $\mathcal{L}$ are*

$$\mathcal{I} = \left(1 + \tau_v \frac{\int_0^1 \frac{t(a)}{\rho(a)^2} da}{\left(\int_0^1 \frac{t(a)}{\rho(a)} da \right)^2} \right)^{-1}, \quad \mathcal{L} = \frac{\int_0^1 \frac{t(a)}{\rho(a)^2} da}{\int_0^1 \frac{t(a)}{\rho(a)} da}.$$

²⁶By “unexpected” we mean that agents do not adjust their demand coefficients β and δ in response to a change in $\bar{\theta}$.

²⁷Our liquidity measure is the inverse of price impact per unit of risk. Price impact is proportional to variance, as shown in [Kyle \(1985\)](#) and [Kyle \(1989\)](#) for the single-asset case, and in [Rostek and Weretka \(2015\)](#) and [Malamud and Rostek \(2017\)](#) for the multi-asset case. Without normality, the price impact is proportional to risk-neutral variance, as demonstrated in [Glebkin et al. \(2023a\)](#). Therefore, scaling by variance, as in (20), allows for better cross-sectional comparisons consistent with empirical practice.

As we see above, information efficiency and liquidity are in closed form. What is more, they are parsimoniously characterized by just three primitives of the economy, two profiles and one parameter: risk tolerances $1/\rho(a)$, precisions $t(a)$, and prior uncertainty $1/\tau_v$. This enables tractable comparative statics that we turn to next.

6 A CRRA–log-normal example

To build intuition for the general comparative statics that follow, we specialize the equilibrium of Theorem 1 to a tractable parametric case. We assume (i) CRRA preferences with common relative risk aversion η , so that $\rho(a) = \eta/W_0(a)$; (ii) common signal precision $t(a) = t$; and (iii) log-normal payoffs $V(v) = e^v$. For the statements about *aggregate dollar demand* below—and the demand elasticity derived in Appendix I.3—we additionally impose (iv) that supply is inelastic in dollar terms: $\bar{\theta} = 0$ and $\theta(P) \propto 1/P$, so that dollar supply $P \Theta(P)$ is constant. The price-informativeness, liquidity, and information-efficiency results below do not rely on (iv) and hold for an arbitrary supply schedule $\Theta(P)$.

Equilibrium in log prices. Under assumptions (i)–(iii), the price function (15) becomes log-linear in the sufficient statistic:

$$\ln P = \frac{\tau_p}{\tau} s_p + \frac{1}{2\tau}. \quad (21)$$

The weighting function (14) reduces to $\omega(a) = W_0(a)/\bar{W}$, where $\bar{W} = \int_0^1 W_0(a) da$ is average wealth: each trader’s weight in the price equals her wealth share.

The dollar trading intensity, $P \beta(a, P)$ —the trading intensity converted from shares to dollars—is price-independent:

$$P \beta(a, P) = \frac{t W_0(a)}{\eta \tau (e^{1/\tau} - 1)}, \quad (22)$$

This price-independence pins down only the signal-loading component of demand. Under the dollar-inelastic supply of (iv), the supply-accommodating component $P \int_0^1 \delta(a, P) da$ is affine in

in P as well, so *total* aggregate dollar demand is linear in the log price, with slope proportional to aggregate risk tolerance $\bar{W}/\eta$; Appendix [I.3](#) carries out this decomposition and derives the resulting demand elasticity.

A convenient feature of the CRRA–log-normal case is that market quality can be expressed in terms of a single, empirically observable statistic: the Herfindahl–Hirschman Index (HHI) of the wealth distribution, which is widely used to measure AUM concentration.

Price informativeness and liquidity via the HHI. Define the Herfindahl–Hirschman Index (HHI) of the wealth distribution as

$$\text{HHI} = \int_0^1 \left(\frac{W_0(a)}{\bar{W}} \right)^2 da = \frac{\overline{W^2}}{\bar{W}^2},$$

where $\overline{W^2} = \int_0^1 W_0(a)^2 da$. By Jensen’s inequality $\text{HHI} \geq 1$, with equality if and only if all agents have the same wealth. Substituting $\rho(a) = \eta/W_0(a)$ and $t(a) = t$ into the expressions in Proposition [1](#):

$$\tau_p = \frac{t}{\text{HHI}}, \quad \mathcal{L} = \frac{\bar{W}}{\eta} \text{HHI}. \quad (23)$$

Information efficiency becomes

$$\mathcal{I} = \frac{t}{t + \tau_v \text{HHI}}. \quad (24)$$

Equations [\(23\)](#)–[\(24\)](#) deliver a sharp message. Greater wealth concentration (higher HHI) reduces price informativeness and information efficiency, because it tilts the price weights toward a smaller effective number of signals. At the same time, greater concentration increases liquidity, because a few wealthy, risk-tolerant traders absorb supply shocks more aggressively. Note that for a fixed wealth distribution, a higher average wealth $\bar{W}$ raises liquidity but leaves informativeness unchanged.

In the next two sections we show that these findings—the opposing effects of concentration on informativeness and liquidity—hold under much more general conditions.

The scaled risk premium $\psi(P)$ also depends on HHI through the posterior precision $\tau = \tau_v + t/\text{HHI}$. Because the dependence of ψ on HHI is mediated by nonlinear conditional moments

of the log-normal payoff, the comparative statics of ψ are richer than those of $\mathcal{I}$ and $\mathcal{L}$; we explore this in Appendix I.

7 Comparative statics

We now explore how market quality is affected by concentration, starting with concentration in wealth, followed by concentration in information. In more concrete terms, we consider an equilibrium object $\mathcal{O}$ —our placeholder notation for either information efficiency $\mathcal{I}$ or liquidity $\mathcal{L}$ —and we change the primitives $W_0(a)$ and $t(a)$ around those associated with the original equilibrium. That is, the task at hand for this section is to carry out comparative statics with respect to functions. The right tool for this job is the Gateaux derivative.

Definition 2. The *Gateaux derivative* of an equilibrium object $\mathcal{O}$ with respect to a parameter $h(a)$ in the direction $h^\Delta(a)$ is

$$\mathcal{O}'(h(a)) [h^\Delta(a)] = \lim_{\varepsilon \rightarrow 0} \frac{\mathcal{O}(h(a) + \varepsilon h^\Delta(a)) - \mathcal{O}(h(a))}{\varepsilon}.$$

To facilitate exposition, we use the following conventions and types of notation in our definition above. The *parameter* (namely, a function) with respect to which we differentiate appears inside round brackets, while the *direction* (another function) along which we perturb the parameter appears inside square brackets. Our notation also distinguishes the parameter from the direction by indicating the direction with a Δ superscript.²⁸

Of particular interest are directions that correspond to reduced concentration. To this end, we refer to a variation of model parameters that makes poor agents better off (either richer or more informed) and rich agents worse off (either poorer or less informed) as a *Robin Hood variation*. We then index our agents in the same order as their wealth, implying that $W_0(a)$ *increases* in a , an assumption maintained hereafter without loss of generality. The poor and rich are defined with respect to thresholds $\underline{a}$ and $\bar{a}$, with the poor lying below $\underline{a}$ and the rich above $\bar{a}$.

²⁸Once we fix a direction, computing a Gateaux derivative can be done with standard calculus. One differentiates $\mathcal{O}(h(a) + \varepsilon h^\Delta(a))$ with respect to the scalar ε , evaluating the result at $\varepsilon = 0$.

Definition 3. A *Robin-Hood variation* of parameter $h(a)$ is a direction $h^\Delta(a)$, bounded over $a \in [0, 1]$, and two associated thresholds $\underline{a} < \bar{a}$ such that

- (i) $h^\Delta(a) \geq 0$ for $a < \underline{a}$,
- (ii) $h^\Delta(a) \leq 0$ for $a > \bar{a}$,
- (iii) $h^\Delta(a) = 0$ for $a \in [\underline{a}, \bar{a}]$, and
- (iv) $h^\Delta(a)$ is not zero for some set of indices a with positive Lebesgue measure.

Returning to our questions on concentration, we can arrive at answers by considering how Robin-Hood variations in wealth affect market quality. As, however, wealth effects may also enter through information acquisition, we must do so with care, isolating different possible effects. We thus first take Robin-Hood variations with respect to wealth keeping precisions fixed, then with respect to precisions keeping wealth fixed, then allowing for both effects by letting precisions depend on wealth.

As we see below, this sequence of comparative statics not only uncovers useful intuition but also reveals some surprising results. Before we proceed, however, we establish an important benchmark for the discussion of later subsections.

7.1 Ideal information aggregation in CHILE

We begin by pointing out that prices reveal the weighted average of all private signals—see equation (13). Revisiting how information is aggregated in our economy, we ask: is there an “informationally ideal” version of the weighting function $\omega(a)$, in the sense that the corresponding price would reveal maximum information to the agents?

We answer this question for a hypothetical weighting function $\omega^H(a)$, using it to define the notion of an *aggregate signal* as

$$s[\omega^H(a)] = \int_0^1 \omega^H(b) ds(b), \quad \omega^H(a) \geq 0, \quad \int_0^1 \omega^H(a) da = 1.$$

Writing $s_p = s[\omega(a)]$ then shows that the equilibrium-price weights $\omega(a)$ are but one possibility for $\omega^H(a)$, with the more general $\omega^H(a)$ yielding an aggregate signal with precision

$$\text{Var}(v|s[\omega^H(a)])^{-1} - \text{Var}(v)^{-1}.$$

The informationally-ideal weighting function maximizes this precision, and is characterized in the following result.

Lemma 4 (Ideal information aggregation). *The precision of any aggregate signal cannot exceed the cumulative precision of the entire economy, that is,*

$$\text{Var}(v|s[\omega^H(a)])^{-1} - \text{Var}(v)^{-1} \leq \int_0^1 t(a)da \quad (25)$$

for any weighting function $\omega^H(a)$. The weighting function that maximizes signal precision satisfies (25) as an equality, and is given by

$$\omega^I(a) = \frac{t(a)}{\int_0^1 t(a)da}. \quad (26)$$

As we can see in (26), if we were to design an aggregate signal with maximizing informativeness as our only goal, we should be weighing each signal in proportion to its precision. Comparing (26) with (14) now highlights the key to understanding information inefficiency: the informationally-ideal weights are proportional to precisions, $\omega^I \propto t(a)$, whereas the price weights ω are distorted by risk tolerances, $\omega \propto t(a)/\rho(a)$. With DARA, it is the wealthier agents that can tolerate more risk. Consequently, DARA preferences yield equilibria where prices overweight the signals of the rich, underweighting those of the poor.

7.2 Wealth concentration

Using the result above as an illuminating benchmark, we examine the effects of transferring wealth from the rich to the poor on market quality holding precisions fixed. To proceed, we need the following technical conditions.

Assumption 2. *The following hold for the profiles of wealth, $W_0(a)$, absolute risk tolerances, $1/\rho(a)$, and relative risk aversions, $\rho(a)W_0(a)$.*

1. *The cross-sectional cdf of wealth is continuous, strictly increasing, and has support $[0, \infty)$;*
2. *The cross-sectional cdf of absolute risk tolerances $1/\rho(a)$ is continuous and strictly increasing;*
3. *There exist constants $\underline{\eta}$ and $\bar{\eta}$ such that $0 < \underline{\eta} \leq \rho(a)W_0(a) \leq \bar{\eta} < \infty$.*²⁹

Condition (iii) restricts *relative* risk aversion, $\rho(a)W_0(a)$, to lie on a compact subset of $(0, \infty)$, bounded away from both 0 and ∞ . It is satisfied by standard preference specifications; in particular, it holds in the CRRA–log-normal example of Section 6, where relative risk aversion is constant, $\rho(a)W_0(a) = \eta$, so that the condition holds with $\underline{\eta} \leq \eta \leq \bar{\eta}$.

We also assume $t(a) > 0$, which means that in our comparative statics exercises, we only consider wealth transfers among informed traders. Changing wealth of uninformed (i.e., those with $t(a) = 0$) has no effect on market quality as is clear from Proposition 1. As in earlier parts of the paper, we maintain the above assumption, from its statement onwards. We next state the main result of this section.

Proposition 2. *Suppose that agent preferences are DARA. Then, there exist thresholds $0 < a_1^W \leq a_2^W < 1$, such that for any Robin Hood variation $W_0^\Delta(a)$ with $\underline{a} \leq a_1^W \leq a_2^W \leq \bar{a}$ we have*

$$\mathcal{I}'(W_0(a))[W_0^\Delta(a)] > 0 \text{ and } \mathcal{I}'(W_0(a))[-W_0^\Delta(a)] < 0; \quad (27)$$

$$\mathcal{L}'(W_0(a))[W_0^\Delta(a)] < 0 \text{ and } \mathcal{L}'(W_0(a))[-W_0^\Delta(a)] > 0. \quad (28)$$

Reducing concentration by transferring wealth from the sufficiently rich to the sufficiently poor leads to higher informational efficiency. The intuition behind this is rooted in the distortion highlighted in the previous section: the price overweights the signals of the rich and underweights the signals of the poor. By redistributing wealth from the wealthy to the poor, this distortion is corrected, thereby improving informational efficiency.

²⁹The cross-sectional cdf of wealth (resp., absolute risk tolerances) is defined as $F_{W_0}(x) = \Lambda(a : W_0(a) \leq x)$ (resp., $F_{1/\rho}(x) = \Lambda(a : 1/\rho(a) \leq x)$). Here $\Lambda(\cdot)$ denotes the Lebesgue measure.

We now turn to the liquidity result (28). There are two channels at play. First, a reduction in concentration is associated with increased information efficiency. When prices are more informative, traders become less willing to provide liquidity—for instance, by selling when prices increase—since higher prices are more likely to reflect stronger fundamentals. Second, there is the uncertainty reduction channel: higher information efficiency leads to less uncertainty about fundamentals, potentially decreasing $\text{Var}[V(v)|P]$ and making traders more willing to provide liquidity. By examining the risk-adjusted measure $\mathcal{L}$, we isolate the first effect. As a result, a reduction in concentration negatively impacts $\mathcal{L}$.

7.2.1 The role and economic interpretation of thresholds a_1^W and a_2^W

To build intuition for the thresholds in Proposition 2, consider the special case of homogeneous preferences. Denoting risk tolerance by $y(a) = 1/\rho(a)$, precision τ_p (monotonically related to $\mathcal{I}$) and liquidity $\mathcal{L}$ become

$$\tau_p = \frac{\left(\int_0^1 t(a)y(a) da\right)^2}{\int_0^1 t(a)y(a)^2 da}, \quad \mathcal{L} = \frac{\int_0^1 t(a)y(a)^2 da}{\int_0^1 t(a)y(a) da}. \quad (29)$$

Index traders so that $y(a)$ is increasing in a ; under DARA with homogeneous preferences, more risk-tolerant traders are also wealthier. A first-order expansion shows that increasing risk tolerance by ϵ for traders in $[a, a + da)$ changes precision and liquidity by

$$\begin{aligned} \delta\tau_p &= t(a) \frac{2\epsilon da}{\mathcal{L}} \left(1 - \frac{y(a)}{\mathcal{L}}\right), \\ \delta\mathcal{L} &= t(a) \frac{2\epsilon da}{\int_0^1 t(x)y(x) dx} \left(y(a) - \frac{\mathcal{L}}{2}\right). \end{aligned}$$

Here $\mathcal{L}$ can be interpreted as a weighted-average level of risk tolerance: it has the same units as y and lies between its minimum and maximum values.

These expressions pin down the thresholds. Precision τ_p increases with risk tolerance for agents below the average level $\mathcal{L}$ and decreases above it, yielding $y(a_2^W) = \mathcal{L}$. Liquidity $\mathcal{L}$ increases with risk tolerance for agents above $\mathcal{L}/2$ and decreases below it, yielding $y(a_1^W) = \mathcal{L}/2$.

Since $a_1^W \leq a_2^W$, a Robin Hood transfer that spans both thresholds improves informational efficiency but reduces liquidity.

Thus, both thresholds are determined by the weighted-average risk tolerance, which in equilibrium coincides with the liquidity measure $\mathcal{L}$. This connection is natural: markets populated by more risk-tolerant agents are, *ceteris paribus*, more liquid, so $\mathcal{L}$ simultaneously captures the representative level of risk-bearing capacity and the ease of trading.

7.3 Information concentration

Here, we examine the effects of changing the distribution of information across agents on market quality holding the wealth profile fixed.

Proposition 3. *Suppose that traders have DARA utilities. Then, there exist thresholds $0 < a_1^t \leq a_2^t < 1$, such that for any Robin Hood variation $t^\Delta(a)$ with $\underline{a} \leq a_1^t \leq a_2^t \leq \bar{a}$*

$$\begin{aligned} \mathcal{I}'(t(a))[t^\Delta(a)] &> 0 \text{ and } \mathcal{I}'(t(a))[-t^\Delta(a)] < 0; \\ \mathcal{L}'(t(a))[t^\Delta(a)] &< 0 \text{ and } \mathcal{L}'(t(a))[-t^\Delta(a)] > 0. \end{aligned} \tag{30}$$

Making the rich less informed and the poor more informed improves information efficiency but reduces liquidity. We discuss the intuition for the information result, from which the liquidity result follows. Recall the key distortion highlighted in Section 7.1: the trading intensities of the rich are too large, while those of the poor are too small. To correct this inefficiency, one needs to increase the trading intensities of the poor relative to the rich. This can be achieved by making the poor more informed and the rich less informed.

Note that the definition of a Robin Hood variation allows for weak inequalities. Therefore, the proposition above applies to a variation that decreases the precisions of the rich while leaving other precisions unchanged.

Corollary 1 (An information-aggregation paradox). *Suppose that traders have DARA utilities. Then, there exists a threshold a_2^t such that for any $t^\Delta(a) \neq 0$ such that $t^\Delta(a) \leq 0$ for $a > a_2^t$, and $t^\Delta(a) = 0$ otherwise, $\mathcal{I}'(t(a))[t^\Delta(a)] > 0$ and $\mathcal{I}'(t(a))[-t^\Delta(a)] < 0$.*

Weakly increasing the information of all traders can hurt information efficiency, while weakly decreasing it can have the opposite effect. [Banerjee, Davis, and Gondhi \(2018\)](#), [Dugast and Foucault \(2018\)](#), and [Glebkina and Kuong \(2023\)](#) also showed that improving the quality of private information can reduce information efficiency. In their studies, the mechanism is that giving more information to some traders invites more noise from another source. In our paper, we focus on a pure information aggregation channel: the only noise comes from the traders' signals themselves. Yet, reducing the noise in some traders' signals can harm information efficiency. This occurs because better information is aggregated less effectively: increasing the precisions of the rich makes them trade more aggressively, causing their signals to be even more overweighted in price, exacerbating the existing distortion.

The aforementioned papers also do not feature wealth effects, so the effect there is not specific to large traders (i.e., traders with large wealth). An implication of our result is that making large investors differentially more informed could harm information efficiency. One of the changes introduced in the MiFID II regulation was to unbundle investment research from trading costs. Simplifying, before MiFID, everyone who traded obtained information. After that, only whoever is willing to pay gets it. Since large traders have a higher value of information (see [Appendix D](#)), such regulation makes large traders differentially more informed, potentially negatively affecting price informativeness.

7.3.1 The role and economic interpretation of thresholds a_1^t and a_2^t

To build intuition for the thresholds in [Proposition 3](#), consider again the case of homogeneous preferences. Using the expressions for τ_p and $\mathcal{L}$ in [\(29\)](#), a first-order expansion shows that increasing precision by ϵ for traders in $[a, a + da)$ changes precision and liquidity by

$$\begin{aligned}\delta\tau_p &= \frac{\epsilon da y(a)}{\mathcal{L}} \left(2 - \frac{y(a)}{\mathcal{L}} \right), \\ \delta\mathcal{L} &= \frac{\epsilon da y(a)}{\int_0^1 t(x)y(x) dx} (y(a) - \mathcal{L}).\end{aligned}$$

These expressions pin down the thresholds. Increasing precision improves τ_p (and hence $\mathcal{I}$)

for agents with risk tolerance below $2\mathcal{L}$ and worsens it above, yielding $y(a_2^t) = 2\mathcal{L}$. Liquidity $\mathcal{L}$ increases with precision for agents above $\mathcal{L}$ and decreases below it, yielding $y(a_1^t) = \mathcal{L}$. Since $a_1^t \leq a_2^t$, a Robin Hood transfer in information that spans both thresholds improves informational efficiency but reduces liquidity.

The sign of $\delta\tau_p$ above the threshold $y(a_2^t) = 2\mathcal{L}$ reflects the information-aggregation paradox. The expression (29) for τ_p can be interpreted as the signal-to-noise ratio of the price signal $\int_0^1 \beta(a) ds(a) \propto v \int_0^1 t(a)y(a) da + \int_0^1 (t(a)y(a))/\sqrt{t(a)} dB(a)$: the numerator $N = \left(\int_0^1 t(a)y(a) da\right)^2$ is proportional to the variance of the signal component, while the denominator $D = \int_0^1 t(a)y(a)^2 da$ captures the variance of the noise. Agent a 's marginal contribution to the signal is proportional to $y(a)$, whereas the contribution to the noise is proportional to $y(a)^2$. For sufficiently risk-tolerant agents, the noise contribution dominates, so increasing their precision worsens price informativeness despite adding information to the economy.

7.4 Wealth concentration redux: the role of information acquisition

Do the results in Proposition 2 hold when the precisions are endogenous (i.e., can depend on $W_0(a)$)? Consider the information efficiency result. The mechanism there is that the signals of the rich are overweighted, while the signals of the poor are underweighted. With endogenous information acquisition, one can expect the rich to acquire higher quality signals: they trade more aggressively and thus have more use for their information, making them value it more.³⁰ Is overweighting the signals of better quality a bad idea?

The transfer of wealth from rich to poor has two effects, one direct and one indirect. Proposition 2 captures the direct one. Proposition 3 captures the indirect one, by decreasing the precisions of the rich and increasing that of the poor, via the dependence of precisions on wealth. Since both effects work in the same direction, the results (27)–(28) (see p.29) are reinforced in the presence of information acquisition. We summarize this in the proposition below.

Proposition 4. *Suppose that Assumption 2 holds. Suppose that traders have DARA utilities.*

³⁰In Appendix D, we consider the information acquisition problem and show that agent a 's precision $t(a)$ is indeed an increasing function of his wealth $W_0(a)$.

Suppose that precisions are a function of wealth $t(W_0(a), a)$ and are differentiable and increasing in $W_0(a)$ with a uniformly bounded marginal response t_W . Then, for any Robin Hood variation $W_0^\Delta(a)$ with $\underline{a} \leq \min\{a_1^t, a_1^W\} \leq \max\{a_2^t, a_2^W\} \leq \bar{a}$, results (27)–(28) hold.

The result of Proposition 4 is perhaps surprising. Since precisions are increasing in wealth, a Robin Hood variation could lead to a smaller overall amount of information being produced in the economy. Why does the information efficiency improve? Similar to our information aggregation paradox (Corollary 1), the effect of better aggregation of information dominates: Less information, but aggregated better results in more information efficiency.

8 Empirical implications

Beyond providing comparative statics, our results connect with several sets of empirical observations. First, regarding secular trends, it is known that while stocks have become more liquid since the beginning of the 20th century (e.g., [Chordia, Roll, and Subrahmanyam, 2001](#)), the informativeness of the average US stock has deteriorated ([Farboodi, Matray, Veldkamp, and Venkateswaran, 2022](#)). This is puzzling given the improved data availability in modern markets.³¹ Our model can explain both trends by appealing to the growing wealth inequality among individual investors ([Saez and Zucman, 2016](#)), or to the unequal distribution of assets under management for institutional investors ([Ben-David et al., 2021](#)).

Second, our results are consistent with recent empirical findings in the working paper by [Xiong, Yang, and Zheng \(2024\)](#), who document a negative relationship between concentration—measured by the Herfindahl-Hirschman Index (HHI) of fund assets under management (AUM) or the ownership share of the top five investors—and information efficiency, as defined in [Bai et al. \(2016\)](#).³² This relationship holds both at the market level and for individual stocks. Notably, what matters in their analysis is the concentration among *active* (i.e., informed) investors—consistent with our model, where uninformed demand can be absorbed into

³¹There is also evidence that the trend in price informativeness is not uniform: [Bai, Philippon, and Savov \(2016\)](#) find that the informativeness of S&P 500 firms has increased over time, while [Farboodi et al. \(2022\)](#) show that the negative average trend is driven by small firms.

³²They also examine the effects of an exogenous increase in inequality driven by fund mergers.

the supply function $\Theta(P)$ and plays no role in determining price informativeness.

Third, our model speaks to the recent rise in retail investor participation. If the smallest market participants are interpreted as retail investors, their private signals are systematically underweighted in equilibrium—a consequence of wealth effects under DARA preferences. Increasing retail participation generates two predictions: price informativeness should increase, while liquidity should decrease.³³ The first prediction is consistent with [Boehmer, Jones, Zhang, and Zhang \(2021\)](#), who find that retail order imbalances positively predict future stock returns, with predictive power that persists beyond what order-flow persistence, contrarian trading, or public news sentiment can explain—indicating that retail orders carry information not yet incorporated into prices. The second prediction is consistent with [Eaton, Green, Roseman, and Wu \(2022\)](#), who exploit brokerage outages as quasi-natural experiments and find that removing small, inexperienced investors from the market is associated with improved liquidity and lower volatility for stocks with high retail interest.

9 A summary of extensions and robustness exercises

This section summarizes additional robustness checks and model extensions, with full analyses presented in the Appendix.

- Appendix [E](#) shows that our main results are unique to the CHILE framework and do not extend to traditional large-economy models based on law-of-large-numbers (LLN) aggregation, such as [Hellwig \(1980\)](#), [Admati \(1985\)](#), and [Peress \(2004\)](#). The key insight is an aggregation result: even when investors differ in wealth, precision, and preferences, the economy remains observationally equivalent to one with homogeneous agents. As a result, Robin Hood variations that preserve aggregate trading intensity have no effect on market quality—unlike in CHILE.
- Appendix [F](#) incorporates noise traders into our main model. The resulting equilibrium

³³Formally, we model the rise of retail participation as a Robin Hood variation that increases the wealth of the smallest market participants while leaving the rest of the distribution unchanged. Because these agents lie below the threshold a_1^W in Proposition [2](#), this variation improves informational efficiency and reduces risk-adjusted liquidity.

captures both the classical mechanism—where inequality improves information efficiency via greater aggregate trading intensity—and the novel mechanism introduced in this paper, where inequality disrupts signal aggregation. The key result is that the latter effect dominates, making our core conclusion—that transferring wealth from rich to poor improves information efficiency but reduces liquidity—robust to the presence of noise.

- Appendix G compares three equilibrium concepts in a transparent CARA-normal setting: (i) Bayes-Nash equilibrium, where traders internalize both their price impact and their impact on the informational content of prices; (ii) rational-expectations equilibrium, where they internalize only the latter; and (iii) the correlation-neglect equilibrium, where both effects are ignored. It shows, in particular, how market power, the role of exogenous noise, and the risk premium depend on the two scaling assumptions that distinguish CHILE from traditional large economies. Appendix H extends the REE and BNE analysis to general preferences and payoff structures, and shows that the *direct* effect of wealth redistribution from rich to poor—an improvement in information efficiency together with a lowering of liquidity—arises under both competitive Rational Expectations and Bayes-Nash, and is not an artifact of the correlation-neglect equilibrium concept.
- Beyond this direct effect, REE and BNE introduce an *indirect* feedback that is absent in the correlation-neglect benchmark: because traders internalize their influence on the price, trading intensities depend on liquidity, and liquidity is itself determined by trading intensities. Our analysis in Appendix H establishes the robustness of certain results—most notably the information-aggregation paradox of Corollary 1—analytically only in the liquid-market limit, where this feedback is weak (Section H.5; Proposition 12 and Corollary 8). Markets are liquid in this limit in the sense that traders perceive the price as nearly uninformative; the main source of illiquidity in our model is adverse selection, so when the price is uninformative the differences among the three equilibrium concepts collapse. Systematically tracing out the regime in which REE and BNE depart substantively from the correlation-neglect equilibrium—thereby isolating the role of market power per se—is an interesting direction we leave for future research.

10 Conclusion

We introduce a new asymmetric-information asset-pricing framework called “Continuous-and-Heterogeneous Information in a Large Economy” (CHILE). In this economy, we study perfect competition with rich agent heterogeneity, arbitrary preferences, and general payoff distributions. A unique equilibrium features all quantities in closed form. Leveraging the tractability of our model and its ability to accommodate wealth effects, we illustrate its use by showing how the concentration of the market affects different aspects of market quality: information efficiency and liquidity.

There are many potentially fruitful extensions. While this paper focuses on a competitive CHILE equilibrium, a fuller treatment of strategic trading—beyond the analysis of Appendix H—would be a natural next step. Moreover, even though we model markets as uniform-price auctions, our techniques can also find applications in discriminatory-price auctions. Finally, our framework can help study several interesting environments, such as those with dynamic trading, feedback effects, and endogenous growth.

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Online Appendix

A Summary of notation

Notation	Explanation
<i>General mathematical notation</i>	
$X = \text{plim}_{n \rightarrow \infty} X_n$ or $X_n \xrightarrow[n \rightarrow \infty]{\mathbb{P}} X$	X_n converges to X in probability
$dB(a)$	An increment of the Brownian motion
$u^{(l)}(\cdot)$	l -th derivative of $u(\cdot)$
a, b	Index of agents in the continuous economy
i, j	Index of agents in the discrete economy
n	Number of agents in the discrete economy
m_i	Mass of an agent in the discrete economy
$\mu = 1/n$	Average mass of an agent in the discrete economy
$\text{Sk}[Y] = \mathbb{E}[(Y - E[Y])^3]$	Third central moment of r.v. Y (unstandardized skewness)
$\Phi(\cdot)$	Standard normal cdf
$\mathcal{O}'(h(a)) [h^\Delta(a)]$	Gateaux derivative of an equilibrium object $\mathcal{O}$ with respect to a parameter $h(a)$ in the direction $h^\Delta(a)$
<i>Model variables</i>	
$v \sim N(0, \tau_v^{-1})$	Fundamental, with ex-ante precision τ_v
$V(v)$	Payoff function
$\Theta(P) = \bar{\theta} + \theta(P)$	Supply of risky asset
$ds(a) = v da + \frac{1}{\sqrt{t(a)}} dB(a)$	Trader a 's signal, with precision $t(a)$
$\mathcal{F}_{b,c} = \sigma(\{s(z) - s(b)\}_{b \leq z < c})$	Information available to agents living in $[b, c)$
$\mathcal{F}_1 = \mathcal{F}_{0,1}$	Information in the entire economy

Notation	Explanation
$dX(a) = \beta(a, P)ds(a) + \delta(a, P)da$	Demand of trader a ; $\beta(a, P)$ is a 's trading intensity
$\rho(a) = -u''(W_0(a), a)/u'(W_0(a), a)$	Absolute risk aversion of trader a
$\pi(a) = -u'''(W_0(a), a)/u''(W_0(a), a)$	Absolute prudence of trader a
$s[\omega^H(a)] = \int_0^1 \omega^H(b)ds(b)$	Aggregate signal; $\omega^H(a)$ is a weighting function
$s_p = s[\omega(a)] = \int_0^1 \omega(a)ds(a)$	Equilibrium sufficient statistic; $\omega(a)$ is equilibrium weighting function
$\mathcal{P}(s_p)$	Equilibrium price function
$h(\cdot)$	The inverse of $\mathcal{P}(\cdot)$

B Heuristic derivation of equilibrium

In this section, we sketch out the derivation of key equilibrium objects, applying familiar heuristics that have become available in CHILE due to its setup.³⁴

B.1 Key stochastic calculus heuristics

We begin by recalling some properties of Brownian Motion and its increments. For $a \neq b \in [0, 1)$, we have

$$dB(a) \perp\!\!\!\perp dB(b) \quad dB(a), dB(b) \sim \mathcal{N}(0, da),$$

allowing us to interpret the signals in (6) as a continuum of pointwise mutually-independent Normal random variables.

In what follows, we employ second-order Taylor expansions ignoring terms of orders higher

³⁴See, e.g., [Cochrane \(2009\)](#), Appendix A.3, for an application of these heuristics over time, instead of over agents.

than da , following a method commonly referred to as the “box calculus” (Steele, 2001, Chapter 8.4):

$$da \cdot da = 0, \quad dB(a) \cdot da = 0, \quad dB(a) \cdot dB(a) = da; \quad dB(a) \cdot dB(b) = 0. \quad (31)$$

B.2 Information continuity linearizes demand

As our Representation Lemma demonstrates, the demands *must* be linear in signals (see (10)). Suppose, by contradiction, that demands are non-linear in signals—for example, assume that (10) contained a squared-signal term. Denoting the coefficient of the quadratic signal as $\zeta(\cdot)$ and denoting the new version of the remaining quantities by adding tildes on top, we have

$$\begin{aligned} dX(a) &= \tilde{\beta}(P, a)ds(a) + \tilde{\delta}(P, a)da + \zeta(P, a)ds^2(a) \\ &= \tilde{\beta}(P, a)ds(a) + \left[\tilde{\delta}(a, P) + \frac{\zeta(a, P)}{t(a)} \right] da, \end{aligned} \quad (32)$$

where the second equality follows by the box-calculus heuristic. What is more, by comparing (32) to (10) we can see that the demand function in (10) can subsume the extra term through its da term without affecting the signal term, giving us a contradiction. To summarize, the functional form in (10) already accounts for squared signals.³⁵

B.3 Price inference

Equation (11) implies that information in the price is summarized by

$$s_p = v + \int_0^1 \omega(a, P) \frac{dB(a)}{\sqrt{t(a)}}.$$

³⁵Restricting attention to quadratic adjustments is without loss of generality, as any sufficiently differentiable non-linear function can be expanded into a polynomial of $ds(a)$, with terms of order three and above becoming zero due to (31).

This signal has a familiar “truth plus normally distributed noise” structure. Once the precision τ_p is known, the inference from prices, summarized in Lemma 3, follows from Bayes’s rule for normal random variables. Here, we derive expression (12) for τ_p heuristically; see Appendix C.4 for a formal proof.

We show below that the equilibrium weight does not in fact depend on the price: the trading intensity factors as $\beta(a, P) = \beta_a(a) \beta_P(P)$ with $\beta_a(a) = t(a)/\rho(a)$, so the common price-dependent factor $\beta_P(P)$ cancels in the normalization $\omega(a, P) = \beta(a, P)/\int_0^1 \beta(b, P) db$, leaving the deterministic weight $\omega(a) = (t(a)/\rho(a))/\int_0^1 t(b)/\rho(b) db$. We therefore write $\omega(a)$ in place of $\omega(a, P)$ from here on.

The “noise” term $\int_0^1 \omega(a)/\sqrt{t(a)} dB(a)$ is normally distributed with mean 0 and variance $\int_0^1 \omega(a)^2/t(a) da$. This follows from the basic properties of stochastic integrals but can also be derived heuristically using the heuristics outlined at the beginning of this section. Indeed, the noise $\int_0^1 \omega(a)/\sqrt{t(a)} dB(a)$ accumulates terms $\omega(a)/\sqrt{t(a)} dB(a)$, each normally distributed with mean zero and variance $\omega(a)^2/t(a) da$, and independent from each other. As an accumulation of normal mean-zero independent random variables, the integral is normally distributed with a mean of zero and a variance that sums the variances of the individual terms, i.e.,

$$\text{Var} \left[\int_0^1 \omega(a) \frac{dB(a)}{\sqrt{t(a)}} \right] = \int_0^1 \omega(a)^2/t(a) da.$$

This variance is the reciprocal of τ_p .

B.3.1 Price inference under signal–price correlation neglect

Note that equation (11b) can be expressed alternatively as:

$$s_p = v + \frac{\omega(a)}{\sqrt{t(a)}} dB(a) + \int_{b \in [0,1] \setminus [a, a+da)} \frac{\omega(b)}{\sqrt{t(b)}} dB(b).$$

However, a trader viewing the price in this manner would not be neglecting the signal–price correlation. Consistent with [Hellwig \(1980\)](#)’s critique, such a trader would “account for the covariance between the ‘noise’ in their own information and the ‘noise’ embedded in the price.” Indeed, using the heuristics presented above, one can obtain:

$$\text{cov}(dB(a), s_p) = \frac{\omega(a)}{\sqrt{t(a)}} da.$$

A trader who neglects the signal–price correlation, believing their signal does not influence the informational content of the price, must perceive this covariance as zero.

We take a page right out of [Hellwig \(1980\)](#)’s critique: our agents assume that the noise in their private information is independent of the noise in the price. Specifically, trader a perceives the price as equivalent to a sufficient statistic $\hat{s}_p$, defined by

$$\hat{s}_p = v + \frac{\omega(a)}{\sqrt{t(a)}} d\hat{B}(a) + \int_{b \in [0,1] \setminus [a, a+da)} \frac{\omega(b)}{\sqrt{t(b)}} dB(b),$$

where the Brownian motions $\hat{B}$ and B are independent. Under this assumption, it is straightforward to see that

$$\text{cov}(dB(a), \hat{s}_p) = 0.$$

This represents the minimal deviation in traders’ beliefs consistent with information taking. Indeed, our traders correctly perceive their private signals $ds(a)$, the marginal distribution of s_p , and the joint distribution of s_p and v . Their only mistake lies in misunderstanding the correlation between the noise in their own signals and the noise embedded in the price. Consequently, the conditional distribution of v given $\hat{s}_p$ coincides with that of v given s_p . In particular, this implies that $\hat{s}_p$ and s_p have identical precision. We also emphasize that the mistakes agents make are infinitesimal. In the limit as $da \rightarrow 0$, their perceived price statistic $\hat{s}_p$ coincides exactly with the actual statistic s_p .

B.4 A market-efficiency result

Consider trader a , who has size da and receives signal ds . Trader a 's optimal demand x satisfies the first-order condition:

$$\mathbb{E}[u'(W_0(a) + (V - P)x)(V - P)|ds(a), \hat{s}_p] = 0, \quad (33)$$

where we have dropped the dependence of the payoff function on v to lighten notation.

We note that demands must go to zero as the number of agents goes to infinity—if that was not the case, aggregate demand would sum up infinitely many terms of finite size, and it would explode. Consequently, $x = 0$ must solve (33) in the large economy limit. Further noting that the information set of $(ds(a), \hat{s}_p)$ collapses to that of just $\hat{s}_p$ in the limit, setting $x = 0$ in (33) yields

$$\mathbb{E}[u'(W_0(a))(V - P)|\hat{s}_p] = 0 \iff P = \mathbb{E}[V|\hat{s}_p].$$

Since the conditional distribution of v given $\hat{s}_p$ coincides with that of v given s_p , we can replace $\hat{s}_p$ with s_p , yielding:

$$P = \mathbb{E}[V|s_p]. \quad (34)$$

This result implies that equilibrium prices in our model are weak-form efficient.

B.5 Deriving trading intensity

As we will show below, market quality is completely determined by the trading-intensity $\beta(a, P)$, on which we now focus. We Taylor-expand (33) over x up to the second order, and then we

replace x by $dX(a)$, obtaining

$$\begin{aligned}
0 = & u'(W_0(a)) \mathbb{E}[(V - P)|ds(a), \hat{s}_p] \\
& + u''(W_0(a)) \mathbb{E}[(V - P)^2|ds(a), \hat{s}_p] dX(a) \\
& + \frac{1}{2} u'''(W_0(a)) \mathbb{E}[(V - P)^3|ds(a), \hat{s}_p] dX(a)^2. \quad (35)
\end{aligned}$$

To solve for trading intensity, we plug (10) into (35) and compare the result with (10). We then solve for $\beta(a, P)$ by matching $ds(a)$ coefficients (for $\delta(a, P)$, we can match da terms).

After some simplifications explained in detail in Appendix C.11, we obtain

$$0 = u'(W_0(a)) \mathbb{E}[(V - P)|ds(a), \hat{s}_p] + u''(W_0(a)) \mathbb{E}[(V - P)^2|s_p] \beta(a, P) ds(a) + \dots, \quad (36)$$

where we have omitted terms that do not contain the signal, as they do not influence our subsequent calculations. Such “non- ds ” terms are denoted by “ $\dots$ ” in the remainder of the subsection.

Next, note that by the market efficiency condition (34) we have $P = E[V|s_p]$ and so $\mathbb{E}[(V - P)^2|s_p] = \text{Var}[V|s_p]$. Note also that we can replace $\rho(a) = -u''(W_0(a))/u'(W_0(a))$, where $\rho(a)$ is the risk-aversion coefficient of agent a . This simplifies the above even further, eventually yielding

$$\beta(a, P) ds(a) = \frac{\mathbb{E}[(V - P)|ds(a), \hat{s}_p]}{\rho(a) \text{Var}(V|P)} + \dots \quad (37)$$

Juxtaposing (37) with (10) brings forth a striking property: even though we use general preferences, the demand function depends on the signal as if preferences were mean-variance. We can think of this property as the converse of linearizing demand—by applying box calculus on the expanded first-order condition, all higher-order signal terms drop out, and thus the only way for signals to influence demand is through the first two terms in (35).³⁶

³⁶In equilibrium, preferences cannot be substituted by a mean-variance tradeoff locally approximating the utility. While higher-order moments of returns and higher-order marginal preferences (i.e. prudence) may not

To pin down $\beta(a, P)$ we need to separate out the $ds(a)$ term on the right-hand side of (37). To this end, we compute the $ds(a)$ term in the conditional expectation $\mathbb{E}[(V - P)|ds(a), \hat{s}_p]$. Note that for small $ds(a)$, this conditional expectation is linear in $ds(a)$ and so can be computed using a familiar linear regression formula³⁷

$$\mathbb{E}[(V - P)|ds(a), \hat{s}_p] = \frac{\mathbb{Cov}(V(v), ds(a)|\hat{s}_p)}{\mathbb{Var}(ds(a)|\hat{s}_p)} ds(a) + \dots = \frac{\mathbb{Cov}(v, ds(a)|\hat{s}_p)}{\mathbb{Var}(ds(a)|\hat{s}_p)} \mathbb{E}[V'(v)|\hat{s}_p] ds(a) + \dots \quad (38)$$

where the second equality follows from Stein's Lemma.³⁸

Moreover, up to terms of order higher than da , we have

$$\mathbb{Cov}(v, ds(a)|\hat{s}_p) = \mathbb{Cov}(v, v da|\hat{s}_p) = da \cdot \mathbb{Var}(v|\hat{s}_p) = \frac{da}{\tau}.$$

The first equality above uses the independence of $dB(a)$ and $\hat{s}_p$, and the third equality applies the identity $\tau \equiv 1/\mathbb{Var}(v|\hat{s}_p)$.

Similarly, we compute

$$\mathbb{Var}(ds(a)|\hat{s}_p) = \mathbb{Var}(dB(a)/\sqrt{t(a)}|\hat{s}_p) = da/t(a).$$

Combining everything, (37) becomes

$$\beta(a, P) ds(a) = \frac{t(a)}{\tau} \frac{\mathbb{E}[V'(v)|\hat{s}_p]}{\rho(a) \mathbb{Var}(V|\hat{s}_p)} ds(a) + \dots \iff \beta(a, P) = \frac{t(a)}{\rho(a)} \frac{\mathbb{E}[V'(v)|s_p]}{\tau \mathbb{Var}(V|s_p)}.$$

In the last step above, we have replaced $\hat{s}_p$ with s_p inside the terms $\mathbb{E}[V'(v)|s_p]$ and $\mathbb{Var}(V|s_p)$, relying on the fact that the conditional distributions of v given $\hat{s}_p$ and v given s_p coincide.

affect $\beta(a, P)$, they still affect $\delta(a, P)$.

³⁷More precisely, because the signal increment $ds(a)$ is infinitesimal, we can isolate its first-order contribution to the conditional expectation. Under Gaussian updating, this leading local coefficient matches the Best Linear Predictor formula (see, e.g., [Goldberger \(1991\)](#), Chapter 5.4.) that is written following the first equality in (38).

³⁸Stein's Lemma states that if X and Y are jointly normally distributed, then for any differentiable function g such that $\mathbb{E}[g'(X)] < \infty$, $\mathbb{Cov}(g(X), Y) = \mathbb{Cov}(X, Y) \cdot \mathbb{E}[g'(X)]$.

C Derivations and proofs

C.1 Technical conditions

The equilibrium construction requires differentiating the first-order conditions for optimality and evaluating the resulting expressions in the large-economy limit ($\mu \rightarrow 0$, $s \rightarrow 0$). We impose the following conditions on the primitives.

Assumption 3 (Technical restrictions on the primitives). *We impose the following restrictions:*

1. *There exist $q \geq 0$ and $C_u < \infty$ such that for all $w \in \mathbb{R}$ and $a \in [0, 1)$, $|u'(w, a)| + |u''(w, a)| + |u'''(w, a)| \leq C_u(1 + |w|^q)$.*
2. *$V(v)$ satisfies $\lim_{v \rightarrow \pm\infty} \frac{\ln|V(v)|}{v^2} \leq 0$.*
3. *There exist $\varepsilon > 0$ and constants $A, k > 0$ such that for all $v \in \mathbb{R}$,*

$$\sup_{|\mu| \leq \varepsilon} |g(v, P, \mu)| \leq Ae^{-kv^2},$$

and the same bounds hold for the partial derivatives:

$$\sup_{|\mu| \leq \varepsilon} |g_\mu(v, P, \mu)| \leq Ae^{-kv^2}, \quad \sup_{|\mu| \leq \varepsilon} |g_P(v, P, \mu)| \leq Ae^{-kv^2}.$$

4. *The price belief $g(v, P, \mu)$ is continuously differentiable in (μ, P) in a neighborhood of $\mu = 0$.*
5. *The agent primitives satisfy the integrability conditions*

$$\int_0^1 \frac{1}{\rho(a)} da < \infty, \quad \int_0^1 \frac{t(a)}{\rho(a)} da < \infty, \quad \int_0^1 \frac{t(a)}{\rho^2(a)} da < \infty, \quad \int_0^1 \frac{t(a)\pi(a)}{\rho^2(a)} da < \infty.$$

6. *The precision profile satisfies $t(a) > 0$ for every $a \in [0, 1)$ and is continuous on $[0, 1)$.*

Parts 1–3 provide the growth and integrability bounds needed to invoke the Dominated Convergence Theorem, allowing us to pass limits as $\mu \rightarrow 0$ and derivatives under the integral sign. Part 4 ensures that the price belief is smooth enough for the relevant derivatives to be well defined. Part 5 ensures that the equilibrium objects—price informativeness τ_p , the cumulative demand integrals, and the market-clearing integrals—are finite. These conditions are easily satisfied when $\rho(a)$ and $\pi(a)$ are continuous on $[0, 1]$. Part 6 guarantees that $t(a)$ is bounded away from zero on every compact subinterval $[0, y] \subset [0, 1)$, as required by the [Aggregation Lemma](#).

Local versus global restrictions on utility. What has economic content in our setting are the *local* properties of $u(\cdot, a)$ near the initial wealth $W_0(a)$. As noted in Section 2.1, preferences enter the model only through the profiles $u^{(l)}(W_0(a), a)$, $l = 0, 1, 2, 3$, and correspondingly all equilibrium objects in Theorem 1 depend on utility only through $\rho(a)$ and $\pi(a)$, both evaluated at $W_0(a)$. The reason is that individual positions vanish in the large-economy limit (Remark 5), so terminal wealth stays in an arbitrarily small neighborhood of $W_0(a)$.

Part 1 of Assumption 3, by contrast, is stated globally, on all of $\mathbb{R}$. This is a matter of technical convenience rather than economic substance: the global bound is used only to construct the integrable majorant that justifies the Dominated Convergence Theorem in the finite- μ problem (5), where positions are unrestricted and terminal wealth can therefore take any value in $\mathbb{R}$. Read literally, the global bound excludes the CRRA specification of Section 6, whose derivatives are unbounded as $w \downarrow 0$ and which is not defined for $w \leq 0$. The exclusion is only apparent. Any utility of interest can be replaced by one that satisfies part 1 globally and coincides with it—up to a positive affine transformation, to which the demand (5) is invariant—on a neighborhood of $W_0(a)$, and hence delivers exactly the same equilibrium. The following remark records one such extension.

Remark 9 (Extending a utility function of interest). Let $\tilde{u}(\cdot, a)$ be the utility function of interest—CRRA, say—assumed increasing, strictly concave, and thrice continuously differ-

entiable on an open interval $\mathcal{W}(a) \ni W_0(a)$; for CRRA, $\mathcal{W}(a) = (0, \infty)$ and $W_0(a) > 0$. Let $\tilde{\rho}(w, a) = -\tilde{u}''(w, a)/\tilde{u}'(w, a)$ denote its absolute risk aversion and fix $\eta(a) > 0$ with $[W_0(a) - \eta(a), W_0(a) + \eta(a)] \subset \mathcal{W}(a)$. Choose any continuously differentiable $\hat{\rho}(\cdot, a) : \mathbb{R} \rightarrow (0, \infty)$ that coincides with $\tilde{\rho}(\cdot, a)$ on $(W_0(a) - \eta(a), W_0(a) + \eta(a))$, is bounded together with its derivative, and satisfies $\int_{-\infty}^{W_0(a)} \hat{\rho}(z, a) dz < \infty$. Such a function always exists: for example, take a smooth cutoff χ that equals one on a smaller neighborhood of $W_0(a)$ and vanishes outside $(W_0(a) - \eta(a), W_0(a) + \eta(a))$, and set $\hat{\rho} = \chi\tilde{\rho} + (1 - \chi)\varphi$ for any positive, bounded, continuously differentiable φ integrable at $-\infty$, such as $\varphi(w) = (1 + w^2)^{-1}$. Now define

$$u(w, a) = \int_{W_0(a)}^w \exp\left(-\int_{W_0(a)}^y \hat{\rho}(z, a) dz\right) dy, \quad w \in \mathbb{R}. \quad (39)$$

Then $u(\cdot, a)$ is defined on all of $\mathbb{R}$, strictly increasing, strictly concave, and thrice continuously differentiable, with $u' = \exp\left(-\int_{W_0(a)}^{\cdot} \hat{\rho}\right)$, $u'' = -\hat{\rho}u'$, and $u''' = (\hat{\rho}^2 - \hat{\rho}')u'$ all bounded—the integrability of $\hat{\rho}$ at $-\infty$ bounds u' there, and $\hat{\rho} > 0$ bounds it by one at $+\infty$. Because individual demand is invariant to positive affine transformations of utility, we can multiply each $u(\cdot, a)$ by an agent-specific positive constant $c(a)$ to ensure these derivative bounds are uniform across all $a \in [0, 1)$. Part 1 of Assumption 3 therefore holds with $q = 0$. At the same time, on $(W_0(a) - \eta(a), W_0(a) + \eta(a))$ we have $u(w, a) = (\tilde{u}(w, a) - \tilde{u}(W_0(a), a)) / \tilde{u}'(W_0(a), a)$, a positive affine transformation of $\tilde{u}$. Consequently $\rho(a)$ and $\pi(a)$ —and with them every object in Theorem 1—are the same whether computed from $\tilde{u}$ or from u . The same device disposes of the concern that unrestricted positions and a log-normal payoff may push terminal wealth to zero or below: $u(\cdot, a)$ in (39) is well defined for every wealth level, so the finite- μ problem (5) is well posed for arbitrary positions, while the equilibrium is governed by the behavior of $\tilde{u}$ near $W_0(a) > 0$ alone. A further benefit is that $u(\cdot, a)$ is strictly concave on all of $\mathbb{R}$, so the objective in (5) is strictly concave in x and the first-order condition used throughout the proofs is necessary and sufficient for global optimality, with the maximizer unique whenever it exists.

C.2 The Aggregation Lemma

Lemma 5. (Aggregation lemma) Consider a sequence of functions $\hat{x}^n(P, s, m, a) : \mathbb{R} \times \mathbb{R} \times [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ with partial derivatives $\hat{x}_s^n(P, s, m, a)$, $\hat{x}_{ss}^n(P, s, m, a)$, and $\hat{x}_m^n(P, s, m, a)$ that are continuous functions of s and m in the neighborhood of $s = m = 0$ for every $P \in \mathbb{R}$ and $a \in (0, 1)$. Suppose that for every $P \in \mathbb{R}$ and every $y \in (0, 1)$, the functions $a \mapsto 1/\hat{t}^n(a)$, $a \mapsto \hat{x}_s^n(P, 0, 0, a)$, $a \mapsto \hat{x}_{ss}^n(P, 0, 0, a)$, and $a \mapsto \hat{x}_m^n(P, 0, 0, a)$ converge uniformly on $[0, y]$. Denote the respective limits by $1/t(a)$, $x_s(P, a)$, $x_{ss}(P, a)$, and $x_m(P, a)$. Define

$$\beta(P, a) \equiv x_s(P, a) \quad \text{and} \quad \delta(P, a) \equiv \frac{1}{2t(a)}x_{ss}(P, a) + x_m(P, a).$$

Assume that for all n and all $a \in (0, 1)$, $\hat{t}^n(a) \geq \varepsilon > 0$ for some constant ε , and that for every $P \in \mathbb{R}$ and every $y \in (0, 1)$ the following conditions hold:

(i) **Lipschitz:** There exists $K > 0$ such that for all n , $a \in [0, y]$, and sufficiently small s, m ,

$$|\hat{x}_\alpha^n(P, s, m, a) - \hat{x}_\alpha^n(P, 0, 0, a)| \leq K(|s| + |m|), \quad \text{for } \alpha \in \{s, ss, m\}.$$

(ii) **Integrability:** For every $P \in \mathbb{R}$ and every $y \in (0, 1)$,

$$\int_0^y \frac{x_s(P, a)^2}{t(a)} da < \infty, \quad \int_0^y x_s(P, a)^2 da < \infty, \quad \int_0^y x_{ss}(P, a)^2 da < \infty, \quad \text{and} \quad \int_0^y x_m(P, a)^2 da < \infty.$$

(iii) **Continuity:** The functions $a \mapsto t(a)$, $a \mapsto x_s(P, a)$, $a \mapsto x_{ss}(P, a)$, and $a \mapsto x_m(P, a)$ are continuous on $[0, y]$.

For a fixed $y \in (0, 1)$, let $[0, y] = \bigcup_i [a_i, a_i + m_i)$ be a sequence of partitions with mesh size $\bar{m} = \max_i m_i \rightarrow 0$. Let the increments be defined as $\Delta s_i = vm_i + [\hat{t}^n(a_i)]^{-1/2}(B(a_i + m_i) - B(a_i))$.

Then, letting s_a denote the process $ds_a = v da + [t(a)]^{-1/2}dB_a$, we have:

$$\sum_{i:a_i < y} (\hat{x}^n(P, \Delta s_i, m_i, a_i) - \hat{x}^n(P, 0, 0, a_i)) \xrightarrow{p} \int_0^y \beta(P, a) ds_a + \int_0^y \delta(P, a) da,$$

along any sequence with $\bar{m} \rightarrow 0$ and $n \rightarrow \infty$.

Proof of Lemma 5.

Fix $P \in \mathbb{R}$ and write $\Delta B_i \equiv B(a_i + m_i) - B(a_i)$, so that

$$\Delta s_i = vm_i + [\hat{t}^n(a_i)]^{-1/2} \Delta B_i.$$

For each i (suppressing (P, a_i) in the notation), a Taylor expansion of $\hat{x}^n(s, m)$ around $(s, m) = (0, 0)$ (second order in s , first order in m) gives

$$\hat{x}^n(\Delta s_i, m_i) - \hat{x}^n(0, 0) = \hat{x}_s^n(0, 0) \Delta s_i + \frac{1}{2} \hat{x}_{ss}^n(0, 0) (\Delta s_i)^2 + \hat{x}_m^n(0, 0) m_i + r_i^n,$$

where the remainder can be written in mean-value form as

$$r_i^n = (\hat{x}_s^n(0, m_i) - \hat{x}_s^n(0, 0)) \Delta s_i + \frac{1}{2} (\hat{x}_{ss}^n(\tilde{s}_i, m_i) - \hat{x}_{ss}^n(0, 0)) (\Delta s_i)^2 + (\hat{x}_m^n(0, \tilde{m}_i) - \hat{x}_m^n(0, 0)) m_i$$

for some $\tilde{s}_i$ between 0 and Δs_i and some $\tilde{m}_i$ between 0 and m_i . By the Lipschitz assumption (i),

$$|r_i^n| \leq C \left(m_i |\Delta s_i| + (|\Delta s_i| + m_i) (\Delta s_i)^2 + m_i^2 \right)$$

for a constant C independent of n, i .³⁹

By assumption, $\hat{t}^n(a) \geq \varepsilon > 0$ for all n and all $a \in (0, y)$, so there is a constant $C_t < \infty$

³⁹Brownian paths are almost surely uniformly continuous on $[0, y]$, hence $\max_i |\Delta B_i| \rightarrow 0$ almost surely as $\bar{m} \rightarrow 0$, and therefore $\max_i |\Delta s_i^n| \rightarrow 0$ almost surely (uniformly in n). Thus, for $\bar{m}$ small enough, the neighborhood required for condition (i) applies almost surely to all increments.

(e.g. $C_t \equiv 1/\varepsilon$) such that

$$\sup_n \sup_{a \in (0, y)} \frac{1}{\hat{t}^n(a)} \leq C_t.$$

Moreover, since $1/\hat{t}^n \rightarrow 1/t$ uniformly on $(0, y)$, we have $1/t(a) \leq C_t$ on $[0, y]$ and hence $t(a) \geq 1/C_t > 0$ there. Since $\Delta B_i \sim N(0, m_i)$ and $(\hat{t}^n(a_i))^{-1} \leq C_t$, there exists $C' > 0$ (independent of n, i) such that

$$\mathbb{E}(\Delta s_i)^2 \leq C' m_i \quad \text{and} \quad \mathbb{E}|\Delta s_i|^3 \leq C' m_i^{3/2}.$$

Therefore,

$$\mathbb{E} \left[\sum_{i: a_i < y} |r_i^n| \right] \leq C'' \sum_{i: a_i < y} (m_i^{3/2} + m_i^2) \leq C''(y\sqrt{\bar{m}} + y\bar{m}) \xrightarrow{\bar{m} \rightarrow 0} 0,$$

and hence by Markov's inequality, for any $\eta > 0$,

$$\mathbb{P} \left(\sum_{i: a_i < y} |r_i^n| > \eta \right) \leq \frac{1}{\eta} \mathbb{E} \left[\sum_{i: a_i < y} |r_i^n| \right] \xrightarrow{\bar{m} \rightarrow 0} 0,$$

In particular,

$$\sum_{i: a_i < y} r_i^n \xrightarrow{p} 0, \quad \text{equivalently} \quad \sum_{i: a_i < y} r_i^n = o_p(1)$$

Define the shorthand coefficients

$$\beta_i^n \equiv \hat{x}_s^n(P, 0, 0, a_i), \quad \gamma_i^n \equiv \hat{x}_{ss}^n(P, 0, 0, a_i), \quad \mu_i^n \equiv \hat{x}_m^n(P, 0, 0, a_i), \quad \tau_i^n \equiv \hat{t}^n(a_i).$$

Summing the expansion gives

$$\sum_{i: a_i < y} (\hat{x}^n(P, \Delta s_i, m_i, a_i) - \hat{x}^n(P, 0, 0, a_i)) = \sum_{i: a_i < y} \beta_i^n \Delta s_i + \frac{1}{2} \sum_{i: a_i < y} \gamma_i^n (\Delta s_i)^2 + \sum_{i: a_i < y} \mu_i^n m_i + o_p(1).$$

Step 1 (linear term). Decompose

$$\sum_{i:a_i < y} \beta_i^n \Delta s_i = v \sum_{i:a_i < y} \beta_i^n m_i + \sum_{i:a_i < y} \beta_i^n (\tau_i^n)^{-1/2} \Delta B_i.$$

Because $\int_0^y \beta(P, a)^2 / t(a) da < \infty$, the limiting integrand $a \mapsto \beta(P, a)t(a)^{-1/2}$ lies in $L^2([0, y])$. By the uniform convergence of $\hat{x}_s^n(P, 0, 0, \cdot)$ to $\beta(P, \cdot)$ and the uniform convergence $(\tau^n)^{-1/2} \rightarrow t^{-1/2}$, the step functions $a \mapsto \beta_i^n (\tau_i^n)^{-1/2}$ (constant on $[a_i, a_i + m_i)$) converge in $L^2([0, y])$ to $a \mapsto \beta(P, a)t(a)^{-1/2}$ along any refining sequence of partitions. By Itô isometry this implies

$$\sum_{i:a_i < y} \beta_i^n (\tau_i^n)^{-1/2} \Delta B_i \xrightarrow{L^2} \int_0^y \frac{\beta(P, a)}{\sqrt{t(a)}} dB_a.$$

Similarly, using $\int_0^y |\beta(P, a)| da < \infty$ and the uniform convergence of $\hat{x}_s^n(P, 0, 0, \cdot)$ to $\beta(P, \cdot)$,

$$v \sum_{i:a_i < y} \beta_i^n m_i \rightarrow v \int_0^y \beta(P, a) da.$$

Combining yields

$$\sum_{i:a_i < y} \beta_i^n \Delta s_i \xrightarrow{p} \int_0^y \beta(P, a) ds_a, \quad ds_a = v da + t(a)^{-1/2} dB_a.$$

Step 2 (quadratic term). Expand

$$(\Delta s_i)^2 = v^2 m_i^2 + 2v m_i (\tau_i^n)^{-1/2} \Delta B_i + (\tau_i^n)^{-1} (\Delta B_i)^2.$$

The $v^2 m_i^2$ contribution is $o(1)$. Indeed,

$$\frac{1}{2} \sum_{i:a_i < y} |\gamma_i^n| v^2 m_i^2 = \frac{v^2}{2} \sum_{i:a_i < y} (|\gamma_i^n| \sqrt{m_i}) m_i^{3/2} \leq \frac{v^2}{2} \left(\sum_{i:a_i < y} (\gamma_i^n)^2 m_i \right)^{1/2} \left(\sum_{i:a_i < y} m_i^3 \right)^{1/2}.$$

Since $m_i^3 \leq \bar{m}^2 m_i$ we have $\sum_{i:a_i < y} m_i^3 \leq \bar{m}^2 \sum_{i:a_i < y} m_i = \bar{m}^2 y \rightarrow 0$. Moreover, using

$\int_0^y x_{ss}(P, a)^2 da < \infty$ and the uniform convergence of $\hat{x}_{ss}^n(P, 0, 0, \cdot)$ to $x_{ss}(P, \cdot)$,

$$\sum_{i:a_i < y} (\gamma_i^n)^2 m_i \rightarrow \int_0^y x_{ss}(P, a)^2 da,$$

and hence the product above vanishes.

The cross term is a martingale. Using independence of the Brownian increments and $\mathbb{E}[(\Delta B_i)^2] = m_i$,

$$\mathbb{E} \left[\left(\sum_{i:a_i < y} \gamma_i^n m_i (\tau_i^n)^{-1/2} \Delta B_i \right)^2 \right] = \sum_{i:a_i < y} (\gamma_i^n)^2 m_i^2 (\tau_i^n)^{-1} \mathbb{E}[(\Delta B_i)^2] \leq C_t \sum_{i:a_i < y} (\gamma_i^n)^2 m_i^3.$$

Since $m_i^3 \leq \bar{m}^2 m_i$ and $\sum_i (\gamma_i^n)^2 m_i \rightarrow \int_0^y x_{ss}(P, a)^2 da < \infty$, this second moment is $O(\bar{m}^2) \rightarrow 0$, hence, by Chebyshev's inequality,

$$\sum_{i:a_i < y} \gamma_i^n m_i (\tau_i^n)^{-1/2} \Delta B_i \xrightarrow{p} 0.$$

For the remaining part, write $(\Delta B_i)^2 = m_i + ((\Delta B_i)^2 - m_i)$ to obtain

$$\frac{1}{2} \sum_{i:a_i < y} \gamma_i^n (\tau_i^n)^{-1} (\Delta B_i)^2 = \frac{1}{2} \sum_{i:a_i < y} \gamma_i^n (\tau_i^n)^{-1} m_i + \frac{1}{2} \sum_{i:a_i < y} \gamma_i^n (\tau_i^n)^{-1} ((\Delta B_i)^2 - m_i).$$

The last term is again a martingale. Since $\text{Var}((\Delta B_i)^2 - m_i) = 2m_i^2$,

$$\mathbb{E} \left[\left(\frac{1}{2} \sum_{i:a_i < y} \gamma_i^n (\tau_i^n)^{-1} ((\Delta B_i)^2 - m_i) \right)^2 \right] = \frac{1}{4} \sum_{i:a_i < y} (\gamma_i^n)^2 (\tau_i^n)^{-2} \text{Var}((\Delta B_i)^2 - m_i) \leq \frac{1}{2} C_t^2 \sum_{i:a_i < y} (\gamma_i^n)^2 m_i^2.$$

Using $m_i^2 \leq \bar{m} m_i$ and the boundedness of $\sum_i (\gamma_i^n)^2 m_i$, this second moment is $O(\bar{m}) \rightarrow 0$, hence the martingale term converges to 0 in probability by Chebyshev's inequality. The first term converges to $\int_0^y \frac{1}{2t(a)} x_{ss}(P, a) da$ by the uniform convergence of $\hat{x}_{ss}^n(P, 0, 0, \cdot)$ to $x_{ss}(P, \cdot)$ and

uniform convergence of $(\tau^n)^{-1}$. Hence

$$\frac{1}{2} \sum_{i: a_i < y} \gamma_i^n (\Delta s_i)^2 \xrightarrow{p} \int_0^y \frac{1}{2t(a)} x_{ss}(P, a) da.$$

Step 3 (the m term). Since $x_m(P, \cdot) \in L^2(0, y) \subset L^1(0, y)$ and $\hat{x}_m^n(P, 0, 0, \cdot) \rightarrow x_m(P, \cdot)$ uniformly on $(0, y)$,

$$\sum_{i: a_i < y} \mu_i^n m_i \rightarrow \int_0^y x_m(P, a) da.$$

Putting Steps 1–3 together and recalling $\delta(P, a) = \frac{1}{2t(a)} x_{ss}(P, a) + x_m(P, a)$ yields the desired convergence in probability. ■

C.3 Proof of Lemma 1 and Lemma 2

Recall that in the n th economy (with $\mu = 1/n$), the cumulative demand is

$$X(y) = \sum_{i: a_i < y} x^*(\Delta s_i, P, a_i, \mu).$$

Define $x^*(P, a) \equiv x^*(0, P, a, 0)$ —the optimal demand when $s = 0$ in the limiting economy. We decompose:

$$X(y) = \sum_{i: a_i < y} (x^*(\Delta s_i, P, a_i, \mu) - x^*(0, P, a_i, 0)) + \sum_{i: a_i < y} x^*(P, a_i).$$

To apply the [Aggregation Lemma](#), we identify $\hat{x}(P, s, m, a) = x^*(s, P, a, m)$, so that s plays the role of the signal increment and $m = \mu$ is the mesh size (which equals the economy parameter, since the partition is uniform). By [Aggregation Lemma](#),

$$\sum_{i: a_i < y} (x^*(\Delta s_i, P, a_i, \mu) - x^*(0, P, a_i, 0)) \xrightarrow[\mu \rightarrow 0]{\mathbb{P}} \int_0^y \beta(P, a) ds_a + \int_0^y \delta(P, a) da, \quad (40)$$

where $\beta(P, a) = x_s^*(0, P, a, 0)$ and $\delta(P, a) = \frac{1}{2t(a)}x_{ss}^*(0, P, a, 0) + x_\mu^*(0, P, a, 0)$.

The hypotheses of the [Aggregation Lemma](#) hold. By Assumption 3, part 6, $t(a)$ is continuous and strictly positive on the compact set $[0, y]$, ensuring it is bounded away from zero, as required by the condition $\hat{t}^n(a) \geq \varepsilon > 0$ of Lemma 5. Continuity of x_s^* , x_{ss}^* , and x_μ^* in (s, μ) near $(0, 0)$ follows from the Implicit Function Theorem applied to the first-order condition (43). Their joint continuity in (a, μ) over the compact set $[0, y]$ guarantees they are uniformly continuous and bounded, which gives the continuity condition (iii) of Lemma 5. This boundedness directly satisfies the required L^2 integrability conditions (ii). Furthermore, because Brownian paths are uniformly continuous on compact intervals, this uniform continuity is sufficient to control the Taylor remainders in probability, fulfilling the operational role of the uniform Lipschitz condition (i).

For $X(y)$ to have a finite limit, the residual sum $\sum_{i: a_i < y} x^*(P, a_i)$ must also be finite. This leads to the efficiency condition.

C.3.1 Efficiency condition

Setting $s = 0$ and $\mu = 0$ in trader a 's first-order condition (43) gives

$$\int u'(W_0(a) + x^*(P, a)(V(v) - P), a)(V(v) - P)g(v, P, 0)dv = 0. \quad (41)$$

The passage to the limit is justified by the Dominated Convergence Theorem: Assumption 3, parts 1–3 imply that the integrand admits an integrable majorant of the form $A \exp(-kv^2)$, and part 4 ensures the smooth convergence $g(\cdot, \mu) \rightarrow g(\cdot, 0)$.

Lemma 6. *For a fixed $y \in (0, 1)$, $\lim_{\mu \rightarrow 0} \sum_{i: a_i < y} x^*(P, a_i) < \infty$ if, and only if, $x^*(P, a) = 0$ for all $a \in [0, y]$ and*

$$\int (V(v) - P)g(v, P, 0)dv = 0. \quad (42)$$

Proof. Suppose $\int (V(v) - P) g(v, P, 0) dv > 0$ (the opposite case is analogous). Then (41) implies $x^*(P, a) > 0$ for every $a \in [0, y]$. Since $x^*(P, \cdot)$ is continuous in a (by the IFT and continuity of primitives), there exist $\epsilon > 0$ and an interval $(a_0 - \delta, a_0 + \delta) \subset [0, y]$ on which $x^*(P, a) > \epsilon$. As $\mu \rightarrow 0$, the number of partition points falling in this interval grows without bound, so $\sum_{i: a_i < y} x^*(P, a_i) \geq \sum_{|a_i - a_0| < \delta} x^*(P, a_i) \rightarrow \infty$ —a contradiction. Conversely, if (42) holds, then $x^*(P, a) = 0$ for all a (the unique solution to (41) when $\mathbb{E}[V - P | P] = 0$, by strict concavity), and the sum is identically zero. ■

Lemma 6 proves Lemma 1. Substituting the efficiency condition into (40) proves Lemma 2, with β and δ characterized explicitly in Lemmas 7 and 8.

Note that (42) and the efficiency condition (9) are equivalent.

C.4 Proof of Lemma 3

Proof of Lemma 3. By the properties of the Ito integral, $\int_0^1 \omega(a) / \sqrt{t(a)} dB(a)$ is distributed normally with a mean of zero. By Ito isometry, the variance of the integral is given by $\int_0^1 \frac{w(a)^2}{t(a)} da$. The statements of the lemma then follow from the standard results on Bayes rule with normal random variables. ■

C.5 Proof of Theorem 1

Consider a trader at location a , living on the interval $[a, a + \mu)$ in the economy with parameter $\mu = 1/n$. The price belief in that economy is $g(v, P, \mu)$, and we denote the trader's signal realization by s . Since the partition is uniform, the mesh size equals μ , so the optimal demand $x^*(s, P, a, \mu)$ is determined by the first-order condition

$$\int u'(W_0(a) + x^*(s, P, a, \mu) (V(v) - P), a) (V(v) - P) \cdot \exp((sv - \mu v^2/2) t(a)) g(v, P, \mu) dv = 0, \quad (43)$$

where we have dropped the normalizing constant from the conditional density (4), as it does not affect the optimization. The cumulative demand in the n th economy is

$$X(y) = \sum_{i: a_i < y} x^*(\Delta s_i, P, a_i, \mu).$$

C.5.1 Deriving $\beta(P, a)$

By the [Aggregation Lemma](#), the trading intensity equals the signal sensitivity of optimal demand evaluated in the large-economy limit: $\beta(P, a) = x_s^*(0, P, a, 0)$. Computing this derivative requires differentiating the first-order condition (43) with respect to s and evaluating at $s = 0$, $\mu = 0$.

Lemma 7. *Suppose that Assumption 3 is satisfied. Then*

$$\beta(P, a) = \frac{t(a) \operatorname{Cov}(V(v), v \mid P)}{\rho(a) \operatorname{Var}(V(v) \mid P)}. \quad (44)$$

In particular, $\beta(P, a)$ is multiplicatively separable: $\beta(P, a) = \beta_a(a) \beta_P(P)$ where

$$\beta_a(a) = \frac{t(a)}{\rho(a)} \quad \text{and} \quad \beta_P(P) = \frac{\operatorname{Cov}(V(v), v \mid P)}{\operatorname{Var}(V(v) \mid P)}.$$

If $v \mid \mathbf{P}_$ is normally distributed with precision τ , then $\beta_P(P) = \frac{\tau^{-1} \mathbb{E}[V'(v) \mid P]}{\operatorname{Var}(V(v) \mid P)}$.⁴⁰*

Proof. The optimal demand x^* satisfies the first-order condition

$$\int_{\mathbb{R}} (V(v) - P) u'(W_0(a) + x^*(V(v) - P)) \exp(t(a) s v) g(v, P, 0) dv = 0.$$

⁴⁰This follows via integration by parts:

$$\begin{aligned} \mathbb{E}[v(V(v) - P) \mid P] &= \int_{\mathbb{R}} (v - \mathbb{E}[v \mid P]) (V(v) - P) \frac{1}{\sqrt{2\pi/\tau}} \exp\left(-\frac{(v - \mathbb{E}[v \mid P])^2 \tau}{2}\right) dv \\ &= \tau^{-1} \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi/\tau}} \exp\left(-\frac{(v - \mathbb{E}[v \mid P])^2 \tau}{2}\right) V'(v) dv. \end{aligned}$$

Dividing by the marginal price density $f_P(P) = \int_{\mathbb{R}} g(v, P, 0) dv > 0$ —which is independent of x^* and s , so the equation is unchanged—we write it as

$$G(x^*, s) \equiv \int_{\mathbb{R}} (V(v) - P) u'(W_0(a) + x^*(V(v) - P)) \exp(t(a) s v) \frac{g(v, P, 0)}{f_P(P)} dv = 0.$$

Here $g(v, P, 0)/f_P(P)$ is the conditional density of v given P , so integrals against it are conditional expectations: $\int_{\mathbb{R}} Y(v) \frac{g(v, P, 0)}{f_P(P)} dv = \mathbb{E}[Y(v) \mid P]$.

By the Implicit Function Theorem, the sensitivity $\beta(P, a)$ at the limit ($s = 0, x^* = 0$) is

$$\beta(P, a) = \frac{\partial x^*}{\partial s} = -\frac{\partial G / \partial s}{\partial G / \partial x^*}.$$

Assumption 3, parts 1–3 imply that the derivatives can be passed under the integral sign and that G is continuously differentiable at $x^* = 0, s = 0$. Furthermore, the strict concavity of the utility function ($u''(W_0) < 0$) and the non-degenerate conditional payoff ($\text{Var}(V(v) \mid P) > 0$) ensure that $\partial G / \partial x^* \neq 0$, validating the Implicit Function Theorem.

Step 1: The Numerator ($\partial G / \partial s$). Differentiating G with respect to s and evaluating at $s = 0, x^* = 0$ (where $x^* = 0$ follows from Lemma 6):

$$\frac{\partial G}{\partial s} = u'(W_0) t(a) \mathbb{E}[(V(v) - P) v \mid P].$$

Using $\mathbb{E}[V(v) - P \mid P] = 0$ (Lemma 6), this conditional expectation equals $\text{Cov}(V(v), v \mid P)$:

$$\frac{\partial G}{\partial s} = u'(W_0) t(a) \text{Cov}(V(v), v \mid P).$$

Step 2: The Denominator ($\partial G / \partial x^*$). Differentiating G with respect to x^* and evaluating at $x^* = 0$. Since the price belief $g(v, P, 0)$ does not depend on x^* (price-taking assumption),

only the utility derivative contributes:

$$\frac{\partial G}{\partial x^*} = u''(W_0) \mathbb{E}[(V(v) - P)^2 \mid P] = u''(W_0) \text{Var}(V(v) \mid P),$$

where the last equality uses $\mathbb{E}[V(v) \mid P] = P$ (Lemma 6).

Step 3: Combining Terms. Substituting into the IFT formula and using $\rho(a) = -u''(W_0)/u'(W_0)$:

$$\beta(P, a) = -\frac{u'(W_0) t(a) \text{Cov}(V(v), v \mid P)}{u''(W_0) \text{Var}(V(v) \mid P)} = \frac{t(a) \text{Cov}(V(v), v \mid P)}{\rho(a) \text{Var}(V(v) \mid P)}.$$

■

C.5.2 Deriving $\delta(P, a)$

By the [Aggregation Lemma](#), the coefficient $\delta(P, a)$ combines the curvature of optimal demand in the signal and the sensitivity to the economy parameter μ :

$$\delta(P, a) = \frac{1}{2 t(a)} x_{ss}^*(0, P, a, 0) + x_{\mu}^*(0, P, a, 0).$$

Lemma 8. *Suppose that Assumption 3 is satisfied. Let $\Sigma_{VV}(P) = \text{Var}(V(v) \mid P)$. Then*

$$\delta(P, a) = \frac{1}{\rho(a) \Sigma_{VV}(P)} \left(\psi(P) + \frac{\beta(P, a)^2}{2 t(a)} \pi(a) \rho(a) \text{Sk}[V(v) \mid P] - \beta(P, a) \rho(a) \mathbb{E}[(V(v) - P)^2 v \mid P] \right), \quad (45)$$

where $\pi(a) = -u'''(W_0)/u''(W_0)$ is the coefficient of absolute prudence and $\psi(P) = \frac{\partial}{\partial \mu} \mathbb{E}[V(v) - P \mid P]$.

Proof. We follow the same strategy as the computation of $\beta(P, a)$. The optimal demand satisfies $G(x^*, s, \mu) = 0$, where μ appears in both the exponential $\exp((sv - \mu v^2/2) t(a))$ and the price belief $g(v, P, \mu)$. Since the first-order condition is homogeneous in the price belief,

we can normalize $g(v, P, \mu)$ into the corresponding conditional density of v given P and express all moments below as conditional expectations. From the proof of Lemma 7, $G_{x^*} = -u'(W_0) \rho(a) \Sigma_{VV}(P)$.

Step 1: Computing x_μ^* . By the Implicit Function Theorem, $x_\mu^* = -G_\mu / G_{x^*}$. Differentiating G with respect to μ at $s = 0$, $\mu = 0$, $x^* = 0$, both the exponential and the price belief contribute:

$$G_\mu = u'(W_0) \left[-\frac{t(a)}{2} \mathbb{E}[(V(v) - P) v^2 \mid P] + \frac{\partial}{\partial \mu} \mathbb{E}[V(v) - P \mid P] \Big|_{\mu=0} \right].$$

By the efficiency condition, $\mathbb{E}[V - P \mid P] = 0$, so the first term equals $-\frac{t(a)}{2} \text{Cov}(V - P, v^2 \mid P)$. Using the earlier definition $\psi(P) = \frac{\partial}{\partial \mu} \mathbb{E}[V(v) - P \mid P]$, we obtain

$$x_\mu^* = \frac{\psi(P) - \frac{t(a)}{2} \text{Cov}(V - P, v^2 \mid P)}{\rho(a) \Sigma_{VV}(P)}. \quad (46)$$

The derivatives under the integral are justified by Assumption 3, parts 3–4.

Step 2: Computing x_{ss}^* . By implicit differentiation, $x_{ss}^* = -\frac{1}{G_{x^*}} [G_{ss} + 2\beta G_{x^*s} + \beta^2 G_{x^*x^*}]$. Since g is independent of x^* (price-taking), the cross-derivatives involve only utility terms:

$$\begin{aligned} G_{ss} &= u'(W_0) t(a)^2 \mathbb{E}[(V(v) - P) v^2 \mid P] = u'(W_0) t(a)^2 \text{Cov}(V - P, v^2 \mid P), \\ G_{x^*s} &= u''(W_0) t(a) \mathbb{E}[(V(v) - P)^2 v \mid P], \\ G_{x^*x^*} &= u'''(W_0) \mathbb{E}[(V(v) - P)^3 \mid P]. \end{aligned}$$

Step 3: Combining. In the combination $\delta = \frac{1}{2t(a)} x_{ss}^* + x_\mu^*$, the $\text{Cov}(V - P, v^2 \mid P)$ contribution from $G_{ss} / (2t G_{x^*})$ exactly cancels the corresponding term in (46). Using $-u'''(W_0) / u''(W_0) = \pi(a)$, $x_s^* = \beta(P, a)$, $\mathbb{E}[(V(v) - P)^3 \mid P] = \text{Sk}[V \mid P]$, and $\mathbb{E}[(V(v) - P)^2 \mid P] = \Sigma_{VV}(P)$, the surviving terms yield the expression in the lemma. ■

C.5.3 Proof of Theorem 1

Proof. The proof proceeds in four steps.

Step 1: Demand coefficients. By Lemma 7, the trading intensity is

$$\beta(P, a) = \frac{t(a) \text{Cov}(V(v), v \mid P)}{\rho(a) \text{Var}(V(v) \mid P)}.$$

Since $\lambda(P) = 0$ in the correlation-neglect case, this is the REE expression (Lemma 11) with $\lambda = 0$. By Lemma 8, the drift coefficient $\delta(P, a)$ is given by (45), which is the REE expression (Lemma 13) with all λ -terms set to zero.

Step 2: Price function. The multiplicative separability $\beta(P, a) = \beta_a(a) \beta_P(P)$ implies that the cumulative demand depends on the price only through the sufficient statistic

$$s_p = \frac{\int_0^1 \beta_a(a) ds(a)}{\int_0^1 \beta_a(a) da} = v + \int_0^1 \frac{\omega(a)}{\sqrt{t(a)}} dB(a),$$

where $\omega(a) = \beta_a(a) / \int_0^1 \beta_a(b) db$. By Lemma 3, $v \mid s_p$ is normally distributed with precision $\tau = \tau_v + \tau_p$ and mean $(\tau_p / \tau) s_p$. The efficiency condition $\mathbb{E}[V(v) - P \mid P] = 0$ (Lemma 6) then yields the equilibrium price function

$$\mathcal{P}(s_p) = \int V\left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}}\right) d\Phi(z) = P, \quad (47)$$

where Φ is the standard normal CDF. The conditional moments of $V(v)$ can be computed from this representation:

$$\mathbb{E}[(V(v) - P)^k \mid P] = \int \left(V\left(\frac{\tau_p}{\tau} h(P) + \frac{z}{\sqrt{\tau}}\right) - P \right)^k d\Phi(z), \quad (48)$$

where $h(P) = \mathcal{P}^{-1}(P)$.

Step 3: Price informativeness. The price informativeness is

$$\tau_p = \frac{\left(\int_0^1 \beta_a(a) da\right)^2}{\int_0^1 \frac{\beta_a(a)^2}{t(a)} da} = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a)} da\right)^2}{\int_0^1 \frac{t(a)}{\rho(a)^2} da}.$$

Unlike in the REE case (Theorem 6), τ_p does not involve a fixed-point equation: $\beta_a(a) = t(a)/\rho(a)$ depends only on primitives, so τ_p is determined in closed form.

Step 4: Supporting price beliefs and uniqueness. It remains to verify that the scaled risk premium selected by market clearing can be supported by a sequence of price beliefs. We construct one such sequence as follows. Let

$$m_0(P) = \int_{\mathbb{R}} g(v, P, 0) dv \quad \text{and} \quad f_0(v | P) = \frac{g(v, P, 0)}{m_0(P)}$$

on the support of the equilibrium price. Thus, $f_0(\cdot | P)$ is the equilibrium conditional density. Throughout this step, $\mathbb{E}_0[\cdot | P]$ and $\text{Cov}_0(\cdot, \cdot | P)$ denote expectation and covariance under $f_0(\cdot | P)$, respectively. By efficiency, $\mathbb{E}_0[V(v) - P | P] = 0$. The bounded transformation and coefficient used below ensure that the perturbed belief remains nonnegative and satisfies the domination condition even when $V(v) - P$ is unbounded; a direct linear perturbation proportional to $V(v) - P$ could become negative in the tails. Put $Y = V(v) - P$ and define

$$\begin{aligned} q(v, P) &= \tanh(Y) - \mathbb{E}_0[\tanh(Y) | P], \\ K(P) &= \mathbb{E}_0[Y q(v, P) | P] = \text{Cov}_0(Y, \tanh(Y) | P) > 0, \\ c(P) &= \frac{\psi(P)}{K(P)}. \end{aligned}$$

The strict inequality follows because $\tanh$ is strictly increasing and Y is nondegenerate (as is already required by $\text{Var}(V(v) | P) > 0$). For μ near zero, set

$$\alpha(\mu, P) = \frac{1}{2} \tanh(2\mu c(P)), \quad f_\mu(v | P) = f_0(v | P)[1 + \alpha(\mu, P)q(v, P)],$$

and define the joint price belief by $g(v, P, \mu) = m_0(P)f_\mu(v | P)$. Since $\mathbb{E}_0[q | P] = 0$, each $f_\mu(\cdot | P)$ integrates to one. Moreover, $|q| < 2$ and $|\alpha| < 1/2$, so it is nonnegative. The construction therefore gives valid joint densities and satisfies $g(v, P, \mu) \rightarrow g(v, P, 0)$. Finally,

$$\frac{1}{\mu} \mathbb{E}_\mu[V(v) - P | P] = \frac{\alpha(\mu, P)}{\mu} K(P) \longrightarrow c(P)K(P) = \psi(P).$$

Equivalently, to first order $f_\mu = f_0 + \mu f_1 + o(\mu)$, where $f_1 = c(P)qf_0$ satisfies

$$\int_{\mathbb{R}} f_1(v | P) dv = 0, \quad \int_{\mathbb{R}} (V(v) - P) f_1(v | P) dv = \psi(P).$$

For the joint density, the corresponding derivative is $g_1 = m_0(P)f_1$, and hence

$$\int_{\mathbb{R}} (V(v) - P) g_1(v, P) dv = m_0(P)\psi(P),$$

exactly accounting for the denominator in (18). Since the underlying supply schedule $\Theta(P)$ is continuously differentiable, this bounded perturbation preserves the domination required in Assumption 3; its stated smoothness conditions give the required differentiability.

Thus a supporting sequence of price beliefs exists. Given τ_p , the price function $\mathcal{P}(\cdot)$ is uniquely determined by (47), and the drift coefficient $\delta(P, a)$ is determined by Lemma 8, with $\psi(P)$ pinned down by the market-clearing condition $\int_0^1 \delta(a, P) da + h(P) \int_0^1 \beta(a, P) da = \Theta(P)$. The equilibrium is therefore unique.

■

C.6 Proof of Proposition 1

Proof of Proposition 1. From definition of $\mathcal{I}$ we have $\mathcal{I} = \tau_p/\tau$. Substituting (16) into the last equation and rearranging, we obtain the stated expression for $\mathcal{I}$.

We turn to deriving expression for $\mathcal{L}$. The market clearing price $\mathcal{P}(s_p, \bar{\theta})$ is P that solves

$$s_p = \frac{\bar{\theta} + \theta(P) - \int \delta(a, P) da}{\int \beta(a, P) da}.$$

Then define

$$\begin{aligned} \text{Liquidity} &\equiv - \left(\frac{\partial}{\partial \bar{\theta}} \mathcal{P}(s_p, \bar{\theta}) \right)^{-1} \\ &= -s_p \frac{\partial}{\partial P} \left(\int \beta(a, P) da \right) - \frac{\partial}{\partial P} \left(\int \delta(a, P) da - \theta(P) \right) \end{aligned}$$

On the other hand,

$$\begin{aligned} \frac{\partial}{\partial s_p} \mathcal{P}(s_p, \bar{\theta}) &= \frac{- \int \beta(a, P) da}{s_p \frac{\partial}{\partial P} \left(\int \beta(a, P) da \right) + \frac{\partial}{\partial P} \left(\int \delta(a, P) da - \theta(P) \right)} \\ &= \text{Liquidity}^{-1} \cdot \int \beta(a, P) da. \end{aligned}$$

From (15) we get that

$$\frac{\partial}{\partial s_p} \mathcal{P}(s_p, \bar{\theta}) = \frac{\tau_p}{\tau} \int V' \left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}} \right) d\Phi(z) = \frac{\tau_p}{\tau} E[V'(v)|P].$$

Combining the two preceding equations and substituting the expression for $\beta(a, P)$ we get

$$\begin{aligned} \frac{\tau_p}{\tau} E[V'(v)|P] &= \text{Liquidity}^{-1} \cdot \int \frac{t(a)}{\rho(a)} \frac{\tau^{-1} \mathbb{E}[V'(v)|P]}{\text{Var}[V(v)|P]} da \implies \\ \text{Liquidity} &= \frac{1}{\tau_p \text{Var}[V(v)|P]} \int \frac{t(a)}{\rho(a)} da. \end{aligned}$$

Multiplying by $\text{Var}[V(v)|P]$ and substituting (16) we get the stated equation for $\mathcal{L}$. ■

C.7 Proof of Lemma 4

Proof of Lemma 4. First, from Lemma 3 we have $\tau_{agg} \equiv \text{Var}(v|s[\omega^H(a)])^{-1} - \text{Var}(v)^{-1} = 1/\left(\int_0^1 \omega^H(a)^2/t(a)da\right)$. Second, we apply the Cauchy-Bunyakovsky-Schwartz inequality

$$\left(\int_0^1 f(a)g(a)da\right)^2 \leq \int_0^1 f(a)^2da \int_0^1 g(a)^2da \quad (49)$$

with $f(a) = \omega^H(a)/\sqrt{t(a)}$ and $g(a) = \sqrt{t(a)}$ to obtain $\tau_{agg}(b(a)) \leq \int_0^1 t(a)da$. The equality in (49) is attained if, and only if, $f(a)$ and $g(a)$ are linearly dependent, i.e., when $\omega^H(a)/\sqrt{t(a)} = c\sqrt{t(a)}$. The constant c is pinned down by the condition $\int_0^1 w(a)da = 1$. ■

C.8 Proof of Proposition 2

Proof of Proposition 2. Denote absolute risk tolerance $y(a) \equiv 1/\rho(a)$. Without loss of generality, index traders such that $y(b)$ increases in b . (This is in contrast to index a , which is such that $W_0(a)$ is increasing in a .) We first compute the Gateaux derivatives of $\mathcal{I}$ and $\mathcal{L}$ with respect to $y(b)$. We then show that under DARA utilities and the technical conditions imposed, the signs of the derivatives with respect to $y(b)$ and $W_0(a)$ are the same. We start by proving the following statement.

Step 1. There exist thresholds $0 < b_1^y \leq b_2^y < 1$, such that for any Robin Hood variation $y^\Delta(b)$ with $\underline{b} \leq b_1^y \leq b_2^y \leq \bar{b}$

$$\begin{aligned} \mathcal{I}'(y(b))[y^\Delta(b)] &> 0 \text{ and } \mathcal{I}'(y(b))[-y^\Delta(b)] < 0; \\ (\mathcal{L})'(y(b))[y^\Delta(b)] &< 0 \text{ and } (\mathcal{L})'(y(b))[-y^\Delta(b)] > 0. \end{aligned} \quad (50)$$

From (16) we obtain

$$\tau_p = \left(\int_0^1 y(b)t(b)db\right)^2 / \left(\int_0^1 y(b)^2 t(b)db\right). \quad (51)$$

Substituting this expression into $\mathcal{I} = \tau_p/(\tau_p + \tau_v)$ and computing the Gateaux derivative (this entails substituting $y(b) + \epsilon y^\Delta(b)$ instead of $y(b)$, differentiating with respect to ϵ , and evaluating the resulting expression at $\epsilon = 0$) yields:

$$\mathcal{I}'(y(b))[y^\Delta(b)] = C_{\mathcal{I}} \int_0^1 t(b) y^\Delta(b) (I_2 - I_1 y(b)) db.$$

Here, $C_{\mathcal{I}} > 0$ is positive (we have the closed-form expressions for $C_{\mathcal{I}}$ via parameters of the model, but it is not important here), $I_1 = \int_0^1 t(b) y(b) db$ and $I_2 = \int_0^1 t(b) y(b)^2 db$. Lemma 9 (to follow) implies that there exists a unique b_y^* such that $I_2 - I_1 y(b) \geq 0$ iff $b \leq b_y^*$. Then, for a $y^\Delta(b)$ that is Robin Hood with $\underline{b} < b_y^* < \bar{b}$, $y^\Delta(b)(I_2 - I_1 y(b)) > 0$ and the first statement of this step follows by letting $b_1^y = b_2^y = b_y^*$.

One can obtain $\mathcal{L} = \int_0^1 t(b) y(b) db / \tau_p$. Substituting (51) in this equation and computing the Gateaux derivative yields

$$(\mathcal{L})'(y(b))[y^\Delta(b)] = C_{\mathcal{L}} \int_0^1 t(b) y^\Delta(b) (2I_1 y(b) - I_2) db.$$

Here, $C_{\mathcal{L}} > 0$ is positive. Because $y(b)$ is increasing, $2I_1 y(b) - I_2$ is increasing in b , so Lemma 9 (to follow) implies that there exists a unique $b_{**}^y < b_y^*$ such that $2I_1 y(b) - I_2 \geq 0$ iff $b \geq b_{**}^y$. Then, for a $y^\Delta(b)$ that is Robin Hood with $\underline{b} < b_{**}^y < \bar{b}$, the product $y^\Delta(b)(2I_1 y(b) - I_2) \leq 0$ on both tails—nonpositive for $b < \underline{b}$ (where $y^\Delta(b) \geq 0$ while $2I_1 y(b) - I_2 < 0$) and for $b > \bar{b}$ (where $y^\Delta(b) \leq 0$ while $2I_1 y(b) - I_2 > 0$), and zero on $[\underline{b}, \bar{b}]$ —so that $(\mathcal{L})'(y(b))[y^\Delta(b)] < 0$. The first three statements of this step follow by letting $b_2^y = b_y^*$ and $b_1^y = b_{**}^y$.

Step 2. There exist thresholds $0 < a_1^W \leq a_2^W < 1$, such that for any Robin Hood variation $W_0^\Delta(a)$ with $\underline{a} \leq a_1^W \leq a_2^W \leq \bar{a}$ (50) follows.

Let a_2^W be the largest solution to $W_0(a) = \bar{\eta} y(b_2^y)$. Note that for any $a > a_2^W$, we have $y(a) = W_0(a)/\eta(a) > W_0(a)/\bar{\eta} > y(b_2^y)$. Thus, by the previous step of the proposition, decreasing risk tolerances for traders $a > a_2^W$ leads to improvement in information efficiency and reduction in

liquidity. Note also that decreasing wealth for traders $a > a_2^W$ induces their risk tolerances to decrease as well (DARA utilities). Similarly, letting a_1^W be the smallest solution to $W_0(a) = \underline{\eta}y(b_1^y)$, we get that for any $a < a_1^W$, $y(a) < y(b_1^y)$ and that increasing wealth for traders $a < a_1^W$ induces their risk tolerances to increase as well (DARA utilities). Then, by the previous step, the statement of the proposition holds. ■

Lemma 9. *Assume $0 < \underline{\eta} \leq \eta(a) \leq \bar{\eta} < \infty$. Assume that absolute risk tolerance $y(a)$ is a continuous and strictly increasing function of a . For any $c > 0$, there exists a unique solution $\hat{a}(c)$ to $y(a) = c$, moreover, $\hat{a}(c)$ increases in c and $0 < \hat{a}(c) < 1$ for any $0 < c < \infty$.*

Proof of Lemma 9. Since $y(a)$ increases in a , at most one solution to $y(a) = c$ exists. Monotonicity also implies $\hat{a}(c)$ increases in c . For risk tolerance $y(a)$ we can write $y(a) = W_0(a)/\eta(a)$. We have $y(a) < W_0(a)/\underline{\eta}$ and so $0 \leq \lim_{a \rightarrow 0} y(a) = \inf y(a) \leq \inf \{W_0(a)\}/\underline{\eta} = 0$. Thus, $\lim_{a \rightarrow 0} y(a) = 0$. One can show analogously that $\lim_{a \rightarrow \infty} y(a) = \infty$. Then, by Intermediate Value Theorem, $\hat{a}(c)$ exists and $0 < \hat{a}(c) < 1$. ■

C.9 Proof of Proposition 3

Proof of Proposition 3. This proof follows the same steps as the proof of Proposition 2. Without loss of generality, index traders such that $y(b)$ increases in b .

Step 1. There exist thresholds $0 < b_1^t \leq b_2^t < 1$, such that for any Robin Hood variation $t^\Delta(b)$ with $\underline{b} \leq b_1^t \leq b_2^t \leq \bar{b}$

$$\mathcal{I}'(t(b))[t^\Delta(b)] > 0 \text{ and } \mathcal{I}'(t(b))[-t^\Delta(b)] < 0;$$

$$(\mathcal{L})'(t(b))[t^\Delta(b)] < 0 \text{ and } (\mathcal{L})'(t(b))[-t^\Delta(b)] > 0.$$

Here, the proof is identical to step 1 of Proposition 2, with the difference of expressions for the Gateaux derivatives, which we reproduce below:

$$\begin{aligned}\mathcal{I}'(t(b))[t^\Delta(b)] &= C_{\mathcal{I}} \int_0^1 y(b)t^\Delta(b)(2I_2 - I_1 y(b)) db; \\ (\mathcal{L})'(t(b))[t^\Delta(b)] &= C_{\mathcal{L}} \int_0^1 y(b)t^\Delta(b)(I_1 y(b) - I_2) db;\end{aligned}$$

Step 2. There exist thresholds $0 < a_1^t \leq a_2^t < 1$, such that for any Robin Hood variation $t^\Delta(a)$ with $\underline{a} \leq a_1^t \leq a_2^t \leq \bar{a}$, (30) holds.

Let a_2^t be the largest solution to $W_0(a) = \bar{\eta}y(b_2^t)$. Note that for any $a > a_2^t$, we have $y(a) > W_0(a)/\bar{\eta} > y(b_2^t)$. Similarly, letting a_1^t be the smallest solution to $W_0(a) = \underline{\eta}y(b_1^t)$, we get that for any $a < a_1^t$, $y(a) < y(b_1^t)$. Then, by the previous step, the statement of the proposition holds. ■

C.10 Proof of Proposition 4

Proof of Proposition 4. With endogenous information acquisition and DARA utilities, the precision of information is increasing in wealth. The transfer of wealth from rich to poor has two effects. The indirect one of decreasing the precisions of the rich and increasing that of the poor (via endogenous information acquisition) is captured by the Proposition 3. Because the marginal response t_W is bounded, the induced precision direction $t_W W_0^\Delta(a)$ remains bounded and integrable, strictly satisfying the requirements of a Robin Hood variation so the functional chain rule applies. Proposition 2 captures the direct one. By taking the thresholds $\underline{a} \leq \min\{a_1^t, a_1^W\}$ and $\bar{a} \geq \max\{a_2^t, a_2^W\}$ we make sure that both propositions apply ■

C.11 Completing heuristic derivation in Appendix B

Here, we justify the transition from (35) to (36). Note that by the stochastic calculus heuristics (31), $dX(a)^2 = \beta(a, P)^2/t(a)da$ and so $\mathbb{E}[(V - P)^3|ds(a), \hat{s}_p] dX(a)^2$ does not contain any $ds(a)$ terms. (Indeed, the terms in $\mathbb{E}[(V - P)^3|ds(a), \hat{s}_p]$ that contain $ds(a)$ will be zeroed out after

multiplying by $dX(a)^2 = \beta(a, P)^2/t(a)da$ and applying (31).)

Similarly, the term $\mathbb{E}[(V - P)^2|ds(a), \hat{s}_p]$ in (35) can be replaced by $\mathbb{E}[(V - P)^2|s_p]$ in (36). The difference $\mathbb{E}[(V - P)^2|ds(a), \hat{s}_p] - \mathbb{E}[(V - P)^2|\hat{s}_p]$ contains a leading $ds(a)$ term, which will become a da term after being multiplied by $\beta(a, P)ds(a)$ in (36). Finally, $\mathbb{E}[(V - P)^2|\hat{s}_p]$ can be replaced by $\mathbb{E}[(V - P)^2|s_p]$ since conditional distributions $v|\hat{s}_p$ and $v|s_p$ are the same.

C.12 Proof of Proposition 5

Proof of Proposition 5. Fix a trader i . Given the price P , his realized utility at time $t = 2$, net of the information acquisition cost, is

$$\mathcal{U}_{i,t=2}(\Delta s_i, m_i) = u(W_0(a_i) - c(t(a_i), a_i)m_i + x(\Delta s_i, m_i; a_i)(V(v) - P)).$$

The Δs_i is a finite increment of a diffusion process

$$ds(b) = vdb + \frac{1}{\sqrt{t(a_i)}}dB(b)$$

between $b = a_i$ and $b = a_i + m_i$. Similarly, $\mathcal{U}_{i,t=2}(\Delta s_i, m_i) - \mathcal{U}_{i,t=2}(0, 0)$ can be viewed as a finite increment of a diffusion process driven by $ds(b)$. Because the cost penalty $c(t(a_i), a_i)m_i$ is deterministic and linear in m_i , its contribution to the first-order drift is simply $-u'(W_0(a_i))c(t(a_i), a_i)$, matching the heuristic derivation in Section D.1.

By Ito's lemma, we can write

$$\mathcal{U}_{i,t=2}(\Delta s_i, m_i) - \hat{\mathcal{U}}_{i,t=2}(0, 0) = \int_0^m \mu_u(b)db + \int_0^m \sigma_u(b)dB,$$

where μ_u and σ_u denote the drift (the “ db ” coefficient) and the diffusion coefficients (the “ $dB(b)$ ” coefficient) of $\mathcal{U}_{i,t=2}$ process.

By Lemma 10 (to follow), the optimal precision solves

$$t(a_i) \in \arg \max_t \left\{ \frac{\partial \mathbb{E}[\mathcal{U}_{i,t=2}]}{\partial m_i} \Big|_{m_i=0} \right\}. \quad (52)$$

Thus, $t(a_i)$ maximizes the expected drift of $\hat{\mathcal{U}}_{i,t=2}$ at $m_i = 0$, $\mathbb{E}[\mu_u(0)]$. Under the uniform integrability, smoothness, and domination conditions established in Assumption 3, the application of the envelope theorem to eliminate the derivatives of optimal demand is justified. Furthermore, these conditions permit the interchange of the expectation and the limit as $m_i \rightarrow 0$. Consequently, the formal expected drift $\mathbb{E}[\mu_u(0)]$ exactly coincides with the limit of the objective derived algebraically in Section D.1, with the first order condition in (52) reducing to (59). ■

Lemma 10. *Consider a continuously differentiable function $f(t, m)$ such that $f(t, 0)$ does not depend on t . Consider $t(m) \in \arg \max_t f(t, m)$. Suppose that $\lim_{m \rightarrow 0} t(m) = t(0)$. Then, $t(0) \in \arg \max_t f_m(t, 0)$.*

Proof of Lemma 10. Suppose, on the contrary, that there exists some $\check{t}$ such that $f_m(\check{t}, 0) > f_m(t(0), 0)$. Because $\lim_{m \rightarrow 0} t(m) = t(0)$, continuity implies that $\lim_{m \rightarrow 0} f_m(t(m), m) = f_m(t(0), 0)$. Therefore, there exists $\bar{m}$ such that

$$f_m(\check{t}, m) > f_m(t(m), m) \text{ for } m < \bar{m}. \quad (53)$$

Integrate (53) with respect to m :⁴¹

$$f(\check{t}, m) - f(\check{t}, 0) > f(t(m), m) - f(t(0), 0).$$

Since $f(t, 0)$ does not depend on t we have $f(t(0), 0) = f(\check{t}, 0)$ and so

$$f(\check{t}, m) > f(t(m), m).$$

⁴¹To integrate the right-hand side we use the Envelope Theorem $\frac{df(t(m), m)}{dm} = f_m(t(m), m)$

We obtained a contradiction with $t(m) \in \arg \max_t f(t, m)$. ■

D Information acquisition

Information acquisition happens at $t = 0$. The precisions $t(a)$ are endogenized by requiring them to be optimal, given an information acquisition cost. We define the notion of optimality of the profile of precisions $t(a)$ similarly to how we defined the optimality of demands: precision profile $t(a)$ is optimal in CHILE if it is a limit of optimal precisions in the discrete economies. We make this definition precise after we describe the information acquisition in the discrete economy. We assume that agents are uncertain about the distribution of wealth $W_0(a)$ (but they know their wealth).⁴² This distribution becomes known before the start of the trade at $t = 1$.

The new primitive in CHILE is the information acquisition cost $c(t, a)$. We assume that the cost of acquiring an infinitesimal signal $ds(a) = vda + 1/\sqrt{t}dB(a)$ for a trader a is $c(t; a)da$, where $c(\cdot)$ is continuous, strictly increasing and convex function of t . Thus, the cost of acquiring a finite signal

$$\Delta s = \int_{a_i}^{a_i+m_i} vda + \int_{a_i}^{a_i+m_i} \frac{1}{\sqrt{t(a)}}dB(a) \text{ is } \int_{a_i}^{a_i+m_i} c(t(a), a_i)da.$$

A finite signal can be split into a collection of infinitesimal ones, with associated costs. Using Jensen's inequality, one can show that it is not optimal to split a finite signal into infinitesimal ones of varying precisions. If a trader wants to get information of precision tm_i , he should acquire a signal

$$\Delta s = \int_{a_i}^{a_i+m_i} vda + \frac{1}{\sqrt{t}}dB(a) = vm_i + \frac{B(a_i + m_i) - B(a_i)}{\sqrt{t}} \text{ at a cost } c(t; a_i)m_i. \quad (54)$$

Thus, without loss of generality, we restrict finite signals in the discrete economy to be of the form (54).

⁴²Done this way, the information choice of agent a will only depend on his wealth, but not the wealth of others, which helps to simplify some of the analysis.

Denote $\mathcal{U}_i(W_0^i, t_i, m_i)$ the maximum utility for a given precision t_i and initial wealth W_0^i .⁴³ Define $c(t_i, a_i)m_i$ as a (monetary) cost of acquiring signal Δs_i . We require that precisions t_i are *optimal*, i.e.

$$t_i = \arg \max_t \left\{ \int \mathcal{U}_i(W_0^i - c(t, a_i)m_i, t, m_i) g(v, P, \mu) dv dP \right\}. \quad (55)$$

Here $g(\cdot)$ denotes the PDF of the joint distribution of $\mathbf{P}_i^n$ and v . We denote $t^n(a)$ the profile of optimal precisions in the discrete economy, $t^n(a) = t_i$, for all $a \in [a_i, a_i + m_i)$, where t_i solves (55).

We introduce some technical restrictions on $c(t; a)$.

Assumption 4. *The information acquisition cost $c(t; a)$ is such that there exists M_t such that for any i , $\hat{t}_i > M_t$ is not optimal.*

The additional technical restriction allows us to make a choice set in (55) compact. Thus, without loss of generality, we assume $t \in [0, M_t]$ everywhere in the sequel. We define the notion of optimal profile of precisions $t(a)$ by continuity.

Definition 4. *A profile $t(a)$ is **optimal** if, for every a , the optimal precision in discrete economy $t^n(a)$ converges to $t(a)$ as $n \rightarrow \infty$.*

Finally, we define the information acquisition equilibrium at $t = 0$.

Definition 5. *A profile $t(a)$ is an information acquisition **equilibrium** if for every a , $t(a)$ is optimal.*

With these definitions at hand, we are ready to state this section's main result in the subsection below.

⁴³Here we follow the setup of the main paper, where the profile of preferences in the discrete and continuous economy are the same. This approach can be generalized to allow them to differ for finite n but coincide in the $n \rightarrow \infty$ limit. Such an alternative approach could allow to lift the technical restrictions from the primitives of CHILE and “shift” those restrictions to the primitives of discrete economies instead.

D.1 Information acquisition: heuristic derivation

It is illuminating to start with the heuristic derivation. Consider a change in trader a 's time-2 realized utility due to trade

$$\begin{aligned} d\mathcal{U}_{t=2}(dX(a); a) &= u(W_0(a) - c(t(a))da + dX(a)(V(v) - P); a) - u(W_0(a); a) \\ &= u'(W_0(a)) \left(dX(a)(V(v) - P) - c(t(a))da - \frac{\rho(a)}{2} dX(a)^2 (V(v) - P)^2 \right) \end{aligned}$$

In the second line, we Taylor expanded the $u(\cdot)$, up to terms of order da . Due to the heuristics of Appendix B, we have $dX(a)^2 = \beta(a, P)^2/t(a)da$. Now substitute $P = \mathbf{P}(a)$ (trader a 's conjecture about the market-clearing price) and take the expectation:

$$\frac{\mathbb{E}[d\mathcal{U}_{t=2}(dX(a); a)]}{u'(W_0(a); a)} = \mathbb{E}[dX(a)(V(v) - \mathbf{P}(a)) - c(t(a))da - \frac{\rho(a)}{2t(a)} \mathbb{E}[\beta(a, \mathbf{P}(a))^2 (V(v) - \mathbf{P}(a))^2]da]. \quad (56)$$

Now simplify

$$\begin{aligned} \mathbb{E}[dX(a)(V(v) - \mathbf{P}(a))] &= da \mathbb{E}[\beta(a, \mathbf{P}(a))v(V(v) - \mathbf{P}(a))] \\ &\quad + 1/\sqrt{t(a)} \mathbb{E}[dB(a)\beta(a, \mathbf{P}(a))(V(v) - \mathbf{P}(a))] \\ &\quad + da \mathbb{E}[\delta(a, \mathbf{P}(a))(V(v) - \mathbf{P}(a))] \\ &= da \mathbb{E}[\beta(a, \mathbf{P}(a))v(V(v) - \mathbf{P}(a))]. \end{aligned} \quad (57)$$

In the first transition, we substituted $dX(a) = \beta(a, P)ds(a) + \delta(a, P)da$. To get (57), we substituted $\mathbb{E}[dB(a)\beta(a, \mathbf{P}(a))(V(v) - \mathbf{P}(a))] = 0$ ($dB(a)$ is independent of $V(v)$ and $\mathbf{P}(a)$ as the traders neglect the signal-price correlation and assume the noise in their signals is independent from that in the price) and $\mathbb{E}[\delta(a, \mathbf{P}(a))(V(v) - \mathbf{P}(a))] = 0$ (market efficiency condition).

To proceed further, we divide (56) by da and pass to $da \rightarrow 0$ limit. In that limit, $\mathbf{P}(a)$ becomes the market clearing price $\mathbf{P}_*$, which for ease of notation we simply denote P going forward.

We can further simplify (57) by noting that $\mathbb{E}[v(V(v) - P)|P] = \beta(a, P)\text{Var}[V(v)|P]\rho(a)/t(a)$ (see (44)). Then, after substituting $\mathbb{E}[(V(v) - P)^2|P] = \text{Var}[V(v)|P]$, and $\beta(a, P) = t(a)/\rho(a)\beta_P(P)$ we can finally obtain

$$\frac{\mathbb{E}[d\mathcal{U}_{t=2}(dX(a); a)]}{u'(W_0(a); a)da} = \frac{t(a)}{2\rho(a)} \mathbb{E}[\beta_P(P)^2 \text{Var}[V(v)|P]] - c(t(a)) \quad (58)$$

The optimal precision maximizes (58) with respect to $t(a)$. We summarize in the Proposition below. The rigorous proof is in the Appendix C.12.

Proposition 5. *Assuming the cost function $c(t, a)$ is differentiable and is such that the optimal precision is interior ($0 < t(a) < M_t$), the precision choice for a trader a solves*

$$c'(t, a) = \frac{1}{2\rho(a)} \mathbb{E}[\beta_P(P)^2 \text{Var}[V(v)|P]]. \quad (59)$$

Here $\beta_P(P) = \frac{\mathbb{E}[v(V(v) - P)|P]}{\text{Var}[V(v)|P]}$. When $V(v) = \exp(v)$, the optimal precision solves

$$c'(t, a) = \frac{1}{2\rho(a)} \mathbb{E}\left[\frac{1}{\tau^2 (\exp(\tau^{-1}) - 1)}\right]. \quad (60)$$

Provided that the equilibrium exists, the equilibrium precision for trader a is an increasing function of his wealth $W_0(a)$ under DARA preferences.

Note that the expectations in (59) and (60) are over prices P , distribution of aggregate signal paths $\{s(a)\}_a$, and potential wealth distributions.

E A benchmark model with LLN aggregation

This section demonstrates that our main results do not obtain if one adopts the traditional approach to modeling a large economy based on law of large numbers (LLN) aggregation, as in [Hellwig \(1980\)](#), [Admati \(1985\)](#), [Peress \(2004\)](#), among others. The key finding here is an aggregation result: an economy with investors differing in wealth, precision, and preferences appears observationally equivalent to one with homogeneous investors. Consequently, in such economies, Robin Hood variations in wealth that leave aggregate trading intensity unchanged have no effect on market quality, unlike in CHILE.

This section derives our main results using the model of [Peress \(2011\)](#), which is, to the best of our knowledge, the only model in the literature with micro-founded wealth effects and heterogeneous information.⁴⁴ We present the [Peress \(2011\)](#) model using notation similar to ours and provide brief derivations of the main results. For more details, we refer the reader to the original paper.

The economy unfolds over two time periods, $t = 1$ and $t = 2$. It consists of a continuum of agents of total mass one. There are two assets: one risk-free and one risky, with the returns of both assets, realized at $t = 2$. The risk-free asset has a perfectly elastic supply and gross return normalized to 1, while the risky asset has a liquidation value of V that is log-normally distributed.⁴⁵

Agents trade at $t = 1$, determining the price P of the risky asset. There are rational traders indexed by $a \in [0, 1]$ —the continuum limit of an economy with $n \rightarrow \infty$ traders—and noise traders. Noise traders submit price-inelastic demand $\theta \sim N(0, \tau_\theta^{-1})$ (measured in units of stocks). Agent a has initial wealth $W_0(a)$, and he trades a total amount of x dollars—i.e. price

⁴⁴It is straightforward to derive the results of this section within a mean-variance preference framework that includes ad-hoc wealth effects, where the "variance" coefficient depends on wealth, as in [Farboodi et al. \(2025\)](#) and [Mihet \(2025\)](#).

⁴⁵It is also straightforward to derive our results for the case of normal distribution of V . However, existing methods are limited in their ability to accommodate the broad range of payoff functions that CHILE supports.

per share times number of shares. His realized utility is

$$u(W, a),$$

where his wealth at time $t = 2$ is

$$W = W_0(a) + x(R - 1),$$

and where

$$R = V/P.$$

Each trader has a private signal $s(a) = v + Z(a)/\sqrt{t(a)}$, where $Z(a)$ are i.i.d. standard Normal random variables.

The analytical tractability is achieved by employing a “small risk” approximation. It is assumed that

$$\ln V = \zeta \cdot v + v_n, \tag{61}$$

where $v \sim N(0, \tau_v^{-1})$ is a learnable component of asset payoff, and $v_n \sim N(0, \zeta \tau_{v,n}^{-1})$ is a non-learnable component.⁴⁶ The approach consists of characterizing the equilibrium asymptotics as $\zeta \rightarrow 0$.

Remark 10. Relative to the model in the paper, the model here features two new ingredients. The first is noise θ . The second one is the unlearnable component v_n of asset payoff. Both are essential for the equilibrium here to be well-defined. Without θ , there will be no noise in the price, and the equilibrium will not exist. Without v_n , the asymptotics of equilibrium as $\zeta \rightarrow 0$ will not be well-behaved (see [Peress \(2011\)](#)). Our model in the paper is well-defined without these two elements, so we did not introduce them there. However, we verified that introducing them does not change our main conclusions. Thus, the comparison to the benchmark here is

⁴⁶This is equivalent to [Peress \(2011\)](#) formulation, saying that agents learn about the mean of log-payoff as opposed to log-payoff itself. Indeed our assumption can equivalently be formulated as $\ln V \sim N(\zeta v, \zeta \tau_{v,n}^{-1})$, the mean $v \sim N(0, \tau_v^{-1})$ is not known, and agents learn about mean v as opposed to about log-payoff $\ln V$.

fair.

Remark 11. The aggregation here will be performed by computing $\int x(a)da$. It should be noted that $\int x(a)da$ represents the average, not aggregate demand. Indeed, the integral is approximately $\sum_a x(a)\frac{1}{n}$, which is average demand. (Here, n is the number of traders in the economy.) In the large economy here, each demand is not infinitesimal $1/x(a) < \infty$, so their sum is infinite. Starting from [Hellwig \(1980\)](#), the literature solves this problem by equalizing the average, not aggregate demand $\int x(a)da$ to the noisy supply θ . Consequently, the actual aggregate noisy supply, $\theta \cdot n$, has infinite variance.

E.1 Equilibrium

Solving for equilibrium involves taking the standard steps. We postulate that for small ζ , traders have the following conjecture about the price function

$$\ln P \equiv p = \hat{p}\zeta, \text{ where } \hat{p} = c_0 + c_1v - c_2\theta. \quad (62)$$

Given such conjecture, agents form their demands, which then results in an equilibrium price function $p^{eq}(\zeta)$. Equilibrium requires traders' conjecture to be consistent in the $\zeta \rightarrow 0$ limit: $p^{eq}(\zeta) = \hat{p}\zeta + o(\zeta)$.

Substituting (61) and conjecture (62) into agent a 's first order condition $E_a[u'(W_0(a) + x(\exp(\ln V - p) - 1)(\exp(\ln V - p) - 1))] = 0$, expanding it up to the terms of order ζ , we obtain $x(p) = \hat{x}(\hat{p}) + o(1)$, where

$$\hat{x}(\hat{p}) = \frac{E_a[v] - \hat{p} + 0.5\tau_{v,n}^{-1}}{\rho(a)\tau_{v,n}^{-1}},$$

where $\rho(a) = -u''(W_0(a))/u'(W_0(a))$ is trader a 's absolute risk aversion and

$$E_a[v] = \frac{t(a)}{\tau_v + \tau_p + t(a)}s(a) + \frac{\tau_p}{\tau_v + \tau_p + t(a)}\frac{\hat{p} - c_0}{c_1}, \text{ and } \tau_p \equiv \text{Var}[v|p]^{-1} - \tau_v = \frac{c_1^2}{c_2^2}\tau_\theta.$$

Applying market clearing, we get

$$v \cdot \tau_{v,n} \int_0^1 \frac{t(a)/\rho(a)}{\tau_v + \tau_p + t(a)} da + (\text{affine function of } p) = \theta,$$

and hence

$$\frac{c_1}{c_2} = \sqrt{\frac{\tau_p}{\tau_\theta}} = \tau_{v,n} \int_0^1 \frac{t(a)/\rho(a)}{\tau_v + \tau_p + t(a)} da.$$

The main result of this section follows.

Theorem 2. *There exists a unique equilibrium. The equilibrium demands can be written as $\hat{x}(\hat{p}) = \alpha(a) + \beta(a)s(a) - \gamma(a)\hat{p}$, where*

$$\beta(a) = \frac{t(a)\tau_{v,n}}{\rho(a)(\tau_v + \tau_p + t(a))}.$$

The price informativeness τ_p can be written as

$$\tau_p = \left(\int_0^1 \beta(a) da \right)^2 \tau_\theta$$

and is the unique solution to

$$\sqrt{\frac{\tau_p}{\tau_\theta}} = \tau_{v,n} \int_0^1 \frac{t(a)/\rho(a)}{\tau_v + \tau_p + t(a)} da.$$

The price function and the aggregate demand are the same as in the economy with homogeneous investors with absolute risk aversion ρ and precision t satisfying $\int_0^1 \frac{t(a)/\rho(a)}{\tau_v + \tau_p + t(a)} da = \frac{t/\rho}{\tau_v + \tau_p + t}$.

The economy with LLN aggregation allows for a “representative agent” formulation. Aggregate quantities, such as the price function and informativeness, remain consistent between the original heterogeneous economy and the equivalent homogeneous one. In this case, heterogeneity is irrelevant, unlike in CHILE. Consequently, we also get the following result, further illustrating the distinctions of CHILE and traditional economies in terms of comparative statics

of information efficiency.

Corollary 2. *Aggregate trading intensity $\int_0^1 \beta(a) da$ is a sufficient statistic of price informativeness. Changes in wealth distribution that leave the aggregate trading intensity unchanged leave price informativeness unchanged.*

In CHILE, under the assumptions of Proposition 2, *any* Robin Hood variation of wealth with thresholds $\underline{a} \leq a_1^W \leq a_2^W \leq \bar{a}$ results in increased information efficiency. This implies that the result holds even for redistributions that leave aggregate trading intensity unchanged. This distinction underscores the difference between CHILE and traditional large economies: changes in parameters that do not affect the parameters of an equivalent homogeneous economy have no impact on traditional economies, but they do in CHILE.

F A model with noise traders

In this section, we introduce noise traders. The resulting equilibrium captures both the mechanism emphasized in previous literature—where inequality enhances information efficiency by increasing the aggregate trading intensity of informed traders—and the new mechanism highlighted previously in this paper, which shows that inequality hinders efficiency by disrupting the optimal aggregation of informed traders’ signals. The main result is that the new mechanism tends to dominate, ensuring that our findings from the main part of the paper remain robust to the introduction of noise.

F.1 Setup

The model follows the setup in Section 3, with the addition of noise traders. There is a measure ν of these traders, and we index all traders by $a \in [0, 1 + \nu)$. Traders in the range $a \in [0, 1)$ are rational, while those in $a \in [1, 1 + \nu)$ are noise traders.

To maintain analytical tractability, we model noise traders following Black (1976), who describes them as traders “trading on noise as if it were information.” Specifically, we assume that noise traders observe

$$ds(a) = u da + \frac{1}{\sqrt{t(a)}} dB(a), \quad a \in [1, 1 + \nu). \quad (63)$$

Here, $u \sim N(0, \tau_v^{-1})$ is distributed identically to v but is independent of it. Although their signals follow (63), noise traders act as if they were receiving signals given by (6). Noise traders are identical to rational traders in all aspects except for the information they receive and how they respond to it.

F.2 Derivation of equilibrium

Derivation identical to that of Theorem 1 yields that the demand of noise traders is given by

$$dX(a) = \beta(a, P)ds(a) + \delta(a, P)da,$$

where

$$\beta(a, P) = \frac{t(a)}{\rho(a)} \frac{\tau^{-1} \mathbb{E}[V'(v)|s_p]}{\text{Var}[V(v)|s_p]}.$$

From market clearing, agents can extract the sufficient statistic

$$s_p = v + \frac{\int_1^{1+\nu} \frac{t(a)}{\rho(a)} da}{\int_0^1 \frac{t(a)}{\rho(a)} da} u + \int_0^{1+\nu} \frac{t(a)/\rho(a)}{\int_0^1 t(a)/\rho(a) da} \frac{dB(a)}{\sqrt{t(a)}}, \quad (64)$$

which has precision

$$\tau_p = \left(\left(\frac{\int_1^{1+\nu} \frac{t(a)}{\rho(a)} da}{\int_0^1 \frac{t(a)}{\rho(a)} da} \right)^2 \tau_v^{-1} + \left(\frac{1}{\int_0^1 t(a)/\rho(a) da} \right)^2 \int_0^{1+\nu} \frac{t(a)}{\rho(a)^2} da \right)^{-1}. \quad (65)$$

As in the main model, prices must be weak-form efficient, which implies

$$P = \mathbb{E}[V(v) | s_p].$$

The conditional distribution of v given P is normal with mean $(\tau_p/\tau) \cdot s_p$ and variance $1/\tau$. Defining the standardized variable $z = \sqrt{\tau}(v - (\tau_p/\tau) \cdot s_p)$, we note that z follows a standard normal distribution.

Thus, we can express v as

$$v = \frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}}.$$

Substituting this into the efficiency condition gives

$$\mathbb{E}[V(v) \mid P] = \mathbb{E} \left[V \left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}} \right) \right] = \int V \left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}} \right) d\Phi(z),$$

where $\Phi(z)$ is the standard normal cumulative distribution function.

The rest of the derivation follows the steps of Theorem 1, leading to the final result.

Theorem 3. *There exists a unique equilibrium. The equilibrium price function has the representation $\mathbf{P}_* = \mathcal{P}(s_p)$, where the equilibrium sufficient statistic s_p is given by (64). The function $\mathcal{P}(x)$ is given by*

$$\mathcal{P}(x) = \int V \left(\frac{\tau_p}{\tau} x + \frac{z}{\sqrt{\tau}} \right) d\Phi(z). \quad (66)$$

Here $\Phi(z)$ denotes the standard normal cumulative distribution function (cdf). Consequently, the price function is completely determined by $V(\cdot)$ and two other quantities, the precision of s_p , given by (65) and the posterior precision of v , given by $\tau = \mathbb{V}ar(v|P)^{-1} = \tau_v + \tau_p$.

The equilibrium cumulative demand function has the representation $dX(a) = \beta(a, P)ds(a) + \delta(a, P)da$, for all $a \in [0, 1 + \nu)$, where

$$\beta(a, P) = \frac{t(a)}{\rho(a)} \frac{\tau^{-1} \mathbb{E}[V'(v)|s_p]}{\text{Var}[V(v)|s_p]}, \text{ and}$$

$$\delta(a, P) = \frac{\beta(a, P)^2}{2t(a)} \pi(a) \frac{\text{Sk}[V(v)|s_p]}{\text{Var}[V(v)|s_p]} - \beta(a, P) \frac{\mathbb{E}[v(V(v) - P)^2|s_p]}{\text{Var}[V(v)|s_p]} + \frac{\psi(P)}{\rho(a) \text{Var}[V(v)|s_p]}.$$

Here $\rho(a)$ and $\pi(a)$ denote the absolute risk aversion and prudence coefficients, defined in (19).

The sufficient statistic s_p is related to the price P as follows:

$$s_p = \mathcal{P}^{-1}(P).$$

Here $\mathcal{P}^{-1}(\cdot)$ is the inverse of the function $\mathcal{P}(\cdot)$ defined in (66). The conditional moments of $V(v)$ and the function $\psi(P)$ are given in the closed form in (48).

F.3 Market quality

Our definitions of liquidity $\mathcal{L}$ and information efficiency $\mathcal{I}$ follow those in the main paper. Using the derivation steps from Proposition 1, we obtain

$$\mathcal{I} = \frac{\tau_p}{\tau_v + \tau_p},$$

and

$$\mathcal{L} = \frac{1}{\tau_p} \int_0^1 \frac{t(a)}{\rho(a)} da.$$

Since τ_p is given by (65), this leads to the following result.

Proposition 6. *The equilibrium information efficiency and liquidity are given by*

$$\mathcal{I} = \frac{\tau_p}{\tau_v + \tau_p}, \quad \mathcal{L} = \frac{1}{\tau_p} \int_0^1 \frac{t(a)}{\rho(a)} da,$$

where τ_p is given by (65).

F.4 Comparative statics

To ensure consistency with previous literature, which assumes exogenous noise remains unaffected by the parameters of interest, we impose that comparative statics do not influence noise traders. Formally, our definition of the Robin Hood variation assumes $h^\Delta(a) = 0$ for all $a \in [1, 1 + \nu)$ and coincides with Definition 3 in all other respects.

F.4.1 Wealth concentration

First, we examine changes in the wealth distribution $W_0(a)$ while keeping the precisions $t(a)$ fixed. This leads to the following proposition, which exactly matches Proposition 2.

Proposition 7. *Suppose that agent preferences are DARA. Then, there exist thresholds $0 <$*

$a_1^W \leq a_2^W < 1$, such that for any Robin Hood variation $W_0^\Delta(a)$ with $\underline{a} \leq a_1^W \leq a_2^W \leq \bar{a}$ we have

$$\mathcal{I}'(W_0(a))[W_0^\Delta(a)] > 0 \text{ and } \mathcal{I}'(W_0(a))[-W_0^\Delta(a)] < 0; \quad (67)$$

$$\mathcal{L}'(W_0(a))[W_0^\Delta(a)] < 0 \text{ and } \mathcal{L}'(W_0(a))[-W_0^\Delta(a)] > 0. \quad (68)$$

The mechanism highlighted in previous literature—where concentration enhances information efficiency by increasing the aggregate trading intensity of informed traders—does not alter Proposition 2 in the presence of noise. However, this does not imply that exogenous noise is irrelevant. Its presence may influence the thresholds a_1^W and a_2^W .

F.4.2 Information concentration

Here, we examine the effects of changing the distribution of information across agents on market quality holding the wealth profile fixed.

Proposition 8. *Suppose that traders have DARA utilities. Then, there exist thresholds $0 < a_1^t \leq a_2^t < 1$, such that for any Robin Hood variation $t^\Delta(a)$ with $\underline{a} \leq a_1^t \leq a_2^t \leq \bar{a}$*

$$\mathcal{I}'(t(a))[t^\Delta(a)] > 0 \text{ and } \mathcal{I}'(t(a))[-t^\Delta(a)] < 0;$$

$$\mathcal{L}'(t(a))[t^\Delta(a)] < 0 \text{ and } \mathcal{L}'(t(a))[-t^\Delta(a)] > 0.$$

The formulation of Proposition 3 remains unchanged. Corollary 1 also holds with noise.

Corollary 3 (An information-aggregation paradox). *Suppose that traders have DARA utilities. Then, there exists a threshold a_2^t such that for any $t^\Delta(a) \neq 0$ such that $t^\Delta(a) \leq 0$ for $a > a_2^t$, and $t^\Delta(a) = 0$ otherwise, $\mathcal{I}'(t(a))[t^\Delta(a)] > 0$ and $\mathcal{I}'(t(a))[-t^\Delta(a)] < 0$.*

F.4.3 The role of information acquisition

Combining the results of Propositions 7 and 8, we conclude that the statement of Proposition 4 remains unchanged.

Proposition 9. *Suppose that Assumption 2 holds. Suppose that traders have DARA utilities. Suppose that precisions are a function of wealth $t(W_0(a), a)$ and are increasing in $W_0(a)$. Then, for any Robin Hood variation $W_0^\Delta(a)$ with $\underline{a} \leq \min\{a_1^t, a_1^W\} \leq \max\{a_2^t, a_2^W\} \leq \bar{a}$, results (67)–(68) hold.*

F.5 Proofs for Appendix F

F.6 Proof of Proposition 7

Proof of Proposition 7. Following the proof of Proposition 2, we denote absolute risk tolerance $y(a) \equiv 1/\rho(a)$. Without loss of generality, index traders such that $y(b)$ increases in b . (This is in contrast to index a , which is such that $W_0(a)$ is increasing in a .) We first compute the Gateaux derivatives of $\mathcal{I}$ and $\mathcal{L}$ with respect to $y(b)$.

Substituting (65) into $\mathcal{I} = \tau_p/(\tau_p + \tau_v)$ and computing the Gateaux derivative (this entails substituting $y(b) + \epsilon y^\Delta(b)$ instead of $y(b)$, differentiating with respect to ϵ , and evaluating the resulting expression at $\epsilon = 0$) yields:

$$\mathcal{I}'(y(b))[y^\Delta(b)] = C_{\mathcal{I}} \int_0^1 t(b) y^\Delta(b) (K_1 - K_2 y(b)) db.$$

Here, $C_{\mathcal{I}} > 0$ is positive (we have the closed-form expressions for $C_{\mathcal{I}}$ via parameters of the model, but it is not important here), $I_1 = \int_0^1 t(b) y(b) db$, $I_2 = \int_0^{1+\nu} t(b) y(b)^2 db$, and $I_3 = \int_1^{1+\nu} t(b) y(b) db$, $K_1 = I_3^2 + I_2 \tau_v$, and $K_2 = I_1 \tau_v$.

Lemma 9 implies the existence of a unique b_*^y such that $K_1 - K_2 y(b) \geq 0$ if and only if $b \leq b_*^y$. The remainder of the proof follows the same steps as Proposition 2 and is omitted for

brevity.

We now turn to the liquidity results. Substituting (65) into $\mathcal{L} = \int_0^1 t(b)y(b)db/\tau_p$ and computing the Gateaux derivative yields

$$(\mathcal{L})'(y(b))[y^\Delta(b)] = C_{\mathcal{L}} \int_0^1 t(b)y^\Delta(b) (2K_2y(b) - K_1) db.$$

Here, $C_{\mathcal{L}} > 0$ is positive, $K_1 = I_3^2 + \tau_v I_2$, and $K_2 = \tau_v I_1$. Lemma 9 implies that there exists a unique b_{**}^y such that $2K_2y(b) - K_1 \geq 0$ iff $b \geq b_{**}^y$. Furthermore, because the information-efficiency threshold b_*^y satisfies $K_2y(b_*^y) - K_1 = 0$, the strict monotonicity of $y(b)$ guarantees that $b_{**}^y < b_*^y$. Then, for a $y^\Delta(b)$ that is Robin Hood with $\underline{b} < b_{**}^y < \bar{b}$, the product $y^\Delta(b) (2K_2y(b) - K_1)$ is nonpositive everywhere and strictly negative on the tails, so $(\mathcal{L})'(y(b))[y^\Delta(b)] < 0$. The statements of this step follow by letting $b_2^y = b_*^y$ and $b_1^y = b_{**}^y$. ■

F.7 Proof of Proposition 8

Proof of Proposition 8.

This proof follows the same steps as the proof of Proposition 2. Without loss of generality, we index traders such that $y(b)$ is increasing in b . The proof remains identical to that of Proposition 2, except for the expressions for the Gateaux derivatives, which we reproduce below:

$$\begin{aligned} \mathcal{I}'(t(b))[t^\Delta(b)] &= C_{\mathcal{I}} \int_0^1 y(b)t^\Delta(b)(K_2 - K_1y(b)) db; \\ \mathcal{L}'(t(b))[t^\Delta(b)] &= C_{\mathcal{L}} \int_0^1 y(b)t^\Delta(b)(k_1y(b) - k_2) db. \end{aligned}$$

Here, $K_2 = 2I_3^2 + 2I_2\tau_v$, $K_1 = I_1\tau_v$, $k_1 = I_1\tau_v$, and $k_2 = K_2/2$. The notation I_1 , I_2 , and I_3 follows the definitions in the proof of Proposition 7. ■

G Scaling, market power, and noise: CHILE vs. traditional large economies

This appendix isolates, in a deliberately simple CARA-normal setting, the economic role of the two scaling assumptions that distinguish CHILE from classic large rational-expectations economies, together with the role of the correlation-neglect assumption. We compare two large economies:

- **CHILE**: each trader’s signal precision shrinks as the economy grows, so that the *total* amount of information in the economy stays finite, while the aggregate supply of the asset is held fixed, so that *per-capita* supply vanishes as the number of traders grows.
- **Traditional large economies** (Hellwig, 1980, Admati, 1985): each trader’s signal precision is fixed, so that total information grows without bound, while aggregate supply grows in proportion to the number of traders, so that *per-capita* supply stays fixed.

We embed both in a single model indexed by one scaling function $c(n)$, characterize the correlation-neglect equilibrium, the competitive rational expectations equilibrium (REE), and the Bayesian Nash equilibrium (BNE), and study their large-economy limits. The two cases formalize the contrast in Remark 1. To keep every expression transparent, we assume throughout that all investors share the same absolute risk aversion ρ and observe signals of the same quality t .

G.1 Setup

We consider a sequence of economies with n traders. Trader i lives at $a_i = (i-1)/n$, has CARA preferences with common absolute risk aversion ρ , and observes a private signal

$$\Delta s_i = v c(n) + \frac{\sqrt{n c(n)}}{\sqrt{t}} \Delta B(a_i), \quad v \sim N(0, \tau_v^{-1}),$$

where $\Delta B(a_i)$ is a Brownian increment with $\text{Var}[\Delta B(a_i)] = m \equiv 1/n$, independent across traders, and $t > 0$ is the common signal quality. Dividing through by $c(n)$ yields the informationally equivalent signal

$$\tilde{s}_i \equiv \frac{\Delta s_i}{c(n)} = v + e_i, \quad e_i \sim N\left(0, \frac{1}{c(n)t}\right),$$

so each trader's signal has precision

$$\tau_s \equiv c(n)t$$

about v , and the *total information* in the economy is

$$\sum_{i=1}^n c(n)t = n c(n)t.$$

In the *traditional* case $c(n) = 1$, each trader observes v plus finite-variance noise (precision t , independent of n) and the total information nt diverges; in the *CHILE* case $c(n) = 1/n$, the individual precision t/n vanishes while the total information t stays finite.

The asset is in aggregate supply

$$\Theta_n = n c(n) (\bar{\theta} + u), \quad u \sim N(0, \tau_u^{-1}),$$

where $\bar{\theta} > 0$ is a constant and u is an exogenous supply (noise-trader) shock; equivalently, *per-capita* supply is $c(n)(\bar{\theta} + u)$. The same scaling function $c(n)$ that governs information also governs supply: the traditional case $c(n) = 1$ gives aggregate supply $n(\bar{\theta} + u)$ that scales with the economy, with a fixed per-capita supply $\bar{\theta} + u$; the CHILE case $c(n) = 1/n$ gives a fixed aggregate supply $\bar{\theta} + u$, with per-capita supply $(\bar{\theta} + u)/n$ vanishing.⁴⁷

⁴⁷Tying the supply noise to the same scaling as the deterministic supply keeps *per-capita* supply noise fixed in the traditional case (its aggregate variance grows like n^2), mirroring the treatment of per-capita signal precision and per-capita supply. This is what guarantees a non-degenerate limit: if the aggregate supply noise grew more slowly, per-capita supply noise would vanish, the price would become fully revealing in the limit, and the risk premium would collapse to zero.

We consider three equilibrium concepts, which differ only in which dimensions of her price influence a trader internalizes: (i) the *correlation-neglect equilibrium*, in which traders disregard both their influence on the price level and on its informational content (treating their signal and the price as conditionally independent given v); (ii) the *competitive REE*, in which traders internalize their influence on the informational content of the price but not on its level; and (iii) the *BNE*, in which traders internalize both.

G.2 Equilibrium and its large-economy limits

Derivation. We conjecture a symmetric linear equilibrium in which each trader uses $x_i = \beta \Delta s_i - \gamma P m$. Clearing the market, $\sum_i x_i = \Theta_n$, the price becomes informationally equivalent to the sufficient statistic

$$s_p = v + \frac{\zeta}{\sqrt{n c(n) t}} - \frac{u}{\beta}, \quad \zeta \equiv \sum_i \Delta B(a_i) \sim N(0, 1), \quad (69)$$

where ζ aggregates the signal noise (it has unit variance because each of the n increments has variance $m = 1/n$) and u is the supply shock. Its precision about v is

$$\tau_p^{-1} = \frac{1}{n c(n) t} + \frac{\tau_u^{-1}}{\beta^2}. \quad (70)$$

The aggregate trading intensity—the loading of aggregate demand on v —is $n c(n) \beta$. We solve the model exactly for finite n and take the large-economy limit at the end. Two forces shape the trading intensity: how a trader *infers* v from her signal and the price, and how she accounts for her own *price impact*.

Inference. Trader i observes her signal $\tilde{s}_i = \Delta s_i / c(n) = v + e_i$, of precision $c(n) t$, and the price, which is informationally equivalent to the statistic s_p in (69). The decisive finite- n feature is that s_p is built from *all* n signals, and so embeds a $1/n$ copy of trader i 's own noise:

writing $\bar{e} \equiv \frac{1}{n} \sum_j e_j$ for the average signal noise the price carries,

$$\text{Cov}(e_i, \bar{e}) = \frac{1}{n c(n) t}. \quad (71)$$

A trader who is rational about this correlation understands that part of what the price “tells” her is an echo of her own signal, and so relies on that signal less than one who treats signal and price as independent. This is the informational channel—the dependence of $\mathbb{E}_i[v]$ on the price.

Demand and price impact. With CARA utility and Gaussian information, optimal demand is mean–variance. A correlation-neglecting or REE trader treats the price level as exogenous,

$$x_i = \frac{\mathbb{E}_i[v] - P}{\rho \text{Var}_i[v]} \quad (\text{correlation-neglect and REE}), \quad (72)$$

while a BNE trader internalizes that her order moves the clearing price, adding her price impact λ_i to the denominator:

$$x_i = \frac{\mathbb{E}_i[v] - P}{\rho \text{Var}_i[v] + \lambda_i} \quad (\text{BNE}). \quad (73)$$

Here $\lambda_i \geq 0$ is the slope of the residual inverse-supply curve she faces, supplied by the *other* $n - 1$ traders (whose demands $x_j = \beta \Delta s_j - \gamma P m$ share the price-coefficient γm),

$$\lambda_i = \frac{1}{(n - 1) \gamma m},$$

and we define the *price impact per unit of risk*

$$\kappa \equiv \frac{\lambda_i}{\text{Var}_i[v]}.$$

Trading intensity. Start from the demand formulas (72)–(73). In all cases, the trading intensity is

$$\beta \equiv \frac{\partial x_i}{\partial \Delta s_i};$$

it is obtained by differentiating the demand with respect to the trader's own signal, not by imposing a separate first-order condition. It is therefore useful to write each posterior mean directly as a linear function of Δs_i and the price statistic s_p .

A *correlation-neglecting trader* treats $\tilde{s}_i$ and s_p as conditionally independent given v . Since $\tilde{s}_i = \Delta s_i / c(n)$, her posterior mean can be written

$$\mathbb{E}_i^{CNE}[v] = w_{\Delta}^{CNE} \Delta s_i + w_p^{CNE} s_p, \quad w_{\Delta}^{CNE} = \frac{t}{\tau_v + \tau_s + \tau_p}, \quad w_p^{CNE} = \frac{\tau_p}{\tau_v + \tau_s + \tau_p}, \quad (74)$$

and

$$\text{Var}_i^{CNE}[v] = \frac{1}{\tau_v + \tau_s + \tau_p}.$$

Differentiating (72) at a fixed price gives

$$\beta^{CNE} = \frac{w_{\Delta}^{CNE}}{\rho \text{Var}_i^{CNE}[v]} = \frac{t}{\rho}.$$

An *REE* trader uses the true joint law of $(\tilde{s}_i, s_p)$, in which (71) makes the two positively correlated beyond v : $\text{Cov}(\tilde{s}_i, s_p) = \tau_v^{-1} + (n \tau_s)^{-1}$. Let

$$D_n \equiv (\tau_v^{-1} + \tau_s^{-1}) (\tau_v^{-1} + \tau_p^{-1}) - (\tau_v^{-1} + (n \tau_s)^{-1})^2.$$

The posterior mean is

$$\mathbb{E}_i^{REE}[v] = w_{\Delta}^{REE} \Delta s_i + w_p^{REE} s_p,$$

with weights

$$\begin{aligned} w_{\Delta}^{REE} &= \frac{\tau_v^{-1} (\tau_p^{-1} - (n \tau_s)^{-1})}{c(n) D_n}, \\ w_p^{REE} &= \frac{\tau_v^{-1} (\tau_s^{-1} - (n \tau_s)^{-1})}{D_n}. \end{aligned} \quad (75)$$

Writing $\text{Var}_i^{REE}[v]$ for the corresponding posterior variance, differentiating (72) gives

$$\beta^{REE} = \frac{w_{\Delta}^{REE}}{\rho \text{Var}_i^{REE}[v]} = \frac{t}{\rho} R_n, \quad R_n \equiv \frac{\tau_p^{-1} - (n \tau_s)^{-1}}{\tau_p^{-1} - (n^2 \tau_s)^{-1}} \in (0, 1). \quad (76)$$

The factor R_n is the finite- n informational discount: the REE trader realizes that the price already contains part of her own signal.

A *BNE* trader has the same posterior mean as the REE trader,

$$\mathbb{E}_i^{BNE}[v] = w_{\Delta}^{REE} \Delta s_i + w_p^{REE} s_p,$$

but differentiates the strategic demand (73). Hence

$$\beta^{BNE} = \frac{w_{\Delta}^{REE}}{\rho \text{Var}_i^{REE}[v] + \lambda_i} = \beta^{REE} \frac{\rho \text{Var}_i^{REE}[v]}{\rho \text{Var}_i^{REE}[v] + \lambda_i} = \beta^{REE} \frac{\rho}{\rho + \kappa}.$$

Thus the BNE trader shades twice: once through the informational discount R_n , and once through the price-impact term λ_i (or, per unit of risk, κ).⁴⁸

Large-economy approximations. The exact finite- n fixed point is algebraically possible but unilluminating, so we take limits before writing the fixed point. The key object is the informational discount R_n in (76). Under CHILE ($c(n) = 1/n$, hence $\tau_s = t/n$),

$$R_n = \frac{\tau_p^{-1} - t^{-1}}{\tau_p^{-1} - (nt)^{-1}} = \left(1 - \frac{\tau_p}{t}\right) + O(n^{-1}).$$

Thus the limiting REE intensity first satisfies

$$\beta^{REE} = \frac{t}{\rho} \left(1 - \frac{\tau_p}{t}\right) = \frac{t - \tau_p}{\rho}. \quad (77)$$

⁴⁸The simplification in (76) follows from the bivariate normal regression formula. The corresponding posterior variance is $\text{Var}_i^{REE}[v] = \tau_v^{-1} - \tau_v^{-2}(\tau_s^{-1} + \tau_p^{-1} - 2(n \tau_s)^{-1})/D_n$; substituting this and (75) into $w_{\Delta}^{REE}/(\rho \text{Var}_i^{REE}[v])$ gives $(t/\rho)R_n$. Replacing $\text{Cov}[\tilde{s}_i, s_p]$ by τ_v^{-1} gives the correlation-neglect weights in (74).

Only at this point do we translate τ_p into price impact. In the limiting normal economy the price is affine in the sufficient statistic s_p , with slope $\tau_p/(\tau_v + \tau_p)$: the constant risk premium does not affect this slope. Since s_p contains the supply shock as $-u/\beta$ while aggregate supply is $n c(n)(\bar{\theta} + u)$, the price response to a unit of aggregate supply is

$$\frac{1}{\tau_v + \tau_p} \frac{\tau_p}{n c(n) \beta}.$$

Dividing by the conditional variance $1/(\tau_v + \tau_p)$ gives the price impact per unit of risk,

$$\kappa = \frac{\tau_p}{n c(n) \beta}.$$

Under CHILE scaling $n c(n) = 1$, $\tau_p = \kappa\beta$ in the limit. Substituting into (77) gives $\rho\beta = t - \kappa\beta$, hence $\beta^{REE} = t/(\rho + \kappa)$. For BNE, the same CHILE limit of R_n applies but the demand denominator also contains λ_i : $\beta = \{t - \kappa\beta\}/(\rho + \kappa)$, hence $\beta^{BNE} = t/(\rho + 2\kappa)$. Therefore, for CHILE,

$$\begin{aligned}\beta^{CNE} &= \frac{t}{\rho}, \\ \beta^{REE} &= \frac{t}{\rho} \frac{\tau_p^{-1} - t^{-1}}{\tau_p^{-1} - (nt)^{-1}} = \frac{t}{\rho + \kappa} + O(n^{-1}), \\ \beta^{BNE} &= \frac{t}{\rho + 2\kappa} + O(n^{-1}).\end{aligned}$$

We now repeat the derivation under the second scaling system. Under the traditional scaling ($c(n) = 1$, hence $\tau_s = t$),

$$R_n = \frac{\tau_p^{-1} - (nt)^{-1}}{\tau_p^{-1} - (n^2t)^{-1}} = 1 + O(n^{-1}),$$

so the informational discount vanishes and

$$\begin{aligned}\beta^{CNE} &= \frac{t}{\rho}, \\ \beta^{REE} &= \frac{t}{\rho} \frac{\tau_p^{-1} - (nt)^{-1}}{\tau_p^{-1} - (n^2t)^{-1}} = \frac{t}{\rho} + O(n^{-1}), \\ \beta^{BNE} &= \frac{t}{\rho} + O(n^{-1}).\end{aligned}$$

Comparing the expressions for trading intensities we can see that the informational discount survives under CHILE but vanishes under the traditional scaling.

The limiting fixed point. In the limit, both λ_i and $\text{Var}_i[v]$ become common across traders. In large economies, the posterior variance approaches $1/(\tau_v + \tau_p + t \lim c(n))$. Because $tc(n) \rightarrow 0$ under CHILE and price impact vanishes under traditional scaling, it follows that in both limits $\kappa \rightarrow 1/\mathcal{L} = \tau_p/(nc(n)\beta)$ (the inverse liquidity of the main paper). The three trading intensities can then be written in the unified limiting form

$$\beta = \frac{t}{\rho + j\kappa}, \quad j \in \{0, 1, 2\},$$

with $j = 0$ for correlation neglect, $j = 1$ for REE, and $j = 2$ for BNE—so, relative to correlation neglect, the REE trader shades by $+\kappa$ (informational channel) and the BNE trader by $+2\kappa$ (informational *and* price-impact channels). Price informativeness, information efficiency, and liquidity become

$$\tau_p = \frac{nc(n)t\kappa}{\rho + j\kappa}, \quad \mathcal{I} = \frac{\tau_p}{\tau_v + \tau_p}, \quad \mathcal{L} = \frac{1}{\kappa}, \quad (78)$$

and substituting $\beta = t/(\rho + j\kappa)$ and (78) into (70) gives the limiting fixed point

$$\kappa = \frac{t(\rho + j\kappa)}{t + nc(n)\tau_u^{-1}(\rho + j\kappa)^2}. \quad (79)$$

For correlation neglect ($j = 0$) the fixed point (79) is explicit, $\kappa = t\rho/(t + n c(n) \tau_u^{-1} \rho^2)$: here κ does not enter $\beta = t/\rho$, but it still records the realized illiquidity $1/\mathcal{L}$.

For the competitive equilibria (correlation neglect and REE, $j \in \{0, 1\}$), the unconditional risk premium is

$$\mathbb{E}[v - P] = \frac{c(n) \rho \bar{\theta}}{\tau_v + c(n) t + \tau_p}. \quad (80)$$

In the BNE ($j = 2$), traders restrict demand due to their price impact, requiring additional compensation. Their unconditional risk premium is therefore $\mathbb{E}[v - P] = c(n) (\rho + \kappa) \bar{\theta} / (\tau_v + c(n) t + \tau_p)$.

The two scaling systems. Everything hinges on the product $n c(n)$ multiplying the noise term in (79): it equals 1 under CHILE ($c(n) = 1/n$) and grows like n under the traditional scaling ($c(n) = 1$). This single contrast drives both theorems below.

Theorem 4 (CHILE limit). *Let $c(n) = 1/n$, so that $n c(n) = 1$ and aggregate supply is fixed. Then, for each concept $j \in \{0, 1, 2\}$, the price impact converges to a strictly positive limit $\kappa_j > 0$, the unique positive solution of*

$$\kappa = \frac{t (\rho + j \kappa)}{t + \tau_u^{-1} (\rho + j \kappa)^2},$$

and the trading intensities converge to the distinct limits

$$\beta^{CNE} = \frac{t}{\rho}, \quad \beta^{REE} = \frac{t}{\rho + \kappa_1}, \quad \beta^{BNE} = \frac{t}{\rho + 2\kappa_2}. \quad (81)$$

Price informativeness $\tau_p \rightarrow t \kappa_j / (\rho + j \kappa_j)$ is finite and bounded away from full revelation, and the risk premium vanishes, $\mathbb{E}[v - P] \rightarrow 0$. For the correlation-neglect equilibrium, in particular, $\kappa_0 = t\rho/(t + \tau_u^{-1} \rho^2)$ and $\tau_p = t^2/(t + \tau_u^{-1} \rho^2)$.

Theorem 5 (Traditional limit). *Let $c(n) = 1$, so that $n c(n) = n \rightarrow \infty$ and aggregate supply scales with the economy. Then, for every concept $j \in \{0, 1, 2\}$, the price impact vanishes,*

$\kappa \rightarrow 0$, and the three equilibria coincide in the limit, with common trading intensity

$$\beta^{CNE} = \beta^{REE} = \beta^{BNE} = \frac{t}{\rho}.$$

Price informativeness converges to the common finite limit

$$\tau_p \rightarrow \tau_u t^2 / \rho^2,$$

and the risk premium converges to the strictly positive constant $\mathbb{E}[v - P] \rightarrow \rho \bar{\theta} / (\tau_v + t + \tau_u t^2 / \rho^2) > 0$.

Why noise is needed for existence. The role of the exogenous noise u is already visible in (79). Setting $\tau_u^{-1} = 0$ reduces the REE/BNE fixed point to $\kappa = \rho + j\kappa$, which for $j \geq 1$ and $\rho > 0$ has no positive solution: the only consistent outcome is $\kappa \rightarrow \infty$, i.e. $\mathcal{L} \rightarrow 0$ and breakdown of the equilibrium. With noise, the term $n c(n) \tau_u^{-1} (\rho + j\kappa)^2$ in the denominator of (79) pulls the right-hand side back below κ at large κ , restoring a finite solution. The correlation-neglect equilibrium is the exception: since κ enters β with weight zero, $\beta(a) = t/\rho$ is well-defined even without noise—provided its price informativeness τ_p stays finite, which is exactly where the scaling system enters. To see this, take the noiseless correlation-neglect limit $\tau_u \rightarrow \infty$ in (79)–(80). The risk premium vanishes in both scaling systems, $\mathbb{E}[v - P] \rightarrow 0$: under CHILE because the prefactor $c(n) = 1/n \rightarrow 0$, and under the traditional scaling because $\tau_p \rightarrow \infty$ in the denominator. Their fates otherwise diverge. Under CHILE ($n c(n) = 1$), the price impact $\kappa = t\rho / (t + \tau_u^{-1} \rho^2) \rightarrow \rho$ and price informativeness $\tau_p \rightarrow t$ both stay strictly positive and finite, so the limiting correlation-neglect equilibrium retains a genuine notion of liquidity (and price impact) and a non-degenerate price informativeness. Under the traditional scaling ($n c(n) = n \rightarrow \infty$), $\kappa \rightarrow 0$ in any case, while $\tau_p = n t \rightarrow \infty$: the price becomes fully revealing and the correlation-neglect equilibrium breaks down completely. Thus a traditional large economy needs noise for *all three* concepts, whereas CHILE needs it only for REE and

BNE.

G.3 Discussion

The closed forms above make four points sharply.

1. Market power survives in CHILE but washes out in traditional large economies.

By Theorem 5, in the traditional large economy the price impact $\kappa \rightarrow 0$ and the correlation-neglect equilibrium, REE, and BNE coincide: each individual trader is negligible, so whether and how she internalizes her influence on the price is immaterial in the limit. Such an economy therefore cannot be used to study the effects of market power. By contrast, Theorem 4 shows that in CHILE the price impact κ remains strictly positive in the limit, and the three concepts yield genuinely different trading intensities (81) and different market quality. The reason is the information scaling: with $c(n) = 1/n$ each trader's signal precision—and hence her individual trading aggressiveness—shrinks at exactly the rate that keeps the aggregate price impact finite even as each trader becomes individually negligible. This is what allows CHILE to deliver a granular analysis of the two channels of price influence that is unavailable in the classic setting.

2. Noise is required for existence in the large economy. Without exogenous noise the REE/BNE fixed point (79) has no finite solution ($\kappa \rightarrow \infty$), so noise is required for the existence of REE and BNE in both scaling systems. The correlation-neglect equilibrium is special. Under CHILE, $\beta(a) = t/\rho$ together with $nc(n) = 1$ keeps $\tau_p = t$ finite even at $\tau_u^{-1} = 0$, so a correlation-neglect equilibrium exists *without* noise. Under the traditional scaling, however, $nc(n) = n \rightarrow \infty$ makes $\tau_p = nt \rightarrow \infty$ in the noiseless correlation-neglect case: the price becomes fully revealing and the equilibrium degenerates. Thus in a traditional large economy *all three* equilibria require noise, whereas in CHILE only REE and BNE do.

3. Risk premia. By (80), the risk premium carries a prefactor $c(n)$ (the per-capita supply scale). Under CHILE ($c(n) = 1/n$) it vanishes, $\mathbb{E}[v - P] \rightarrow 0$: with fixed aggregate supply, per-capita exposure to aggregate risk shrinks to zero, so the asset commands no per-share risk premium in the limit. This vanishing limiting risk premium is exactly what the limiting correlation-neglect equilibrium of the main text requires (cf. the role of a zero limiting risk premium in Lemma 3 and Theorem 1). Under the traditional scaling ($c(n) = 1$), per-capita supply is fixed, each trader bears non-trivial aggregate risk, and the per-share risk premium converges to a strictly positive constant (Theorem 5).

4. The two scaling channels. Although a single scaling function $c(n)$ now indexes both economies, the two economic channels of Remark 1 remain conceptually distinct, and they are governed by different functions of $c(n)$. Informativeness and market power are governed by the joint scaling of total information $nc(n)t$ and supply noise. With $c(n) = 1/n$ they stay finite, individual aggressiveness shrinks, and the price impact κ stays positive (so the equilibrium concept matters and the price is not fully revealing). With $c(n) = 1$ total information diverges, aggressiveness is nonvanishing, and—absent the compensating supply noise—the price would fully reveal v . The risk premium, by contrast, is governed by the per-capita supply scale $c(n)$ itself: with $c(n) = 1/n$ the per-share premium vanishes, with $c(n) = 1$ it stays finite. CHILE selects $c(n) = 1/n$ and the classic economies of Hellwig (1980) and Admati (1985) select $c(n) = 1$. Table 2 collects the limiting outcomes.

	CHILE	Traditional
	$c(n) = 1/n$	$c(n) = 1$
Total information $n c(n) t$	finite (t)	diverges (nt)
Per-capita supply $c(n)(\bar{\theta} + u)$	vanishes	fixed
Noise term $n c(n) \tau_u^{-1}$	τ_u^{-1} (constant)	$n \tau_u^{-1} \rightarrow \infty$
Price impact $\kappa = 1/\mathcal{L}$	> 0	$\rightarrow 0$
CNE, REE, BNE	distinct	coincide
Risk premium $\mathbb{E}[v - P]$	$\rightarrow 0$	$\rightarrow$ positive constant
Noise needed for existence	REE, BNE only	all three

Table 2: Large-economy limits in the two scaling systems.

H Competitive REE and BNE: General Case

In this section, we show how the CHILE framework accommodates alternative equilibrium concepts—specifically, competitive rational expectations equilibrium (REE) and Bayesian Nash equilibrium (BNE)—under general preferences and payoff structures. The goals are threefold: (i) to demonstrate that the modeling approach extends naturally to these equilibrium concepts; (ii) to show that REE and BNE do not exist without noise, reinforcing the intuition that the small belief perturbations inherent in correlation-neglect equilibrium are what enable existence in the noiseless limit; and (iii) to establish that the information aggregation paradox can arise under alternative equilibrium concepts and is not an artifact of correlation-neglect equilibrium.

The setup is the same as in the main text, except that supply now includes a noise component:

$$\Theta(P) = \bar{\theta} + \theta(P) + u \tilde{\theta}(P).$$

Here supply consists of a price-inelastic component $\bar{\theta}$, a price-elastic component $\theta(P)$, and a noisy component $u \tilde{\theta}(P)$, where $u \sim N(\bar{u}, \sigma_u^2)$. The mean supply shock $\bar{u}$ is a fixed parameter that is unknown to traders; they hold a prior $\bar{u} \sim N(0, \sigma_{\bar{u}}^2)$. We write $\tau_u^{-1} \equiv \sigma_u^2 + \sigma_{\bar{u}}^2$ for the total ex-ante supply uncertainty as perceived by traders. As we demonstrate below, this supply

uncertainty is essential for equilibrium existence.

The two variances σ_u^2 and $\sigma_{\bar{u}}^2$ play conceptually distinct roles. The variance σ_u^2 is a feature of the data-generating process: it is the variance of the contemporaneous supply shock u around its true (but unknown) mean $\bar{u}$. By contrast, $\sigma_{\bar{u}}^2$ is a measure of *epistemic* uncertainty: the mean supply $\bar{u}$ is a fixed parameter of the environment, and $\sigma_{\bar{u}}^2$ captures the tightness of traders' prior over its value, reflecting the fact that traders do not know all parameters of the environment in which they operate. Both forms of uncertainty matter for how traders update from prices and hence for equilibrium existence, but they need not coincide: σ_u^2 can be small while $\sigma_{\bar{u}}^2$ is large, in which case the data-generating process features little supply noise but traders nevertheless perceive prices as nearly uninformative.

The equilibrium construction proceeds as in the main text. First, by the Representation Lemma, cumulative demand X takes the form

$$dX = \beta(P, a) ds + \delta(P, a) da,$$

where β and δ are coefficients to be determined.

Aggregation consistency and optimality link β and δ to the derivatives of optimal demand x^* :

$$\begin{aligned}\beta(P, a) &= x_s^*(P, a; 0, 0), \\ \delta(P, a) &= \frac{1}{2t(a)} x_{ss}^*(P, a; 0, 0) + x_\mu^*(P, a; 0, 0).\end{aligned}$$

The optimal demand x^* solves the utility maximization problem

$$x_i^*(s, P) = \arg \max_x \int_{\mathbb{R}} u(W_0(a_i) + x(V(v) - P), a_i) f(v, s, P, \mu) dv. \quad (82)$$

The key difference from correlation-neglect equilibrium lies in the definition of optimal demand. In both REE and BNE, traders account for the correlation between the noise in their

private signal and the noise embedded in the price. Formally, the optimization problem (82) differs from its correlation-neglect counterpart in that we no longer assume conditional independence of the signal and the price when decomposing the density $f_{v|\Delta s_i, \mathbf{P}_i^n}(v, s, P)$.

The central object that allows us to isolate the price information uncorrelated with a given trader's signal is the *residual demand distribution*. Following Wilson (1979), for a given price level P —treated as a parameter (i.e., the input to the demand schedule)—define

$$H(x, v, P, \mu) \equiv \mathbb{P}\left(\sum_{j \neq i} x_j^*(s_j, P, \mu) < x \mid v\right). \quad (83)$$

The parameter μ captures the dependence of residual demand on the number of traders. The distribution H plays a role analogous to the price belief $g(v, P, \mu)$ in the construction of correlation-neglect equilibrium. We require that $H(x, v, P, \mu)$ converge to $H(x, v, P, 0)$ pointwise in v for any fixed x and P as $\mu \rightarrow 0$. By the representation result, the limiting distribution is

$$H(x, v, P, 0) = \mathbb{P}\left(\int_0^1 \beta(a, P) ds(a) + \int_0^1 \delta(a, P) da < x \mid v\right). \quad (84)$$

Consistent with our global maintenance of Assumption 1, we restrict attention to the regular class of equilibria in which the finite-economy aggregate excess demand is strictly decreasing in P . This ensures a unique market-clearing price for any realization of the shocks, so that $\tilde{P} < P$ if and only if cumulative demand at P falls short of supply. Conditioning on the realized supply shock u , we therefore have

$$\mathbb{P}(\tilde{P} < P \mid s_i, v, u) = \mathbb{P}\left(\sum_{j \neq i} x_j^*(s_j, P) < \Theta(P) - x_i^*(P, s_i) \mid v, u\right) = H(\Theta(P) - x_i^*(P, s_i), v, P, \mu).$$

The corresponding density is

$$h(P; x_i^*, v, u, \mu) = \frac{\partial}{\partial P} H(\Theta(P) - x_i^*, v, P, \mu).$$

We can then decompose the joint density as

$$f(v, s_i, P, u, \mu) = \underbrace{h(P; x_i^*(P, s_i), v, u, \mu)}_{\text{density of } P|s_i, v, u} \cdot \underbrace{\exp\left(t(a_i)\left(s_i v - \frac{\mu}{2} v^2\right)\right)}_{\propto \text{density of } s_i|v} \cdot \underbrace{f(v)}_{\text{density of } v} \cdot \underbrace{f_u(u)}_{\text{density of } u}. \quad (85)$$

Dropping terms that do not depend on the choice variable, the optimal demand can be formulated as follows.

REE. In competitive REE, each trader maximizes expected utility taking the equilibrium price distribution as given. The optimal demand solves

$$x^*(P, a; s, \mu) = \arg \max_y \int_{\mathbb{R}} \int_{\mathbb{R}} u(W_0(a) + y(V(v) - P), a) \\ \times h(P; x^*(\cdot), v, u, \mu) \exp\left(t(a)\left(sv - \frac{\mu}{2} v^2\right)\right) f(v) f_u(u) dv du. \quad (86)$$

The trader correctly recognizes that the equilibrium price distribution is determined by the equilibrium demand x^* , thereby accounting for how her information is incorporated into the price. However, she treats the density h as invariant to the off-equilibrium demand choice y —that is, she ignores the price impact of her own trade.

BNE. In BNE, each trader additionally internalizes that her demand choice affects the market-clearing price and hence the density h . The optimal demand schedule solves

$$x^*(P, a; s, \mu) = \arg \max_{y(\cdot)} \int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\mathbb{R}} u(W_0(a) + y(P)(V(v) - P), a) \\ \times h(P; y(\cdot), v, u, \mu) \exp\left(t(a)\left(sv - \frac{\mu}{2} v^2\right)\right) f(v) f_u(u) dP dv du. \quad (87)$$

The key distinction from REE is that the density h now depends on the choice variable $y(\cdot)$ rather than on the equilibrium demand x^* , reflecting the trader's internalization of her price impact.

Before formally defining the equilibrium, we state the technical conditions required for the equilibrium construction.

Assumption 5 (Technical restrictions on the primitives). *The following technical conditions are satisfied:*

1. *There exist $q \geq 0$ and $C_u < \infty$ such that for all $w \in \mathbb{R}$, $|u'(w, a)| + |u''(w, a)| + |u'''(w, a)| \leq C_u(1 + |w|^q)$.*
2. *$V(v)$ is absolutely continuous, weakly increasing and satisfies $\lim_{v \rightarrow \pm\infty} \frac{\ln|V(v)|}{v^2} \leq 0$.*
3. *There exist $\varepsilon > 0$ and constants $A, k > 0$ such that for all $v, u \in \mathbb{R}$,*

$$\sup_{|\mu| \leq \varepsilon} |h(P; x, v, u, \mu) f(v) f_u(u)| \leq A e^{-k(v^2 + u^2)}.$$

Moreover, denoting partial derivatives by subscripts (e.g., $h_x = \partial h / \partial x$), the same bounds hold for the derivatives h_l and H_l

$$\sup_{|\mu| \leq \varepsilon} |h_l(P; x, v, u, \mu) f(v) f_u(u)| \leq A e^{-k(v^2 + u^2)}, \quad \text{for } l \in \{x, xx, \mu, P, Px\};$$

$$\sup_{|\mu| \leq \varepsilon} |H_l(x, v, P, \mu) f(v)| \leq M(P) e^{-kv^2} \quad \text{for } l \in \{x, xx, P, Px\}.$$

4. *The density $h(P; x, v, u, \mu)$ is twice continuously differentiable with respect to x and continuously differentiable with respect to (μ, P) , with a continuous mixed partial derivative h_{Px} , in a neighborhood of $x = 0, \mu = 0$. In addition, $H(x, v, P, \mu)$ is twice continuously differentiable in x and continuously differentiable in (P, μ) , with continuous mixed partials needed above, e.g., H_{Px} , in a neighborhood of $x = 0, \mu = 0$.*

The equilibrium construction requires differentiating the first-order conditions for optimality and evaluating the resulting expressions in the large-economy limit ($\mu \rightarrow 0, s \rightarrow 0$). Part 4 ensures that the equilibrium objects—in particular, the residual demand density h —are smooth

enough for these derivatives to be well defined. Parts 1–3 provide the growth and integrability bounds needed to invoke the Dominated Convergence Theorem, allowing us to pass limits and derivatives under the integral sign.

We now state the equilibrium definition formally.

Definition 6. *The cumulative demand $X(a, P)$ constitutes a competitive REE (respectively, a BNE) if it is a limit of cumulative demands constructed from $x^*(P, a; s, \mu)$, where x^* satisfies the optimality condition (86) for the REE (respectively, (87) for the BNE), the distribution $H(x, v, P, \mu)$ satisfies the technical conditions in Assumption 5, and $H(x, v, P, \mu)$ converges pointwise in v to $H(x, v, P, 0)$ given in (84) for any fixed x and P .*

Before deriving competitive REE and BNE equilibria separately, we define the market quality objects.

H.1 Market quality

Our definition of informational efficiency follows the main text:

$$\mathcal{I} = 1 - \frac{\mathbb{V}\text{ar}(v|P)}{\mathbb{V}\text{ar}(v)}.$$

Because $\bar{u}$ is a fixed parameter of the data-generating process, the conditional variance $\mathbb{V}\text{ar}(v|P)$ is evaluated under the actual DGP, in which the noise variance of supply is σ_u^2 (the variance of u around its true mean $\bar{u}$) rather than the traders' perceived variance $\tau_u^{-1} = \sigma_u^2 + \sigma_{\bar{u}}^2$. Thus, while the equilibrium trading behavior—and in particular the fixed-point system for κ —is governed by the perceived noise τ_u^{-1} , the informational efficiency $\mathcal{I}$ is determined by the actual noise σ_u^2 . We elaborate on this distinction in Section H.5.

Similarly, we define liquidity based on how much price-inelastic supply shocks can move prices. We assume that these shocks are unexpected, in that demand coefficients do not adjust

to changes in $\bar{\theta}$. The corresponding market-clearing price is denoted $\mathcal{P}(\bar{\theta})$. We use three interrelated measures.

Price impact is

$$\lambda(P) = -\frac{\partial \mathcal{P}(\bar{\theta})}{\partial \bar{\theta}};$$

The price impact per unit of risk is

$$\kappa = \frac{\lambda(P)}{\mathbb{V}\text{ar}(V|P)};$$

And liquidity is

$$\mathcal{L} = 1/\kappa(P).$$

Hereafter, we focus on equilibria with price independent κ .

H.2 Competitive REE

We start with a formal derivation; a heuristic derivation is in Section [H.3](#). We proceed as in the main text, deriving the coefficients $\beta(P, a)$ and $\delta(P, a)$ from the [Aggregation Lemma](#) and the first-order conditions for optimal demand, evaluated in the large-economy limit. We then impose market clearing to characterize the equilibrium.

H.2.1 Deriving $\beta(P, a)$

By [Aggregation Lemma](#), the trading intensity equals the signal sensitivity of optimal demand evaluated in the large-economy limit: $\beta(P, a) = x_s(P, a; 0, 0)$. Computing this derivative requires differentiating the first-order condition (86) with respect to s and evaluating at $\mu = 0$, $s = 0$. Proceeding this way yields the following.

Lemma 11. *Suppose that Assumption 5 is satisfied and that $0 < \lambda(P) < \infty$ for every P . Then:*

$$\beta(P, a) = \frac{t(a) \operatorname{Cov}(V(v), v \mid P)}{\rho(a) \operatorname{Var}(V(v) \mid P) + \lambda(P)}.$$

H.2.2 Equilibrium

The coefficient $\beta(P, a)$ is the central object for the equilibrium characterization: it determines both market quality (informational efficiency and liquidity) and the fixed-point system that pins down the aggregate quantities κ and τ_p . The companion coefficient $\delta(P, a)$, which governs the deterministic component of demand, is more involved—reflecting prudence, the conditional third central moment (“skewness”), and the price-sensitivity of Kyle’s lambda—but does not affect the fixed-point system. Its characterization is stated in Lemma 13 and proved in Appendix H.6.

The key challenge is that $\beta(P, a)$ depends on the endogenous price informativeness τ_p and Kyle’s lambda $\lambda(P)$, which must be determined in equilibrium. We introduce the following assumption to ensure tractability.

Assumption 6 (Noise Calibration). *The noise supply function $\tilde{\theta}(P)$ satisfies*

$$\frac{\mathbb{E}[V'(v) \mid P]}{\operatorname{Var}(V(v) \mid P) \tilde{\theta}(P)} = \xi \tag{88}$$

for some $\xi > 0$ that does not depend on P .⁴⁹ Furthermore, we assume that ξ , viewed as a function of the endogenous precision τ_p , is continuous at $\tau_p = 0$ with a finite, strictly positive limit, which we denote by ξ_0 .

While Assumption 6 may appear restrictive, it is in fact quite natural, as the following two leading cases illustrate. For linear payoffs ($V(v) = v$), we have $\mathbb{E}[V' \mid P] = 1$ and $\operatorname{Var}(V \mid P) =$

⁴⁹The conditional moments $\mathbb{E}[V'(v) \mid P]$ and $\operatorname{Var}(V(v) \mid P)$ are determined by τ_p and P , since $v \mid P \sim N(\frac{\tau_p}{\tau} \mathcal{P}^{-1}(P), \tau^{-1})$ with $\tau = \tau_v + \tau_p$. The assumption eliminates the dependence on P , so ξ can depend on τ_p but on no other endogenous quantity. When $\xi = \xi(\tau_p)$, the fixed-point system (92)–(93) must be solved taking this dependence into account.

τ^{-1} , both independent of P , so the assumption is automatically satisfied for any constant $\tilde{\theta}$ —the standard normally distributed price-inelastic noisy supply specification in the literature. For log-normal payoffs ($V(v) = e^v$), we have $V'(v) = V(v)$, so $\mathbb{E}[V'|P] = P$ and $\text{Var}(V|P) = P^2(e^{1/\tau} - 1)$. Here $1/\tau = \mathbb{V}\text{ar}(v|P)$. The assumption then requires $\tilde{\theta}(P) \propto 1/P$, which means that the *dollar* noise supply $P \cdot u \tilde{\theta}(P)$ does not depend on P —i.e., noise traders submit normally distributed price-inelastic dollar demand.

With this assumption at hand, we are ready to characterize the equilibrium.

Theorem 6. *Suppose that Assumptions 5 and 6 are satisfied. Then, any such equilibrium is characterized as follows.*

The equilibrium price function has the representation $P = \mathcal{P}(s_p)$, where

$$\mathcal{P}(x) = \int V\left(\frac{\tau_p}{\tau}x + \frac{z}{\sqrt{\tau}}\right) d\Phi(z), \quad (89)$$

Φ denotes the standard normal cumulative distribution function, $\tau = \tau_v + \tau_p$ is the posterior precision of v , and s_p is the equilibrium sufficient statistic related to the price by $s_p = \mathcal{P}^{-1}(P)$.

The sufficient statistic has the representation

$$s_p = \int_0^1 \omega(a) ds(a) - \frac{u \tilde{\theta}(P)}{\int_0^1 \beta(a, P) da},$$

where the weighting function is

$$\omega(a) = \frac{t(a)/(\rho(a) + \kappa)}{\int_0^1 t(b)/(\rho(b) + \kappa) db}.$$

The precision of s_p is given by

$$\tau_p = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a) + \kappa} da\right)^2}{\int_0^1 \frac{t(a)}{(\rho(a) + \kappa)^2} da + \frac{\tau^2}{\xi^2 \tau_u}}. \quad (90)$$

The equilibrium cumulative demand has the representation $dX(a) = \beta(a, P) ds(a) + \delta(a, P) da$, where

$$\beta(a, P) = \frac{t(a)}{\rho(a) + \kappa} \cdot \frac{\mathbb{E}[V'(v) \mid P]}{\tau \text{Var}(V(v) \mid P)}, \quad (91)$$

and $\delta(a, P)$ is given by Lemma 13, with $D(P, a) = (\rho(a) + \kappa) \text{Var}(V(v) \mid P)$.

The equilibrium is fully determined by the aggregate quantities κ and τ_p , which solve the system

$$\kappa = \frac{\tau_p}{\int_0^1 \frac{t(a)}{\rho(a) + \kappa} da}, \quad (92)$$

$$\tau_p = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a) + \kappa} da \right)^2}{\int_0^1 \frac{t(a)}{(\rho(a) + \kappa)^2} da + \frac{(\tau_v + \tau_p)^2}{\xi^2 \tau_u}}. \quad (93)$$

Substituting (92) into (93), the system reduces to a single fixed-point equation for κ :

$$\kappa = \frac{\int_0^1 \frac{t(a)}{\rho(a) + \kappa} da}{\int_0^1 \frac{t(a)}{(\rho(a) + \kappa)^2} da + \frac{(\tau_v + \kappa \int_0^1 \frac{t(a)}{\rho(a) + \kappa} da)^2}{\xi^2 \tau_u}}. \quad (94)$$

The next result shows that noise is essential for equilibrium existence.

Proposition 10 (Non-existence without noise). *No competitive REE within the class characterized in Theorem 6 exists when $1/\tau_u = 0$.*

The result above can be understood as follows. Without noise, for any realization of the price, the environment reduces to one in which the no-trade theorem applies, precluding information-based trade. This requires that traders do not load on their private information, meaning $\beta(a, P) = 0$ for all a . But $\beta(a, P) = 0$ for all a cannot be an equilibrium either, since in that case the price conveys no additional information and traders would find it optimal to trade on their signals.

This observation also illuminates the existence result for the correlation-neglect equilibrium.

The key difference is that correlation-neglecting traders make small “mistakes” by ignoring the correlation between the noise in their own signal and the noise embedded in the price. These mistakes aggregate across the continuum of traders, generating a sufficient amount of endogenous noise for the equilibrium to exist. Crucially, it is these small mistakes in *inference*—not the negligible price impact of each trader—that sustain the equilibrium.

H.2.3 Log-normal payoffs

We now specialize the general characterization in Theorem 6 to the leading case of log-normal payoffs, $V(v) = e^v$. We assume that noise traders submit price-inelastic dollar demand, so that

$$\tilde{\theta}(P) = \frac{1}{P}.$$

The dollar noise supply is then $P \cdot u \tilde{\theta}(P) = u$, which does not depend on P .

Verifying the noise calibration. For $V(v) = e^v$, we have $V'(v) = e^v = V(v)$, so $\mathbb{E}[V'(v) | P] = \mathbb{E}[V(v) | P] = P$. The conditional variance of the payoff is

$$\text{Var}(V(v) | P) = P^2(e^{1/\tau} - 1),$$

where $\tau = \tau_v + \tau_p$ and we use the log-normal variance formula. Substituting into the noise calibration condition (88):

$$\xi = \frac{\mathbb{E}[V'(v) | P]}{\text{Var}(V(v) | P) \tilde{\theta}(P)} = \frac{P}{P^2(e^{1/\tau} - 1) \cdot 1/P} = \frac{1}{e^{1/\tau} - 1}.$$

Since $\tau = \tau_v + \tau_p$, the parameter ξ depends on the endogenous τ_p but not on P , confirming that Assumption 6 is satisfied.

Equilibrium characterization. With the noise calibration verified, we can specialize the general characterization of Theorem 6 to obtain closed-form expressions for all equilibrium objects.

Corollary 4 (Log-normal payoffs). *Suppose that $V(v) = e^v$, that Assumption 5 holds, and that $\tilde{\theta}(P) = 1/P$. Then the equilibrium of Theorem 6 has the following explicit form.*

1. **Price function.** *The price function is log-linear in the sufficient statistic:*

$$P = \mathcal{P}(s_p) = \exp\left(\frac{\tau_p}{\tau} s_p + \frac{1}{2\tau}\right),$$

so that $\ln P = \frac{\tau_p}{\tau} s_p + \frac{1}{2\tau}$, or equivalently $s_p = \frac{\tau}{\tau_p} (\ln P - \frac{1}{2\tau})$.

2. **Trading intensity.** *The trading intensity is*

$$\beta(a, P) = \frac{t(a)}{\rho(a) + \kappa} \cdot \frac{1}{\tau P (e^{1/\tau} - 1)}. \quad (95)$$

The dollar trading intensity $P \beta(a, P) = \frac{t(a)}{(\rho(a) + \kappa) \tau (e^{1/\tau} - 1)}$ is independent of P .

3. **Price impact.** *Kyle's lambda is $\lambda(P) = \kappa P^2 (e^{1/\tau} - 1)$, which is proportional to P^2 .*

4. **Price informativeness.** *The precision of the sufficient statistic is*

$$\tau_p = \frac{A(\kappa)^2}{B(\kappa) + \frac{\tau^2 (e^{1/\tau} - 1)^2}{\tau_u}}, \quad (96)$$

where $A(\kappa) = \int_0^1 \frac{t(a)}{\rho(a) + \kappa} da$ and $B(\kappa) = \int_0^1 \frac{t(a)}{(\rho(a) + \kappa)^2} da$.

5. **Equilibrium system.** *The aggregate quantities κ and τ_p are determined by $\kappa = \tau_p / A(\kappa)$ together with (96), where $\tau = \tau_v + \tau_p$. Substituting yields a single fixed-point equation for κ :*

$$\kappa = \frac{A(\kappa)}{B(\kappa) + \frac{(\tau_v + \kappa A(\kappa))^2 (e^{1/(\tau_v + \kappa A(\kappa))} - 1)^2}{\tau_u}}. \quad (97)$$

H.2.4 Normal payoffs

We now specialize the general characterization in Theorem 6 to the benchmark case of normal payoffs, $V(v) = v$. This is the standard specification in the noisy rational expectations literature. We assume that noise traders submit a price-inelastic quantity, so that

$$\tilde{\theta}(P) = 1.$$

Verifying the noise calibration. For $V(v) = v$, we have $V'(v) = 1$, so $\mathbb{E}[V'(v) \mid P] = 1$. The conditional variance of the payoff is $\text{Var}(V(v) \mid P) = \text{Var}(v \mid P) = 1/\tau$, where $\tau = \tau_v + \tau_p$. Substituting into the noise calibration condition (88):

$$\xi = \frac{\mathbb{E}[V'(v) \mid P]}{\text{Var}(V(v) \mid P) \tilde{\theta}(P)} = \frac{1}{1/\tau \cdot 1} = \tau.$$

Since $\tau = \tau_v + \tau_p$, the parameter ξ depends on the endogenous τ_p but not on P , confirming that Assumption 6 is satisfied.

Equilibrium characterization.

Corollary 5 (Normal payoffs, REE). *Suppose that $V(v) = v$, that Assumption 5 holds, and that $\tilde{\theta}(P) = 1$. Then the equilibrium of Theorem 6 has the following explicit form.*

1. **Price function.** *The price function is linear in the sufficient statistic:*

$$P = \mathcal{P}(s_p) = \frac{\tau_p}{\tau} s_p,$$

so that $s_p = \frac{\tau}{\tau_p} P$.

2. **Trading intensity.** The trading intensity is

$$\beta(a, P) = \frac{t(a)}{\rho(a) + \kappa},$$

and is independent of P .

3. **Price impact.** Kyle's lambda is $\lambda(P) = \kappa/\tau$, which is constant in P .

4. **Price informativeness.** The precision of the sufficient statistic is

$$\tau_p = \frac{A(\kappa)^2}{B(\kappa) + \tau_u^{-1}}, \quad (98)$$

where $A(\kappa) = \int_0^1 \frac{t(a)}{\rho(a) + \kappa} da$ and $B(\kappa) = \int_0^1 \frac{t(a)}{(\rho(a) + \kappa)^2} da$.

5. **Equilibrium system.** The aggregate quantities κ and τ_p are determined by $\kappa = \tau_p/A(\kappa)$ together with (98). Substituting yields a single fixed-point equation for κ :

$$\kappa = \frac{A(\kappa)}{B(\kappa) + \tau_u^{-1}}. \quad (99)$$

The equilibrium system (99) reduces, in the CARA-normal case, to the standard competitive REE fixed point, confirming that the general framework recovers the CARA-normal competitive REE as a special case.

H.3 Heuristic derivation of equilibrium

Following the heuristic approach used for the correlation-neglect case, one can obtain:

$$\beta(a, P) ds(a) = \frac{\mathbb{E}[V(v) - P \mid ds(a), s_p]}{\rho(a) \text{Var}(V(v) \mid P)} + \dots \quad (100)$$

where s_p denotes the sufficient statistic for the information conveyed by the price and “...” denotes terms that do not depend on $ds(a)$.

To pin down $\beta(a, P)$, we need to separate out the $ds(a)$ term in $\mathbb{E}[V(v) - P \mid ds(a), s_p]$. We proceed in two steps: first deriving $\mathbb{E}[v \mid ds(a), s_p]$, and then passing to $V(v)$.

Step 1: Linear projection of v . We rewrite the conditional expectation as:

$$\mathbb{E}[v \mid ds(a), s_p] = \mathbb{E}[v \mid ds(a), s_p - \omega(a) ds(a)],$$

where $\omega(a) = \beta(a, P) / \int_0^1 \beta(b, P) db$. Since $ds(a)$ and $s_p - \omega(a) ds(a)$ are conditionally independent given v , and $v \mid s_p$ is normally distributed, the standard linear projection formula gives:

$$\begin{aligned} \mathbb{E}[v \mid ds(a), s_p - \omega(a) ds(a)] &= \frac{t(a)}{\tau} ds(a) + \frac{\tau_p}{\tau} (s_p - \omega(a) ds(a)) \\ &= \frac{t(a) - \tau_p \omega(a)}{\tau} ds(a) + \dots, \end{aligned}$$

where $\tau = \tau_v + \tau_p$ and “ $\dots$ ” denotes terms that do not depend on $ds(a)$.

Step 2: From v to $V(v)$ via Stein’s lemma. Since adding one infinitesimal signal $ds(a)$ does not change the posterior variance of v (which remains $1/\tau$ to leading order), the effect on $\mathbb{E}[V(v)]$ is captured entirely by the shift in the posterior mean. By Stein’s lemma,

$$\text{Cov}(V(v), v \mid s_p) = \frac{\mathbb{E}[V'(v) \mid s_p]}{\tau},$$

so the coefficient of $ds(a)$ in $\mathbb{E}[V(v) \mid ds(a), s_p]$ is:

$$\mathbb{E}[V'(v) \mid s_p] \cdot \frac{t(a) - \tau_p \omega(a)}{\tau}.$$

Matching the coefficients of $ds(a)$ in (100):

$$\beta(a, P) = \frac{\mathbb{E}[V'(v) \mid P]}{\tau \rho(a) \text{Var}(V(v) \mid P)} (t(a) - \tau_p \omega(a)). \quad (101)$$

Step 3: Solving for $\beta(a, P)$. Since $\omega(a) = \beta(a, P) / \int_0^1 \beta(b, P) db$ and the common factor $\eta(P) \equiv \mathbb{E}[V'(v) \mid P] / (\tau \text{Var}(V(v) \mid P))$ cancels in the ratio, we define $g(a) = \beta(a, P) / \eta(P)$ so that $\omega(a) = g(a) / \int_0^1 g(b) db$. Then (101) simplifies to:

$$g(a) = \frac{1}{\rho(a)} \left(t(a) - \tau_p \cdot \frac{g(a)}{\int_0^1 g(b) db} \right).$$

To solve for $g(a)$, we introduce the shorthand:

$$\kappa \equiv \frac{\tau_p}{\int_0^1 g(b) db}.$$

Then:

$$g(a) = \frac{1}{\rho(a)} (t(a) - \kappa g(a)) \quad \Longleftrightarrow \quad g(a) = \frac{t(a)}{\rho(a) + \kappa}.$$

Thus, once κ is determined, $\beta(a, P)$ is fully characterized:

$$\beta(a, P) = \frac{t(a)}{\rho(a) + \kappa} \cdot \frac{\mathbb{E}[V'(v) \mid P]}{\tau \text{Var}(V(v) \mid P)},$$

which is equation (91).

Step 4: Computing τ_p and the role of noise calibration. The sufficient statistic has the representation:

$$s_p = v + \int_0^1 \frac{\omega(a)}{\sqrt{t(a)}} dB(a) - \frac{u \tilde{\theta}(P)}{\int_0^1 \beta(a, P) da}.$$

By Itô isometry, the inverse precision of s_p is:

$$\tau_p^{-1} = \int_0^1 \frac{\omega(a)^2}{t(a)} da + \frac{\tilde{\theta}(P)^2}{\tau_u \left(\int_0^1 \beta(a, P) da \right)^2}.$$

Substituting $\omega(a) = \beta(a, P) / \int_0^1 \beta(b, P) db$ and using $\beta(a, P) = \eta(P) \cdot t(a) / (\rho(a) + \kappa)$, the $\eta(P)$ factors cancel in the first term. Under Assumption 6, $\tilde{\theta}(P) = (\tau/\xi) \eta(P)$, so $\tilde{\theta}(P)/\eta(P) =$

τ/ξ is P -independent, and the $\eta(P)$ factors cancel in the second term as well. We obtain:

$$\tau_p = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a)+\kappa} da \right)^2}{\int_0^1 \frac{t(a)}{(\rho(a)+\kappa)^2} da + \frac{\tau^2}{\xi^2 \tau_u}},$$

which is equation (90).

Step 5: Fixed-point equation for κ . From the definition $\kappa = \tau_p / \int_0^1 g(b) db = \tau_p / \int_0^1 \frac{t(a)}{\rho(a)+\kappa} da$ and the expression for τ_p above:

$$\kappa = \frac{\int_0^1 \frac{t(a)}{\rho(a)+\kappa} da}{\int_0^1 \frac{t(a)}{(\rho(a)+\kappa)^2} da + \frac{\tau^2}{\xi^2 \tau_u}}.$$

Since $\tau = \tau_v + \tau_p = \tau_v + \kappa \int_0^1 \frac{t(a)}{\rho(a)+\kappa} da$, this yields the single fixed-point equation (94) for κ . Once κ is determined, τ_p , $\beta(a, P)$, and the price function $P = \mathcal{P}(s_p)$ are all determined as in Theorem 6.

H.4 BNE

We start with a formal derivation; a heuristic derivation is in Section H.4.6. We now derive the Bayesian Nash equilibrium for the general case, following the structure of the competitive REE analysis above. The key difference is that each trader internalizes the effect of her demand on the market-clearing price, as formalized in the optimization problem (87).

H.4.1 First-order condition

The optimization problem (87) is a calculus of variations problem: the density $h(P; y(\cdot), v, u, \mu)$ depends on both $y(P)$ and $y'(P)$, since the total P -derivative of the residual demand CDF involves the slope of the demand schedule.⁵⁰ Consequently, one cannot optimize pointwise in

⁵⁰Explicitly, $h(P; y(\cdot), v, u, \mu) = H_x(\Theta(P) - y(P), v, P, \mu)(\Theta'(P) - y'(P)) + H_P(\Theta(P) - y(P), v, P, \mu)$, where H_x and H_P denote partial derivatives of the residual demand CDF (83).

$y(P)$; the first-order condition must be derived via the Euler–Lagrange equation.

Euler condition (finite μ). Following [Wilson \(1979\)](#), at the equilibrium schedule $y(\cdot) = x^*(\cdot)$, the Euler condition reduces to the algebraic condition:

$$\int_{\mathbb{R}} \int_{\mathbb{R}} u'(W_0(a) + x^*(V(v) - P), a) \left[(V(v) - P) H_P(\Theta(P) - x^*, v, P, \mu) - x^* H_x(\Theta(P) - x^*, v, P, \mu) \right] \times \mathcal{E}(v, s, \mu) f(v) f_u(u) dv du = 0, \quad (102)$$

where $\mathcal{E}(v, s, \mu) = \exp(t(a)(sv - \frac{\mu}{2}v^2))$. Here, $H_P(\Theta(P) - x^*, v, P, \mu)$ is the partial derivative of the residual demand CDF H , with respect to P , holding the demand level y fixed, and $H_x = H_x(\Theta(P) - x^*, v, P, \mu)$ is the density of the residual demand at the market-clearing point.

The Euler condition (102) is valid for any μ and does not require additive separability of the residual demand in price and noise. Compared with the competitive REE first-order condition, the key new term is $-x^* H_x$, which captures the cost of price impact: the trader’s demand displaces probability mass in the residual demand distribution, raising the clearing price.

H.4.2 Deriving $\beta(P, a)$

By the [Aggregation Lemma](#), $\beta(P, a) = x_s(P, a; 0, 0)$. As in the competitive REE, we compute this derivative by differentiating the first-order condition (102) with respect to s and applying the Implicit Function Theorem at the limit point $s = 0, \mu = 0, x^* = 0$.

Lemma 12. *Suppose that Assumptions 5 and 6 are satisfied and that $0 < \lambda(P) < \infty$ for every P . Then:*

$$\beta(P, a) = \frac{t(a) \text{Cov}(V(v), v \mid P)}{\rho(a) \text{Var}(V(v) \mid P) + 2\lambda(P)}.$$

The only difference from the REE expression (Lemma 11) is the factor of 2 multiplying $\lambda(P)$ in the denominator. In the proof (see below), the two copies of $\lambda(P)$ arise from distinct terms in Wilson’s Euler condition (102): the first from the density derivative h_{x^*} (the inference

channel, identical to REE), and the second from the residual demand density H_x (the trader's internalization of price impact, new in BNE).

H.4.3 Equilibrium

As in the competitive REE, the coefficient $\beta(P, a)$ fully determines market quality and the fixed-point system for κ and τ_p . The coefficient $\delta(P, a)$ enters only through the market-clearing condition; its characterization is stated in Lemma 14 and proved in Appendix H.6.

We now characterize the BNE equilibrium. The structure parallels the competitive REE (Theorem 6).

Theorem 7. *Suppose that Assumptions 5 and 6 are satisfied. Then, any such equilibrium is characterized as follows.*

The equilibrium price function has the same representation as in Theorem 6: $P = \mathcal{P}(s_p)$, where

$$\mathcal{P}(x) = \int V\left(\frac{\tau_p}{\tau}x + \frac{z}{\sqrt{\tau}}\right) d\Phi(z), \quad (103)$$

$\tau = \tau_v + \tau_p$, and $s_p = \mathcal{P}^{-1}(P)$.

The sufficient statistic is

$$s_p = \int_0^1 \omega(a) ds(a) - \frac{u \tilde{\theta}(P)}{\int_0^1 \beta(a, P) da},$$

where the weighting function is

$$\omega(a) = \frac{t(a)/(\rho(a) + 2\kappa)}{\int_0^1 t(b)/(\rho(b) + 2\kappa) db}.$$

The precision of s_p is

$$\tau_p = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a) + 2\kappa} da\right)^2}{\int_0^1 \frac{t(a)}{(\rho(a) + 2\kappa)^2} da + \frac{\tau^2}{\xi^2 \tau_u}}. \quad (104)$$

The trading intensity is

$$\beta(a, P) = \frac{t(a)}{\rho(a) + 2\kappa} \cdot \frac{\mathbb{E}[V'(v) \mid P]}{\tau \text{Var}(V(v) \mid P)}, \quad (105)$$

and $\delta(a, P)$ is given by Lemma 14, with $D(P, a) = (\rho(a) + 2\kappa) \text{Var}(V(v) \mid P)$.

Here, $\kappa = \lambda(P)/\text{Var}(V(v) \mid P) \geq 0$ denotes the price impact ratio, where $\lambda(P) = -\frac{\partial P}{\partial \theta}$ is Kyle's lambda as defined in Section H.1. Under the assumptions, κ does not depend on P .

An equilibrium is characterized by the aggregate quantities κ and τ_p , which solve the system

$$\kappa = \frac{\tau_p}{\int_0^1 \frac{t(a)}{\rho(a) + 2\kappa} da}, \quad (106)$$

$$\tau_p = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a) + 2\kappa} da \right)^2}{\int_0^1 \frac{t(a)}{(\rho(a) + 2\kappa)^2} da + \frac{(\tau_v + \tau_p)^2}{\xi^2 \tau_u}}. \quad (107)$$

Substituting (106) into (107), the system reduces to a single fixed-point equation for κ :

$$\kappa = \frac{\int_0^1 \frac{t(a)}{\rho(a) + 2\kappa} da}{\int_0^1 \frac{t(a)}{(\rho(a) + 2\kappa)^2} da + \frac{(\tau_v + \kappa \int_0^1 \frac{t(a)}{\rho(a) + 2\kappa} da)^2}{\xi^2 \tau_u}}. \quad (108)$$

Every expression in the BNE equilibrium is obtained from the corresponding REE expression (Theorem 6) by replacing $\rho(a) + \kappa$ with $\rho(a) + 2\kappa$. This clean mapping extends the relationship between REE and BNE established in the CARA-normal model to the general setting.

Proposition 11 (Non-existence without noise). *No BNE within the class characterized in Theorem 7 exists when $1/\tau_u = 0$.*

Proof. Setting $1/\tau_u = 0$ in (106)–(107) and proceeding as in the proof of Proposition 10, the fixed-point equation requires $A(\kappa) = \kappa B(\kappa)$ where $A(\kappa) = \int_0^1 \frac{t(a)}{\rho(a) + 2\kappa} da$ and $B(\kappa) =$

$\int_0^1 \frac{t(a)}{(\rho(a)+2\kappa)^2} da$. However,

$$A(\kappa) - \kappa B(\kappa) = \int_0^1 \frac{t(a)(\rho(a) + \kappa)}{(\rho(a) + 2\kappa)^2} da > 0$$

for all $\kappa \geq 0$, so no solution exists. ■

H.4.4 Log-normal payoffs

Specializing to $V(v) = e^v$ and $\tilde{\theta}(P) = 1/P$, Theorem 7 yields the following closed-form characterization.

Corollary 6 (Log-normal payoffs, BNE). *Suppose that $V(v) = e^v$, that Assumption 5 holds, and that $\tilde{\theta}(P) = 1/P$. Then the equilibrium of Theorem 7 has the following explicit form.*

1. **Price function.** *The price function is the same as in REE:*

$$P = \mathcal{P}(s_p) = \exp\left(\frac{\tau_p}{\tau} s_p + \frac{1}{2\tau}\right).$$

2. **Trading intensity.** *The trading intensity is*

$$\beta(a, P) = \frac{t(a)}{\rho(a) + 2\kappa} \cdot \frac{1}{\tau P (e^{1/\tau} - 1)}.$$

The dollar trading intensity $P \beta(a, P) = \frac{t(a)}{(\rho(a)+2\kappa)\tau(e^{1/\tau}-1)}$ is independent of P .

3. **Price impact.** *Kyle's lambda is $\lambda(P) = \kappa P^2(e^{1/\tau} - 1)$, proportional to P^2 .*

4. **Price informativeness.** *The precision of the sufficient statistic is*

$$\tau_p = \frac{A(\kappa)^2}{B(\kappa) + \frac{\tau^2 (e^{1/\tau} - 1)^2}{\tau_u}}, \quad (109)$$

where $A(\kappa) = \int_0^1 \frac{t(a)}{\rho(a)+2\kappa} da$ and $B(\kappa) = \int_0^1 \frac{t(a)}{(\rho(a)+2\kappa)^2} da$.

5. **Equilibrium system.** The aggregate quantities κ and τ_p are determined by $\kappa = \tau_p/A(\kappa)$ together with (109), where $\tau = \tau_v + \tau_p$. Substituting yields a single fixed-point equation for κ :

$$\kappa = \frac{A(\kappa)}{B(\kappa) + \frac{(\tau_v + \kappa A(\kappa))^2 (e^{1/(\tau_v + \kappa A(\kappa))} - 1)^2}{\tau_u}}.$$

Proof. The proof is identical to that of Corollary 4, with $\rho(a) + \kappa$ replaced by $\rho(a) + 2\kappa$ throughout. ■

H.4.5 Normal payoffs

Specializing to $V(v) = v$ and $\tilde{\theta}(P) = 1$, Theorem 7 yields the following closed-form characterization.

Corollary 7 (Normal payoffs, BNE). *Suppose that $V(v) = v$, that Assumption 5 holds, and that $\tilde{\theta}(P) = 1$. Then the equilibrium of Theorem 7 has the following explicit form.*

1. **Price function.** The price function is linear:

$$P = \mathcal{P}(s_p) = \frac{\tau_p}{\tau} s_p.$$

2. **Trading intensity.** The trading intensity is

$$\beta(a, P) = \frac{t(a)}{\rho(a) + 2\kappa},$$

and is independent of P .

3. **Price impact.** Kyle's lambda is $\lambda(P) = \kappa/\tau$, constant in P .

4. **Price informativeness.** The precision of the sufficient statistic is

$$\tau_p = \frac{A(\kappa)^2}{B(\kappa) + \tau_u^{-1}}, \tag{110}$$

where $A(\kappa) = \int_0^1 \frac{t(a)}{\rho(a)+2\kappa} da$ and $B(\kappa) = \int_0^1 \frac{t(a)}{(\rho(a)+2\kappa)^2} da$.

5. **Equilibrium system.** The aggregate quantities κ and τ_p are determined by $\kappa = \tau_p/A(\kappa)$ together with (110). Substituting yields a single fixed-point equation for κ :

$$\kappa = \frac{A(\kappa)}{B(\kappa) + \tau_u^{-1}}. \quad (111)$$

The equilibrium system (111) reduces, in the CARA-normal case, to the standard BNE fixed point, confirming that the general framework recovers the CARA-normal BNE as a special case. As in the REE case, the only difference between the REE and BNE equilibria is the replacement of $\rho(a) + \kappa$ by $\rho(a) + 2\kappa$.

H.4.6 Heuristic derivation of the BNE

Following the heuristic approach used for the competitive REE (Section H.3), we derive the BNE trading intensity and equilibrium system. The starting point is Wilson’s Euler condition (102), which—under Assumption 6—reduces in the large-economy limit to pointwise optimization along the residual supply curve. The only modification to the REE heuristic is the inclusion of the price-impact cost $\lambda(P)$ in the first-order condition.

The BNE analogue of equation (100) is:

$$\beta(a, P) ds(a) = \frac{\mathbb{E}[V(v) - P \mid ds(a), s_p]}{(\rho(a) + \kappa) \text{Var}(V(v) \mid P)} + \dots \quad (112)$$

where $\kappa = \lambda(P)/\text{Var}(V(v) \mid P)$ and the “...” denotes terms independent of $ds(a)$. The denominator now includes κ from the price impact in addition to $\rho(a)$ from risk aversion.

Step 1: Linear projection of v . Proceeding exactly as in Step 1 of Section H.3, the

coefficient of $ds(a)$ in $\mathbb{E}[v \mid ds(a), s_p]$ is:

$$\frac{t(a) - \tau_p \omega(a)}{\tau},$$

where $\omega(a) = \beta(a, P) / \int_0^1 \beta(b, P) db$.

Step 2: From v to $V(v)$ via Stein's lemma. Exactly as in Step 2, the coefficient of $ds(a)$ in $\mathbb{E}[V(v) \mid ds(a), s_p]$ is:

$$\mathbb{E}[V'(v) \mid s_p] \cdot \frac{t(a) - \tau_p \omega(a)}{\tau}.$$

Step 3: Solving for $\beta(a, P)$. Matching the coefficients of $ds(a)$ in (112):

$$\beta(a, P) = \frac{\mathbb{E}[V'(v) \mid P]}{\tau (\rho(a) + \kappa) \text{Var}(V(v) \mid P)} (t(a) - \tau_p \omega(a)).$$

Defining $\eta(P) \equiv \mathbb{E}[V'(v) \mid P] / (\tau \text{Var}(V(v) \mid P))$ and $g(a) = \beta(a, P) / \eta(P)$, we obtain (as in Step 3 of Section H.3):

$$g(a) = \frac{1}{\rho(a) + \kappa} (t(a) - \kappa g(a)) \quad \Longleftrightarrow \quad g(a) = \frac{t(a)}{\rho(a) + 2\kappa}.$$

Here, the left-hand side has $\rho(a) + \kappa$ in the denominator (from the FOC, which includes both risk aversion and price impact), and the right-hand side produces $\rho(a) + 2\kappa$ after absorbing the additional κ from the inference channel (via $\omega(a)$). This is the source of the factor 2: one κ from price impact, one from inference.

Thus:

$$\beta(a, P) = \frac{t(a)}{\rho(a) + 2\kappa} \cdot \frac{\mathbb{E}[V'(v) \mid P]}{\tau \text{Var}(V(v) \mid P)},$$

which is equation (105).

Step 4: Computing τ_p . The sufficient statistic and its precision are computed identically

to Steps 4–5 of Section H.3, with $\rho(a) + \kappa$ replaced by $\rho(a) + 2\kappa$. Under Assumption 6, the $\eta(P)$ factors cancel as before, yielding

$$\tau_p = \frac{\left(\int_0^1 \frac{t(a)}{\rho(a)+2\kappa} da \right)^2}{\int_0^1 \frac{t(a)}{(\rho(a)+2\kappa)^2} da + \frac{\tau^2}{\xi^2 \tau_u}},$$

which is equation (104).

Step 5: Fixed-point equation for κ . From $\kappa = \tau_p / \int_0^1 g(b) db = \tau_p / \int_0^1 \frac{t(a)}{\rho(a)+2\kappa} da$ and the expression for τ_p above:

$$\kappa = \frac{\int_0^1 \frac{t(a)}{\rho(a)+2\kappa} da}{\int_0^1 \frac{t(a)}{(\rho(a)+2\kappa)^2} da + \frac{\tau^2}{\xi^2 \tau_u}}.$$

Since $\tau = \tau_v + \tau_p = \tau_v + \kappa \int_0^1 \frac{t(a)}{\rho(a)+2\kappa} da$, this yields the single fixed-point equation (108) for κ .

H.5 Price informativeness in liquid markets

The main complication in extending the comparative statics of Propositions 2 and 3 to REE and BNE is that the price-impact parameter κ is itself an equilibrium object. Changes in primitives—the distributions of $t(a)$ and $W_0(a)$ —affect trading intensities not only directly, through $\rho(a)$ and $t(a)$, but also indirectly through κ . We now show that in sufficiently liquid markets this indirect channel vanishes, and the comparative statics from the correlation-neglect benchmark carry over to both REE and BNE. Because we take the market to its perfectly liquid limit ($\kappa \rightarrow 0$), liquidity itself becomes trivially perfect in this limit; accordingly, we focus exclusively on price informativeness.

We parameterize market liquidity through the traders' uncertainty about mean supply, σ_u^2 . When σ_u^2 is large, traders attribute substantial noise to the price, which depresses price impact and makes markets liquid.

Proposition 12 (Liquid-market limit). *Suppose that Assumptions 5 and 6 hold. In both the*

REE and BNE equilibria, as $\sigma_u^2 \rightarrow \infty$ (with σ_u^2 fixed):

1. $\kappa \rightarrow 0$ and $\mathcal{L} \rightarrow \infty$.
2. The actual precision of the price signal converges to

$$\tau_p^* \equiv \frac{A(0)^2}{B(0) + c_0 \sigma_u^2}, \quad (113)$$

where $A(0) = \int_0^1 \frac{t(a)}{\rho(a)} da$, $B(0) = \int_0^1 \frac{t(a)}{\rho(a)^2} da$, and $c_0 = \tau_v^2 / \xi_0^2 > 0$, where ξ_0 is the noise-calibration parameter evaluated in the uninformative-price limit ($\tau_p = 0$), which is finite and strictly positive by Assumption 6.⁵¹ Informational efficiency converges to $\mathcal{I}^* = \tau_p^* / (\tau_v + \tau_p^*)$.

The limiting precision (113) is $\tau_p^* = A^2 / (B + c)$, where A and B are defined as in the correlation-neglect equilibrium and $c = c_0 \sigma_u^2 \geq 0$. Letting $y(a) = 1/\rho(a)$, the Gateaux derivative of τ_p^* with respect to a risk-tolerance variation $y^\Delta(a)$ is proportional to

$$\int_0^1 t(a) y^\Delta(a) [B + c - Ay(a)] da.$$

This kernel crosses zero at a unique, shifted threshold $y = (B + c)/A$. Similarly, the derivative with respect to a precision variation $t^\Delta(a)$ is proportional to

$$\int_0^1 y(a) t^\Delta(a) [2(B + c) - Ay(a)] da,$$

which crosses zero at the shifted threshold $y = 2(B + c)/A$. Because risk tolerance is strictly increasing in wealth and spans a broad range under Assumption 2, these shifted single-crossing conditions translate into valid wealth-index thresholds. Consequently, Robin Hood variations spanning these adjusted thresholds retain the qualitative signs established in Propositions 2–3

⁵¹For normal payoffs ($V(v) = v$), $c_0 = 1$. For log-normal payoffs ($V(v) = e^v$), $c_0 = \tau_v^2 (e^{1/\tau_v} - 1)^2$.

and Corollary 1. Finally, since these comparative statics yield strict inequalities in the liquid limit ($\kappa \rightarrow 0$), by continuity, the same signs are preserved for all finite but sufficiently large σ_u^2 .

Corollary 8 (Information efficiency in REE and BNE). *Suppose that trader preferences are DARA and that Assumptions 5 and 6 hold. For all σ_u^2 sufficiently large:*

1. (Wealth concentration.) *The comparative statics of Proposition 2 hold in both REE and BNE: there exist thresholds such that a Robin Hood transfer in wealth from the sufficiently rich to the sufficiently poor improves informational efficiency.*
2. (Information concentration.) *The comparative statics of Proposition 3 hold in both REE and BNE: a Robin Hood transfer in information precision from the rich to the poor improves informational efficiency.*
3. (Information-aggregation paradox.) *The paradox of Corollary 1 arises in both REE and BNE: weakly increasing the precision of all traders can reduce informational efficiency.*

Intuitively, when σ_u^2 is large, traders attribute most of the price variation to supply noise, making the price nearly uninformative from their perspective. This drives κ to zero, eliminating the wedge between the three equilibrium concepts. As the trading intensities

$$\beta^{CNE}(a) \propto \frac{t(a)}{\rho(a)}, \quad \beta^{REE}(a) \propto \frac{t(a)}{\rho(a) + \kappa}, \quad \beta^{BNE}(a) \propto \frac{t(a)}{\rho(a) + 2\kappa}$$

make clear, when $\kappa \rightarrow 0$ all three coincide. The direct effects of changing $t(a)$ or $W_0(a)$ on market quality—which are the novel economic forces identified in the main text—therefore dominate, and the information-aggregation paradox carries over to REE and BNE.

H.6 Proofs for Appendix H

Note on the u integration. Throughout the proofs below, the first-order condition G is a double integral over (v, u) , reflecting the conditioning on the supply shock u in the density

decomposition (85). In the limit $s = 0$, $\mu = 0$, $x^* = 0$, the density $h(P; 0, v, u, 0)$ equals the conditional density $f(P \mid v, u)$, and $h f(v) f_u(u) = f(v, u \mid P) f(P)$ by Bayes' rule. Since all payoff-relevant integrands depend on u only through h and its derivatives, integrating over u yields the marginal density $f(v \mid P)$. We therefore write all such intermediate expressions as single integrals over v , with $f(P)$ denoting the unconditional price density. In the BNE proofs, the residual-demand quantities $H_x(\Theta(P), v, P, 0)$ and $H_{xx}(\Theta(P), v, P, 0)$ additionally depend on u through the supply $\Theta(P)$; those terms are handled by averaging against $f_u(u)$ explicitly (see Eq. (120) and its proof).

H.6.1 Proof of Lemma 11

Proof. The optimal demand x^* satisfies the first-order condition (FOC) of the maximization problem:

$$G(x^*, s) \equiv \int_{\mathbb{R}} \int_{\mathbb{R}} (V(v) - P) u' (W_0(a) + x^* (V(v) - P)) \Psi(v, P, s, x^*, u) f(v) f_u(u) dv du = 0$$

where for this proof only we denote the density by:

$$\Psi(P; s, x^*, v, u, \mu) = h(P; x^*, v, u, \mu) \exp \left(t(a) \left(sv - \frac{\mu}{2} v^2 \right) \right).$$

The FOC is both necessary and sufficient due to the concavity of the utility function.

By the Implicit Function Theorem, the sensitivity $\beta(P, a)$ at the limit ($s = 0, \mu = 0, x^* = 0$) is:

$$\beta(P, a) = \frac{\partial x^*}{\partial s} = - \frac{\partial G / \partial s}{\partial G / \partial x^*}.$$

Assumption 5, parts 1–3 imply that the derivatives can be passed under the integral sign, which together with part 4 implies that G is continuously differentiable at $x^* = 0$ and $s = 0$. Part 4 further ensures that $\partial G / \partial x^* \neq 0$, and part 2 guarantees that it is finite. Furthermore, as explicitly calculated in Step 2 below, the strict concavity of the utility function ($u''(W_0) < 0$)

and the positive price impact ($\lambda(P) > 0$) jointly ensure that $\partial G / \partial x^* \neq 0$. Together, these conditions allow us to invoke the Implicit Function Theorem.

Step 1: The Numerator ($\partial G / \partial s$). Differentiating G with respect to s and evaluating at the limit (where the exponential term is 1):

$$\frac{\partial G}{\partial s} = \int_{\mathbb{R}} \int_{\mathbb{R}} (V(v) - P) u'(W_0) \left[h \cdot \frac{\partial}{\partial s} \exp(t(a)sv) \right] f(v) f_u(u) dv du$$

At $s = 0$, the derivative of the exponential is $t(a)v$. In the limit, h becomes the conditional density $f(P | v, u)$. By the convention above, integrating against $f_u(u)$ and applying Bayes' rule $f(P | v, u) f(v) f_u(u) = f(v, u | P) f(P)$ reduces the expression to a single integral over v :

$$\frac{\partial G}{\partial s} = f(P) u'(W_0) t(a) \int_{\mathbb{R}} (V(v) - P) v f(v|P) dv.$$

Here, $f(P)$ denotes the density of P in the limit economy. Using the condition $E[V(v)|P] = P$, the integral represents the covariance:

$$\frac{\partial G}{\partial s} = f(P) u'(W_0) t(a) \text{Cov}(V(v), v | P).$$

We can pass the derivative under the integral and take the limits due to Assumption 5, parts 1–3.

Step 2: The Denominator ($\partial G / \partial x^*$). Differentiating G with respect to x^* involves both the utility function and the density h . Evaluating at the limit where h becomes $f(P | v, u)$ and reducing via the convention above:

$$f(P)^{-1} \frac{\partial G}{\partial x^*} = u''(W_0) \int_{\mathbb{R}} (V(v) - P)^2 f(v|P) dv + u'(W_0) \int_{\mathbb{R}} (V(v) - P) \frac{\partial h}{\partial x^*} f(v) dv.$$

Taking into account that $E[V(v)|P] = P$, the first integral becomes $\text{Var}(V(v) | P)$. The second integral involves the sensitivity of the conditional expectation to the demand x^* . The x^*

enters via the conditioning event in the residual demand distribution: $\mathcal{F} = \sigma(\int_0^1 \beta(P, a) ds(a) + \int_0^1 \delta(P, a) da + x^* = \bar{\theta} + \theta(P) + u\tilde{\theta}(P))$. Using the relationship

$$\frac{\partial}{\partial x^*} E[V(v) | \mathcal{F}] = -\frac{\partial}{\partial \theta} E[V(v) | \mathcal{F}] = -\lambda(P),$$

we have:

$$\int_{\mathbb{R}} (V(v) - P) \frac{\partial h}{\partial x^*} f(v) dv = f(P) \frac{\partial}{\partial x^*} E[V(v) - P | P] = -f(P) \lambda(P).$$

This simplifies to:

$$f(P)^{-1} \frac{\partial G}{\partial x^*} = u''(W_0) \text{Var}(V(v) | P) - u'(W_0) \lambda(P).$$

As in the previous step, we can pass the derivative under the integral and take the limits due to Assumption 5.

Step 3: Combining Terms. Substituting the numerator and denominator into the implicit function formula:

$$\beta(P, a) = -\frac{u'(W_0) t(a) \text{Cov}(V(v), v | P)}{u''(W_0) \text{Var}(V(v) | P) - u'(W_0) \lambda(P)}.$$

Dividing the numerator and denominator by $u'(W_0)$ and using $\rho(a) = -u''(W_0)/u'(W_0)$:

$$\beta(P, a) = \frac{t(a) \text{Cov}(V(v), v | P)}{\rho(a) \text{Var}(V(v) | P) + \lambda(P)}.$$

■

H.6.2 Statement and proof of Lemma 13

By the [Aggregation Lemma](#), the coefficient $\delta(P, a)$ combines the curvature of optimal demand in the signal (x_{ss}) and the sensitivity to market size (x_μ):

$$\delta(P, a) = \frac{1}{2t(a)} x_{ss}(P, a; 0, 0) + x_\mu(P, a; 0, 0).$$

Lemma 13. *Suppose that Assumption 5 is satisfied and that $0 < \lambda(P) < \infty$ for every P . Let $\Sigma_{XY}(P) = \text{Cov}(X, Y \mid P)$. Then:*

$$\begin{aligned} \delta(P, a) = \frac{1}{D(P, a)} & \left(\psi(P) + \beta(P, a) \left[-\lambda(P) \Sigma'_{Vv}(P) - \rho(a) \mathbb{E}[(V - P)^2 v \mid P] - \lambda(P) \mathbb{E}[v \mid P] \right] \right. \\ & \left. + \frac{\beta(P, a)^2}{2t(a)} [\pi(a) \rho(a) \text{Skew}(V \mid P) + 2\rho(a) \lambda(P) \Sigma'_{VV}(P) + \lambda(P) \lambda'(P)] \right), \end{aligned}$$

where $D(P, a) = \rho(a) \Sigma_{VV}(P) + \lambda(P)$, $\pi(a) = -u'''(W_0)/u''(W_0)$, and $\psi(P) = \frac{\partial}{\partial \mu} E[V - P \mid P]$.

Proof. We follow steps similar to the computation of $\beta(P, a)$. The optimal demand x^* satisfies the first-order condition $G(x^*, s, \mu) = 0$. From the derivation of $\beta(P, a)$, we know that $f(P)^{-1} G_x = -u'(W_0) D(P, a)$.

Step 1: Computing x_μ . By the Implicit Function Theorem, $x_\mu = -G_\mu/G_x$. Differentiating G with respect to μ at the limit ($s = 0, \mu = 0$) yields:

$$G_\mu = \int_{\mathbb{R}} \int_{\mathbb{R}} (V - P) u'(W_0) \left[\frac{\partial h}{\partial \mu} - h \frac{t(a)v^2}{2} \right] f(v) f_u(u) dv du.$$

Using $hf(v) = f(P)f(v|P)$ and $E[V|P] = P$, the second term is proportional to $\text{Cov}(V, v^2 \mid P)$. Defining $\psi(P) \equiv \frac{1}{f(P)} \int (V - P) \frac{\partial h}{\partial \mu} f(v) dv$, we have:

$$f(P)^{-1} G_\mu = u'(W_0) \left[\psi(P) - \frac{t(a)}{2} \text{Cov}(V, v^2 \mid P) \right].$$

Thus,

$$x_\mu = \frac{\psi(P) - \frac{t(a)}{2} \text{Cov}(V, v^2 \mid P)}{D(P, a)}.$$

We can pass the derivative under the integral and take the limits due to Assumption 5.

Step 2: Computing x_{ss} . Using the second derivative of the implicit function, we have $x_{ss} = -\frac{1}{G_x} [G_{ss} + 2\beta G_{xs} + \beta^2 G_{xx}]$. Let $\mathcal{E} \equiv \exp\left(t(a)\left(sv - \frac{\mu}{2}v^2\right)\right)$ denote the exponential part of the density. At the limit $s = 0, \mu = 0$, we have $\mathcal{E} = 1$, $\frac{\partial \mathcal{E}}{\partial s} = t(a)v$, and $\frac{\partial^2 \mathcal{E}}{\partial s^2} = t(a)^2 v^2$.

The term G_{ss} is given by:

$$f(P)^{-1} G_{ss} = u'(W_0) \int (V - P) f(v \mid P) \frac{\partial^2 \mathcal{E}}{\partial s^2} dv = u'(W_0) t(a)^2 \text{Cov}(V, v^2 \mid P).$$

Its contribution to $\delta(P, a)$ via $\frac{1}{2t(a)} x_{ss}$ is:

$$\frac{1}{2t(a)} \frac{G_{ss}}{-G_x} = \frac{t(a)}{2D(P, a)} \text{Cov}(V, v^2 \mid P).$$

This term exactly cancels the $-\frac{t(a)}{2} \text{Cov}(V, v^2 \mid P)$ term from x_μ .

The term G_{xs} is:

$$G_{xs} = \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{\partial}{\partial x^*} [(V - P)u'h] t(a) v f(v) f_u(u) dv du.$$

Expanding the derivative and using $\rho(a) = -u''/u'$:

$$f(P)^{-1} G_{xs} = u'(W_0) t(a) \left[-\rho(a) \mathbb{E}[(V - P)^2 v \mid P] + \frac{\partial}{\partial x^*} E[(V - P)v \mid P] \right].$$

The term G_{xx} is:

$$G_{xx} = \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{\partial^2}{\partial (x^*)^2} [(V - P)u'h] f(v) f_u(u) dv du.$$

Expanding the second derivative and using $\pi(a) = -u'''/u''$ (so $-u''' = \pi(a)u'' = -\pi(a)\rho(a)u'$):

$$f(P)^{-1}G_{xx} = u'(W_0) \left[\pi(a)\rho(a) \text{Skew}(V \mid P) - 2\rho(a) \frac{\partial}{\partial x^*} E[(V - P)^2 \mid P] + \frac{\partial^2}{\partial (x^*)^2} E[V - P \mid P] \right].$$

The technical conditions in Assumption 5 ensure that we can differentiate under the integral sign and evaluate the limits for all second-order terms.

We now simplify the derivative terms. Recall that the conditioning event in the residual demand distribution h depends on x^* : $\mathcal{F} = \sigma(\int_0^1 \beta(P, a) ds(a) + \int_0^1 \delta(P, a) da + x^* = \bar{\theta} + \theta(P) + u\tilde{\theta}(P))$. Throughout, all x^* -derivatives are taken at a *fixed* observed price P , since P is a parameter of the trader's demand schedule in the first-order condition. Because x^* and $\bar{\theta}$ enter the event $\mathcal{F}$ only through the difference $\bar{\theta} - x^*$, differentiating with respect to the trader's own demand is the same as differentiating with respect to $-\bar{\theta}$:

$$\frac{\partial}{\partial x^*} E[Y \mid \mathcal{F}] \Big|_P = -\frac{\partial}{\partial \bar{\theta}} E[Y \mid \mathcal{F}] \Big|_P. \quad (114)$$

Economically, raising x^* while the observed transaction price stays at P means that the rest of the market must have generated less demand, which shifts the inferred fundamental downward.

To convert the right-hand side of (114) into a derivative in P , write the market-clearing price as $\mathcal{P}(s_p, \bar{\theta})$, where s_p is the equilibrium sufficient statistic; at $x^* = 0$ the conditioning set $\mathcal{F}$ is $\sigma(s_p)$, the price is strictly increasing in s_p , and $\partial \mathcal{P} / \partial \bar{\theta} = -\lambda(P)$ by the definition of price impact. Holding the observed price fixed at P , the statistic inferred from it solves $\mathcal{P}(s_p, \bar{\theta}) = P$, so the implicit function theorem gives

$$\frac{\partial s_p}{\partial \bar{\theta}} \Big|_P = -\frac{\partial \mathcal{P} / \partial \bar{\theta}}{\partial \mathcal{P} / \partial s_p} = \frac{\lambda(P)}{\partial \mathcal{P} / \partial s_p} > 0 :$$

a larger inelastic supply consistent with the *same* price must have been met by better news.

At fixed $\bar{\theta}$ we have $E[Y \mid P] = E[Y \mid s_p]$ with $s_p = \mathcal{P}^{-1}(P)$, so $\frac{\partial}{\partial P}E[Y \mid P] = (\frac{\partial}{\partial s_p}E[Y \mid s_p])/(\partial\mathcal{P}/\partial s_p)$. Combining the last two displays,

$$\frac{\partial}{\partial \bar{\theta}}E[Y \mid P]\Big|_P = \frac{\partial E[Y \mid s_p]}{\partial s_p} \frac{\partial s_p}{\partial \bar{\theta}}\Big|_P = \lambda(P) \frac{\partial}{\partial P}E[Y \mid P],$$

and hence, by (114), the operator relationship for conditional moments whose integrand does not depend explicitly on P :

$$\frac{\partial}{\partial x^*}E[Y \mid P] = -\lambda(P) \frac{\partial}{\partial P}E[Y \mid P]. \quad (115)$$

Two clarifications about (115). First, the left-hand side is a fixed-price derivative of the conditional law; it is *not* the total derivative of $E[Y \mid \mathcal{P}(s_p, \bar{\theta})]$ along the price path traced out by a supply shift, which vanishes identically because s_p does not depend on $\bar{\theta}$. Second, the operator acts through the conditioning channel only: when the integrand Y contains an explicit P , that dependence is differentiated separately, as we do below. Using this relationship, we can express the terms as derivatives with respect to P . First, using $E[(V - P)^2 \mid P] = \Sigma_{VV}(P)$, we have:

$$\frac{\partial}{\partial x^*}E[(V - P)^2 \mid P] = -\lambda(P)\Sigma'_{VV}(P).$$

Second, writing $(V - P)v = Vv - Pv$, the first term depends on P only through the conditioning event, while the second contributes an explicit P -derivative. Hence

$$\frac{\partial}{\partial x^*}E[(V - P)v \mid P] = -\lambda(P)\Sigma'_{Vv}(P) - \lambda(P)\mathbb{E}[v \mid P].$$

Second, for the second derivative, we have:

$$\frac{\partial^2}{\partial (x^*)^2}E[V - P \mid P] = \lambda(P)\lambda'(P).$$

Step 3: Combining Terms. Substituting these expressions into the terms for G_{xs} and G_{xx} , and combining the surviving terms yields the expression in the lemma. ■

H.6.3 Proof of Theorem 6

Proof. The proof proceeds in five steps. Steps 1–2 establish the demand coefficients using Lemmas 11 and 13. Step 3 derives the price informativeness from the market clearing condition. Step 4 shows that, under Assumption 6, τ_p and κ are price-independent. Step 5 derives the price function via the efficiency condition and closes the system.

Step 1: Demand coefficients from Lemmas 11 and 13. By Lemma 11, the trading intensity is

$$\beta(P, a) = \frac{t(a) \operatorname{Cov}(V(v), v \mid P)}{\rho(a) \operatorname{Var}(V(v) \mid P) + \lambda(P)}. \quad (116)$$

Since $v \mid P$ is normally distributed, Stein's Lemma gives

$$\operatorname{Cov}(V(v), v \mid P) = \operatorname{Var}(v \mid P) \cdot \mathbb{E}[V'(v) \mid P] = \frac{1}{\tau} \mathbb{E}[V'(v) \mid P].$$

Defining the price impact ratio $\kappa(P) \equiv \lambda(P)/\operatorname{Var}(V(v) \mid P)$, we can rewrite (116) as

$$\beta(P, a) = \eta(P) \cdot \frac{t(a)}{\rho(a) + \kappa(P)}, \quad \text{where } \eta(P) \equiv \frac{\mathbb{E}[V'(v) \mid P]}{\tau \operatorname{Var}(V(v) \mid P)}. \quad (117)$$

The coefficient $\delta(P, a)$ is given by Lemma 13.

Step 2: Market clearing and the sufficient statistic. The market clearing condition reads

$$\int_0^1 \beta(a, P) ds(a) + \int_0^1 \delta(a, P) da = \bar{\theta} + \theta(P) + u \tilde{\theta}(P). \quad (118)$$

Using $ds(a) = v da + dB(a)/\sqrt{t(a)}$ and rearranging, we define the sufficient statistic

$$s_p \equiv \frac{\int_0^1 \beta(a, P) ds(a) - u \tilde{\theta}(P)}{\int_0^1 \beta(a, P) da} = v + \int_0^1 \frac{\omega(a)}{\sqrt{t(a)}} dB(a) - \frac{u \tilde{\theta}(P)}{\int_0^1 \beta(a, P) da},$$

where $\omega(a) = \beta(a, P) / \int_0^1 \beta(b, P) db$. Under our restriction to equilibria with price-independent κ and under Assumption 6, the price-dependent factor $\eta(P)$ cancels out, making the normalized weights $\omega(a)$ and the supply-shock coefficient deterministic. Because the two noise terms are independent, the precision of s_p as a signal about v is

$$\tau_p^{-1} = \frac{\int_0^1 \frac{\beta(a, P)^2}{t(a)} da + \frac{\tilde{\theta}(P)^2}{\tau_u}}{\left(\int_0^1 \beta(a, P) da \right)^2}. \quad (119)$$

The sufficient statistic s_p is equivalently recovered from the price: using (118),

$$s_p = \frac{\bar{\theta} + \theta(P) - \int_0^1 \delta(a, P) da}{\int_0^1 \beta(a, P) da} \equiv h(P).$$

Step 3: Price-independence of τ_p and κ under Assumption 6. We conjecture an equilibrium in which the aggregate price-impact ratio κ and the price precision τ_p are independent of the realized price P . Under this conjecture, the cross-sectional integrals $A \equiv \int_0^1 \frac{t(a)}{\rho(a) + \kappa} da$ and $B \equiv \int_0^1 \frac{t(a)}{(\rho(a) + \kappa)^2} da$ are scalar constants. From (117), substituting $\beta(a, P) = \eta(P) \cdot t(a) / (\rho(a) + \kappa)$ into (119), we obtain

$$\tau_p^{-1} = \frac{\eta(P)^2 B + \tilde{\theta}(P)^2 / \tau_u}{\eta(P)^2 A^2} = \frac{B}{A^2} + \frac{\tilde{\theta}(P)^2}{\tau_u \eta(P)^2 A^2}.$$

The first term is independent of P . The second term is independent of P if and only if $\tilde{\theta}(P) / \eta(P)$ does not depend on P . Under Assumption 6,

$$\tilde{\theta}(P) = \frac{\mathbb{E}[V'(v) \mid P]}{\xi \text{Var}(V(v) \mid P)} = \frac{\tau}{\xi} \eta(P),$$

so $\tilde{\theta}(P) / \eta(P) = \tau / \xi$, which is indeed P -independent. This verifies our conjecture, provided that the constants τ_p and κ simultaneously solve

$$\tau_p = \frac{A^2}{B + \tau^2/(\xi^2\tau_u)}, \quad \text{and} \quad \kappa = \frac{\tau_p}{A}.$$

Step 4: Price function via the efficiency condition. Since τ_p is P -independent, $v \mid s_p \sim N\left(\frac{\tau_p}{\tau} s_p, \tau^{-1}\right)$. The efficiency condition $\mathbb{E}[V(v) \mid P] = P$ (up to the infinitesimal risk premium) gives

$$P = \mathbb{E}\left[V\left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}}\right)\right] = \int V\left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}}\right) d\Phi(z) = \mathcal{P}(s_p),$$

where $z = \sqrt{\tau} \left(v - \frac{\tau_p}{\tau} s_p\right)$ has a standard normal distribution.

Step 5: Closing the system. The equilibrium is determined once κ and τ_p are known. From Steps 3 and 4, these satisfy the system

$$\begin{aligned} \kappa &= \frac{\tau_p}{A(\kappa)}, \\ \tau_p &= \frac{A(\kappa)^2}{B(\kappa) + (\tau_v + \tau_p)^2/(\xi^2\tau_u)}, \end{aligned}$$

where $A(\kappa) = \int_0^1 \frac{t(a)}{\rho(a)+\kappa} da$ and $B(\kappa) = \int_0^1 \frac{t(a)}{(\rho(a)+\kappa)^2} da$. ■

H.6.4 Proof of Proposition 10

Proof. Setting $1/\tau_u = 0$ in the fixed-point system (92)–(93) reduces it to

$$\kappa = \frac{\tau_p}{A(\kappa)}, \quad \tau_p = \frac{A(\kappa)^2}{B(\kappa)},$$

where $A(\kappa) \equiv \int_0^1 \frac{t(a)}{\rho(a)+\kappa} da$ and $B(\kappa) \equiv \int_0^1 \frac{t(a)}{(\rho(a)+\kappa)^2} da$. Substituting the second equation into the first yields the requirement $\kappa = A(\kappa)/B(\kappa)$, or equivalently, $A(\kappa) - \kappa B(\kappa) = 0$. However,

$$A(\kappa) - \kappa B(\kappa) = \int_0^1 t(a) \left[\frac{1}{\rho(a) + \kappa} - \frac{\kappa}{(\rho(a) + \kappa)^2} \right] da = \int_0^1 \frac{t(a) \rho(a)}{(\rho(a) + \kappa)^2} da > 0$$

for all $\kappa \geq 0$, since $t(a) > 0$ and $\rho(a) > 0$. Therefore $A(\kappa)/B(\kappa) > \kappa$ for every $\kappa \geq 0$, and the fixed-point equation has no solution. ■

H.6.5 Proof of Corollary 4

Proof. We specialize each component of Theorem 6 to $V(v) = e^v$ and $\tilde{\theta}(P) = 1/P$.

Item 1. Since $v \mid s_p \sim N\left(\frac{\tau_p}{\tau} s_p, \tau^{-1}\right)$, the price function (89) becomes

$$\mathcal{P}(s_p) = \int \exp\left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}}\right) d\Phi(z) = \exp\left(\frac{\tau_p}{\tau} s_p + \frac{1}{2\tau}\right),$$

using the moment-generating function of the standard normal.

Item 2. The common factor from (117) is

$$\eta(P) = \frac{\mathbb{E}[V'(v) \mid P]}{\tau \text{Var}(V(v) \mid P)} = \frac{P}{\tau P^2(e^{1/\tau} - 1)} = \frac{1}{\tau P(e^{1/\tau} - 1)}.$$

Substituting into (91) gives (95). Multiplying by P cancels the price dependence.

Item 3. From $\kappa = \lambda(P)/\text{Var}(V(v) \mid P)$ and $\text{Var}(V(v) \mid P) = P^2(e^{1/\tau} - 1)$, we obtain $\lambda(P) = \kappa P^2(e^{1/\tau} - 1)$.

Item 4. The general formula (93) gives $\tau_p = A(\kappa)^2/(B(\kappa) + \tau^2/(\xi^2\tau_u))$. Substituting $\xi = 1/(e^{1/\tau} - 1)$:

$$\frac{\tau^2}{\xi^2 \tau_u} = \frac{\tau^2 (e^{1/\tau} - 1)^2}{\tau_u},$$

which yields (96).

Item 5. Using $\kappa = \tau_p/A(\kappa)$ to eliminate τ_p and $\tau = \tau_v + \kappa A(\kappa)$ to express τ in terms of κ , equation (96) reduces to the single equation (97). ■

H.6.6 Proof of Corollary 5

Proof. We specialize each component of Theorem 6 to $V(v) = v$ and $\tilde{\theta}(P) = 1$.

For $V(v) = v$, the relevant conditional moments are $\mathbb{E}[V'(v) \mid P] = 1$, $\text{Var}(V(v) \mid P) = 1/\tau$, and $\text{Cov}(V(v), v \mid P) = 1/\tau$, where $\tau = \tau_v + \tau_p$.

Item 1. Since $v \mid s_p \sim N\left(\frac{\tau_p}{\tau} s_p, \tau^{-1}\right)$, the price function (89) becomes

$$\mathcal{P}(s_p) = \int \left(\frac{\tau_p}{\tau} s_p + \frac{z}{\sqrt{\tau}} \right) d\Phi(z) = \frac{\tau_p}{\tau} s_p.$$

Item 2. The common factor from (117) is $\eta(P) = 1/(\tau \cdot 1/\tau) = 1$. Substituting into (91) gives $\beta(a, P) = t(a)/(\rho(a) + \kappa)$, which is independent of P .

Item 3. From $\kappa = \lambda(P)/\text{Var}(V(v) \mid P)$ and $\text{Var}(V(v) \mid P) = 1/\tau$, we obtain $\lambda(P) = \kappa/\tau$, which is constant.

Item 4. The noise calibration gives $\xi = \tau$, so $\tau^2/(\xi^2 \tau_u) = 1/\tau_u$. The general formula (93) then yields (98).

Item 5. Using $\kappa = \tau_p/A(\kappa)$ to eliminate τ_p , equation (98) reduces to the single equation (99). ■

H.6.7 Proof of Lemma 12

Proof. The proof follows the same strategy as the proof of Lemma 11, applied to Euler condition (102).

Because the $y'(P)$ terms perfectly cancel out in the Euler–Lagrange equation, the first-order condition reduces to an algebraic equation $G(x^*, s, \mu) = 0$ in the demand level x^* . Let $H_P(\Theta(P) - x^*, v, P, \mu) \equiv H_x \Theta'(P) + H_P^{(3)}$ denote the partial derivative of $H(\Theta(P) - x^*, v, P, \mu)$

with respect to P holding the demand level x^* fixed. Then we define:

$$G(x^*, s, \mu) \equiv \int_{\mathbb{R}} \int_{\mathbb{R}} u'(W_0(a) + x^*(V(v) - P), a) \left[(V(v) - P) H_P(\Theta(P) - x^*, v, P, \mu) \right. \\ \left. - x^* H_x(\Theta(P) - x^*, v, P, \mu) \right] \mathcal{E}(v, s, \mu) f(v) f_u(u) dv du = 0,$$

where $\mathcal{E}(v, s, \mu) = \exp(t(a)(sv - \frac{\mu}{2} v^2))$.

By the Implicit Function Theorem, $\beta(P, a) = -G_s/G_{x^*}$ evaluated at $s = 0, \mu = 0, x^* = 0$.

Step 1: Numerator (G_s). At $x^* = 0$, the term $-x^* H_x$ vanishes. Moreover, at $x^* = 0$ (and thus $x^{*'} = 0$), the fixed-demand derivative H_P coincides perfectly with the true density $h(P)$. The computation is therefore identical to Step 1 in the proof of Lemma 11:

$$f(P)^{-1} G_s = u'(W_0) t(a) \text{Cov}(V(v), v \mid P).$$

Step 2: Denominator (G_{x^*}). We differentiate G with respect to the scalar x^* (since G is an algebraic function of x^* and does not depend on $x^{*'}$). Evaluating at $x^* = 0, s = 0, \mu = 0$ (so $\mathcal{E} = 1$):

$$G_{x^*} = \underbrace{\int u''(W_0) (V - P)^2 H_P f dv + \int u'(W_0) (V - P) \frac{\partial H_P}{\partial x^*} f dv}_{\text{from the } (V - P) H_P \text{ term}} \\ - \underbrace{\int u'(W_0) H_x f dv}_{\text{from the } -x^* H_x \text{ term (new in BNE)}}.$$

Since $H_P = h(P)$ at $x^* = 0$, the first integral is identical to the REE derivation. For the second integral, notice that:

$$\frac{\partial H_P}{\partial x^*} = \frac{\partial}{\partial x^*} [H_x(\Theta - x^*)\Theta' + H_P^{(3)}(\Theta - x^*)] = -H_{xx}\Theta' - H_{Px}^{(3)}.$$

This perfectly matches the partial derivative of the density $h(P; x^*, x^{*'})$ with respect to its level argument x^* evaluated at $x^{*'} = 0$. Thus, evaluating this term at $x^* = 0$ yields exactly the h_{x^*} term from the REE derivation. The first two integrals combine exactly as in the proof of Lemma 11, yielding

$$f(P)[u''(W_0) \text{Var}(V(v) \mid P) - u'(W_0) \lambda(P)].$$

For the new term, we claim

$$\int_{\mathbb{R}} \int_{\mathbb{R}} H_x(\Theta(P), v, P, 0) f(v) f_u(u) dv du = \lambda(P) f(P). \quad (120)$$

Proof of (120). Define $F(x, P) = \int H(x, v, P, 0) f(v) dv = \mathbb{P}(R_{-i}(P) < x)$. For a fixed realization of u , $\mathbb{P}(\tilde{P} \leq P \mid u) = F(\Theta(P), P)$. Since $\Theta(P) = \bar{\theta} + \theta(P) + u \tilde{\theta}(P)$ and $F(x, P)$ does not depend on $\bar{\theta}$:

$$\frac{\partial}{\partial \bar{\theta}} \mathbb{P}(\tilde{P} \leq P \mid u) = F_x(\Theta(P), P) = \int H_x(\Theta(P), v, P, 0) f(v) dv.$$

Under Assumption 6, the price response to supply, $\partial \tilde{P} / \partial \bar{\theta}$, is a deterministic function of $\tilde{P}$: the noise-trader terms cancel from the excess-supply slope (because $\tilde{\theta}(P) \propto \eta(P)$). Since $\partial \tilde{P} / \partial \bar{\theta} = -\lambda(P)$ at $\tilde{P} = P$:

$$\frac{\partial}{\partial \bar{\theta}} \mathbb{P}(\tilde{P} \leq P \mid u) = f(P \mid u) \lambda(P).$$

Averaging over u :

$$\int_{\mathbb{R}} \int_{\mathbb{R}} H_x(\Theta(P), v, P, 0) f(v) f_u(u) dv du = \lambda(P) \int f(P \mid u) f_u(u) du = \lambda(P) f(P). \quad \square$$

Combining all terms:

$$f(P)^{-1} G_{x^*} = u''(W_0) \text{Var}(V(v) \mid P) - 2 u'(W_0) \lambda(P).$$

The first copy of λ arises from h_{x^*} (the inference channel, identical to REE); the second from H_x (the trader's internalization of price impact, new in BNE).

Step 3: Combining.

$$\beta(P, a) = -\frac{G_s}{G_{x^*}} = \frac{t(a) \operatorname{Cov}(V(v), v \mid P)}{\rho(a) \operatorname{Var}(V(v) \mid P) + 2\lambda(P)}.$$

■

H.6.8 Statement and proof of Lemma 14

The derivation of $\delta(P, a)$ follows the same steps as in the competitive REE (Lemma 13), applied to Wilson's Euler condition (102). As with β , the $-x^* H_x$ term in the Euler condition generates additional contributions compared with the REE.

Lemma 14. *Suppose that Assumptions 5 and 6 are satisfied and that $0 < \lambda(P) < \infty$ for every P . Let $\Sigma_{XY}(P) = \operatorname{Cov}(X, Y \mid P)$. Then:*

$$\begin{aligned} \delta(P, a) = & \frac{1}{D(P, a)} \left(\psi(P) + \beta(P, a) \left[-\lambda(P) \Sigma'_{Vv}(P) - \rho(a) \mathbb{E}[(V-P)^2 v \mid P] - 2\lambda(P) \mathbb{E}[v \mid P] \right] \right. \\ & \left. + \frac{\beta(P, a)^2}{2t(a)} \left[\pi(a) \rho(a) \operatorname{Skew}(V \mid P) + 2\rho(a) \lambda(P) \Sigma'_{Vv}(P) + \lambda(P) \lambda'(P) + 2\Lambda(P) \right] \right), \end{aligned}$$

where $D(P, a) = \rho(a) \Sigma_{VV}(P) + 2\lambda(P)$, $\pi(a) = -u'''(W_0)/u''(W_0)$, $\psi(P) = \frac{\partial}{\partial \mu} \mathbb{E}[V-P \mid P]$, and

$$\Lambda(P) \equiv \frac{\lambda(P)}{f(P)} \frac{d}{dP} [\lambda(P) f(P)] = \lambda(P) \lambda'(P) - \lambda(P)^2 \left(s_p \frac{\tau_v}{\mathbb{E}[V'(v) \mid P]} + \frac{\mathbb{E}[V''(v) \mid P]}{\mathbb{E}[V'(v) \mid P]^2} \right),$$

where $f(P)$ is the unconditional density of the clearing price, $\tau = \tau_v + \tau_p$, and $s_p = \mathcal{P}^{-1}(P)$ is the sufficient statistic.

Compared with the REE expression (Lemma 13), the BNE δ has three modifications, all traceable to the $-x^* H_x$ term in Wilson's Euler condition: (i) the denominator is $\rho \Sigma_{VV} + 2\lambda$

(not $\rho \Sigma_{VV} + \lambda$); (ii) the β -linear bracket acquires a second $-\lambda \mathbb{E}[v \mid P]$ term from the H_x channel. When rewritten using covariance, this extra term combines with the expected utility expansion to yield $-(\rho \Sigma_{VV} + 2\lambda) \mathbb{E}[v \mid P]$, perfectly matching the BNE denominator; (iii) the β -quadratic bracket acquires $2\Lambda(P)$, which can be written entirely in terms of local conditional moments and structural parameters.

Proof. We follow the same strategy as the proof of Lemma 13, applied to Euler condition (102). As in the proof of Lemma 12, we define the algebraic equation:

$$G(x^*, s, \mu) \equiv \int_{\mathbb{R}} \int_{\mathbb{R}} u'(W_0 + x^*(V - P)) \left[(V - P) H_P - x^* H_x \right] \mathcal{E} f(v) f_u(u) dv du = 0,$$

where $H_P = H_P(\Theta(P) - x^*, v, P, \mu)$ is the fixed-demand partial derivative of the residual demand CDF with respect to P , $H_x = H_x(\Theta(P) - x^*, v, P, \mu)$, and $\mathcal{E} = \exp(t(a)(sv - \frac{\mu}{2}v^2))$.

From the proof of Lemma 12, we know $f(P)^{-1} G_{x^*} = -u'(W_0) D(P, a)$ where $D(P, a) = \rho(a) \Sigma_{VV}(P) + 2\lambda(P)$.

The coefficient δ is given by $\delta = \frac{1}{2t} x_{ss} + x_\mu$ where $x_\mu = -G_\mu / G_{x^*}$ and $x_{ss} = -\frac{1}{G_{x^*}} [G_{ss} + 2\beta G_{x^*s} + \beta^2 G_{x^*x^*}]$.

Step 1: G_μ (identical to REE). At $x^* = 0$, the $-x^* H_x$ term vanishes. The computation is identical to Step 1 of the proof of Lemma 13:

$$f(P)^{-1} G_\mu = u'(W_0) \left[\psi(P) - \frac{t(a)}{2} \text{Cov}(V, v^2 \mid P) \right].$$

Step 2: G_{ss} (identical to REE). At $x^* = 0$, only the $(V - P) h$ term survives:

$$f(P)^{-1} G_{ss} = u'(W_0) t(a)^2 \text{Cov}(V, v^2 \mid P).$$

The contribution $\frac{1}{2t} \frac{G_{ss}}{-G_{x^*}} = \frac{t(a)}{2D} \text{Cov}(V, v^2 \mid P)$ cancels the $-\frac{t(a)}{2} \text{Cov}(V, v^2 \mid P)$ from x_μ , exactly as in REE.

Step 3: G_{x^*s} (new BNE term). Differentiating G_{x^*} with respect to s and evaluating at $x^* = 0, s = 0, \mu = 0$ ($\mathcal{E}_s = t(a)v$):

$$\begin{aligned} G_{x^*s} &= t(a) \int \{u''(W_0)(V-P)^2 H_P + u'(W_0)(V-P) \frac{\partial H_P}{\partial x^*} - u'(W_0) H_x\} v f(v) dv \\ &= G_{xs}^{\text{REE}} - t(a) u'(W_0) \int H_x v f(v) dv, \end{aligned}$$

where G_{xs}^{REE} is the cross-derivative from the REE proof.

Now note that at the limit $H_x(\Theta(P), v, P, 0) = f(P | v, u) \lambda(P)$ (by the same supply-shift argument used in the proof of (120)). After averaging against $f_u(u)$ (per the convention above):

$$\int_{\mathbb{R}} \int_{\mathbb{R}} H_x(\Theta(P), v, P, 0) v f(v) f_u(u) dv du = \lambda(P) f(P) \mathbb{E}[v | P].$$

Combining with the REE expression from the proof of Lemma 13:

$$\begin{aligned} f(P)^{-1} G_{x^*s} &= u'(W_0) t(a) \\ &\quad [-\rho(a) \text{Cov}((V-P)^2, v | P) - \lambda(P) \Sigma'_{Vv}(P) - (\rho(a) \Sigma_{VV} + 2\lambda(P)) \mathbb{E}[v | P]]. \end{aligned}$$

The first two terms are identical to REE; the third is new and arises from H_x .

Step 4: $G_{x^*x^*}$ (new BNE term). Write $G = \int \Phi(A+B) \mathcal{E} f dv$ with $\Phi = u'(W_0 + x^*(V-P))$, $A = (V-P) H_P$, $B = -x^* H_x$. The second derivative is $G_{x^*x^*} = \int [\Phi''(A+B) + 2\Phi'(A'+B') + \Phi(A''+B'')] \mathcal{E} f dv$, where primes denote x^* -derivatives.

At $x^* = 0$ ($\mathcal{E} = 1$): $B = 0$, $B' = -H_x$, and $B'' = 2H_{xx}(\Theta, v, P)$ (because $\frac{d^2}{d(x^*)^2}[-x^* H_x(\Theta - x^*)] = 2H_{xx}$ at $x^* = 0$).

The terms involving A and its derivatives reproduce G_{xx}^{REE} exactly. The new BNE terms

are:

$$\Delta G_{x^*x^*} = -2u''(W_0) \int (V-P) H_x f dv + 2u'(W_0) \int H_{xx} f dv.$$

For the first integral, substituting $H_x(\Theta(P), v, P, 0) = f(P | v, u) \lambda(P)$ and averaging over u gives $\int \int (V-P) H_x f f_u dv du = \lambda(P) f(P) \mathbb{E}[V-P | P] = 0$ (efficiency).

For the second integral, we differentiate identity (120) in $\bar{\theta}$. Since $\lambda(P)$ is independent of $\bar{\theta}$ under Assumption 6 and the clearing-price density satisfies the transport equation $\frac{\partial f(P)}{\partial \bar{\theta}} = \frac{d}{dP}[\lambda(P) f(P)]$ (for the price shift $\partial \tilde{P} / \partial \bar{\theta} = \lambda(P)$):

$$\int_{\mathbb{R}} \int_{\mathbb{R}} H_{xx}(\Theta, v, P, 0) f(v) f_u(u) dv du = \frac{\partial}{\partial \bar{\theta}}[\lambda(P) f(P)] = \lambda(P) \frac{d}{dP}[\lambda(P) f(P)] = \Lambda(P) f(P),$$

where $\Lambda(P) \equiv \frac{\lambda(P)}{f(P)} \frac{d}{dP}[\lambda(P) f(P)]$.

Using the equilibrium characterization that the price is a deterministic function of the normally distributed sufficient statistic $P = \mathcal{P}(s_p)$, we can apply the change-of-variables formula $f(P) = f_S(s_p) / \mathcal{P}'(s_p)$ to express the logarithmic derivative $f'(P) / f(P)$ entirely in terms of local conditional moments. Since $f_S(s_p)$ is the density of $\mathcal{N}(0, \frac{\tau}{\tau_v \tau_p})$ and $\mathcal{P}'(s_p) = \frac{\tau_p}{\tau} \mathbb{E}[V'(v) | P]$, a straightforward calculation gives:

$$\frac{f'(P)}{f(P)} = \frac{1}{\mathcal{P}'(s_p)} \left[\frac{f'_S(s_p)}{f_S(s_p)} - \frac{\mathcal{P}''(s_p)}{\mathcal{P}'(s_p)} \right] = -s_p \frac{\tau_v}{\mathbb{E}[V'(v) | P]} - \frac{\mathbb{E}[V''(v) | P]}{\mathbb{E}[V'(v) | P]^2},$$

because $\frac{f'_S(s_p)}{f_S(s_p)} = -s_p \frac{\tau_v \tau_p}{\tau}$ and $\mathcal{P}''(s_p) = \left(\frac{\tau_p}{\tau}\right)^2 \mathbb{E}[V''(v) | P]$. Therefore,

$$\Lambda(P) = \lambda(P) \lambda'(P) + \lambda(P)^2 \frac{f'(P)}{f(P)} = \lambda(P) \lambda'(P) - \lambda(P)^2 \left(s_p \frac{\tau_v}{\mathbb{E}[V'(v) | P]} + \frac{\mathbb{E}[V''(v) | P]}{\mathbb{E}[V'(v) | P]^2} \right).$$

Combining:

$$f(P)^{-1} \Delta G_{x^*x^*} = 2 u'(W_0) \Lambda(P).$$

So the full BNE expression is

$$f(P)^{-1} G_{x^*x^*} = u'(W_0) \left[\pi(a) \rho(a) \text{Skew}(V \mid P) + 2\rho(a) \lambda(P) \Sigma'_{VV}(P) \right. \\ \left. + \lambda(P) \lambda'(P) + 2\Lambda(P) \right].$$

Step 5: Combining terms. Assembling $\delta = x_\mu + \frac{1}{2t}x_{ss}$ with $x_{ss} = -\frac{1}{G_{x^*}}[G_{ss} + 2\beta G_{x^*s} + \beta^2 G_{x^*x^*}]$, and using $-1/G_{x^*} = 1/(u'(W_0) f(P) D)$, the $\text{Cov}(V, v^2 \mid P)$ terms cancel (as in REE), and the surviving terms yield the expression in the lemma. ■

H.6.9 Proof of Theorem 7

Proof. The proof proceeds identically to the proof of Theorem 6, with $\rho(a) + \kappa$ replaced by $\rho(a) + 2\kappa$ throughout. We sketch the key steps.

Step 1. By Lemma 12 and Stein's lemma, the trading intensity factorizes as

$$\beta(P, a) = \eta(P) \cdot \frac{t(a)}{\rho(a) + 2\kappa(P)}, \quad \eta(P) = \frac{\mathbb{E}[V'(v) \mid P]}{\tau \text{Var}(V(v) \mid P)}.$$

The coefficient $\delta(P, a)$ is given by Lemma 14.

Step 2. Market clearing defines the sufficient statistic s_p exactly as in Theorem 6, with weights $\omega(a) = \beta(a, P) / \int_0^1 \beta(b, P) db$.

Step 3. Substituting the factorized form of β into the precision formula (119), the $\eta(P)$ factors cancel in the signal-noise term. Under Assumption 6, $\tilde{\theta}(P)/\eta(P) = \tau/\xi$ is P -independent, so the $\eta(P)$ factors cancel in the supply-noise term as well:

$$\tau_p = \frac{A(\kappa)^2}{B(\kappa) + \tau^2/(\xi^2 \tau_u)},$$

where $A(\kappa) = \int_0^1 \frac{t(a)}{\rho(a)+2\kappa} da$ and $B(\kappa) = \int_0^1 \frac{t(a)}{(\rho(a)+2\kappa)^2} da$. Thus τ_p is P -independent.

Step 4. Standard differentiation of the market-clearing condition yields $\lambda(P) = \frac{\tau_p}{\tau} \mathbb{E}[V'(v) | P]/(\eta(P) A(\kappa))$. Substituting $\eta(P)$ and using $\lambda = \kappa \text{Var}(V|P)$ gives $\kappa = \tau_p/A(\kappa)$, which is P -independent.

Step 5. The efficiency condition $\mathbb{E}[V(v) | P] = P$ determines the price function (103).

Step 6. Combining $\kappa = \tau_p/A(\kappa)$ with the expression for τ_p , and writing $\tau = \tau_v + \kappa A(\kappa)$, yields the single fixed-point equation (108). ■

H.6.10 Proof of Corollary 7

Proof. The proof is identical to that of Corollary 5, with $\rho(a) + \kappa$ replaced by $\rho(a) + 2\kappa$ throughout. ■

H.6.11 Proof of Proposition 12

Proof. We give the argument for REE; the BNE case is identical with $\rho(a) + \kappa$ replaced by $\rho(a) + 2\kappa$ throughout.

Part 1. In the fixed-point equation (94), the denominator includes the term $(\tau_v + \kappa A(\kappa))^2/(\xi^2 \tau_u)$. Since $\tau_u^{-1} = \sigma_u^2 + \sigma_u^2 \rightarrow \infty$ as $\sigma_u^2 \rightarrow \infty$, this term diverges for any $\kappa > 0$, while the numerator $A(\kappa)$ remains bounded. Hence $\kappa \rightarrow 0$. It follows that $A(\kappa) \rightarrow A(0)$, $B(\kappa) \rightarrow B(0)$, $\tau_p = \kappa A(\kappa) \rightarrow 0$, $\tau \rightarrow \tau_v$, and $\xi \rightarrow \xi_0$.

Part 2. As noted in the Market quality subsection, the actual precision of the price signal is obtained by replacing τ_u^{-1} with σ_u^2 in the derivation of Theorem 6 (Step 3 of the proof). This yields

$$\tau_p^*(\kappa) = \frac{A(\kappa)^2}{B(\kappa) + \frac{\tau^2 \sigma_u^2}{\xi^2}}.$$

Substituting the limits from Part 1 gives (113). ■

H.6.12 Proof of Corollary 8

Proof. The result follows from the argument given in Section H.5, in the discussion immediately preceding the statement of the corollary. ■

I Scaled risk premium and demand elasticity in the CRRA–log-normal example

We derive two objects for the CRRA–log-normal specialization of Section 6 that have no direct CARA–normal analogue, because they depend on the higher moments of the payoff and on higher-order utility curvature (prudence): the scaled risk premium $\psi(P)$ (Sections I.1–I.2) and the elasticity of aggregate dollar demand with respect to the price (Section I.3).

I.1 Derivation of $\psi(P)$

With $V(v) = e^v$, $\rho(a) = \eta/W_0(a)$, $t(a) = t$, the conditional moments appearing in the demand coefficient $\delta(a, P)$ from (17) specialize as follows. The posterior $v \mid P \sim N(\mu_P, \tau^{-1})$ with $\mu_P = \ln P - 1/(2\tau)$ and $\tau = \tau_v + \tau_p$. Standard log-normal moment formulas give

$$\text{Var}[V(v) \mid P] = P^2(e^{1/\tau} - 1), \quad (121)$$

$$\begin{aligned} \frac{\text{Sk}[V(v) \mid P]}{\text{Var}[V(v) \mid P]} &= P(e^{1/\tau} - 1)(e^{1/\tau} + 2), \\ \frac{\mathbb{E}[v(V(v) - P)^2 \mid P]}{\text{Var}[V(v) \mid P]} &= \ln P + \frac{3}{2\tau}. \end{aligned} \quad (122)$$

From (22), $\beta(a, P) = W_0(a) c(P)/\eta$ where $c(P) = t/(\tau P(e^{1/\tau} - 1))$. Prudence is $\pi(a) = (\eta+1)/W_0(a)$. Integrating $\delta(a, P)$ over a , each of the three terms in (17) picks up $\int_0^1 W_0(a) da = \bar{W}$, because $\beta^2\pi \propto W_0(a)$, $\beta \propto W_0(a)$, and $1/\rho \propto W_0(a)$. The market clearing condition

$\int_0^1 \delta(a, P) da + s_p \int_0^1 \beta(a, P) da = \Theta(P)$ then yields, after dividing by $\bar{W}/\eta$,

$$\frac{\psi(P)}{\text{Var}[V | P]} = -s_p c(P) - \frac{(\eta + 1) c(P)^2}{2t\eta} \frac{\text{Sk}[V | P]}{\text{Var}[V | P]} + c(P) \frac{\mathbb{E}[v(V - P)^2 | P]}{\text{Var}[V | P]} + \frac{\eta \Theta(P)}{\bar{W}}.$$

Substituting (121)–(122) and $s_p = \tau(\ln P - 1/(2\tau))/\tau_p$, one obtains

$$\psi(P) = P \left[\frac{t}{\tau} \left(\ln P + \frac{3}{2\tau} \right) - \text{HHI} \left(\ln P - \frac{1}{2\tau} \right) \right] - \frac{(\eta + 1) t P (e^{1/\tau} + 2)}{2\eta \tau^2} + \frac{\eta \Theta(P)}{\bar{W}} P^2 (e^{1/\tau} - 1), \quad (123)$$

where we have used $\tau_p = t/\text{HHI}$ from (23).

I.2 Dependence on HHI

The scaled risk premium (123) depends on HHI through two channels:

1. *Direct channel*: the term $-\text{HHI} P(\ln P - 1/(2\tau))$, which appears because the sufficient statistic s_p depends on $\tau_p = t/\text{HHI}$ (so $t/\tau_p = \text{HHI}$);
2. *Indirect channel*: the posterior precision $\tau = \tau_v + t/\text{HHI}$ enters every conditional moment through $e^{1/\tau}$.

Differentiating (123) with respect to HHI, the direct channel decreases the signed scaled premium ψ when $\ln P > 1/(2\tau)$ (i.e., when the asset price exceeds its unconditional median) and increases it otherwise. The indirect channel operates through $\partial\tau/\partial\text{HHI} = -t/\text{HHI}^2 < 0$: a higher HHI lowers the posterior precision, which raises the conditional variance $P^2(e^{1/\tau} - 1)$ and the conditional third central moment, generating competing effects on ψ . In contrast to $\mathcal{I}$ and $\mathcal{L}$ —whose comparative statics with respect to concentration are unambiguous (Proposition 2)—the comparative statics of ψ with respect to HHI are richer and can be nonmonotone.

I.3 Elasticity of dollar demand

A growing empirical literature on demand-system asset pricing summarizes investors' demand by the price elasticity of their dollar holdings (Koijen and Yogo, 2019; Gabaix and Koijen, 2021). The natural model counterpart is the *elasticity of aggregate dollar demand*. Writing $D(P)$ for aggregate dollar demand as a function of the transaction price P (holding the realized signal statistic s_p fixed), we define

$$\varepsilon \equiv \frac{P}{\vartheta} \frac{\partial D(P)}{\partial P}, \quad (124)$$

where ϑ is the equilibrium level of aggregate dollar demand. By market clearing, ϑ coincides with aggregate dollar supply. We compute under $\bar{\theta} = 0$ and dollar-inelastic supply $\theta(P) \propto 1/P$, so that $P\Theta(P) = P\theta(P) \equiv \vartheta$ is constant. The CRRA-log-normal example then delivers a sharp, scale-free prediction: ε is constant (price-independent) and is governed by the cross-sectional distribution of wealth, not merely by aggregate risk tolerance.

Aggregate dollar demand schedule. Recall from (22) that, with $V(v) = e^v$, $\rho(a) = \eta/W_0(a)$, and $t(a) = t$, the signal-trading intensity is $\beta(a, P) = W_0(a) c(P)/\eta$ with $c(P) = t/(\tau P (e^{1/\tau} - 1))$, where $\tau = \tau_v + \tau_p$ is the posterior precision. Hence the *dollar* trading intensity is price-independent, and aggregating over traders (using $\int_0^1 W_0(a) da = \bar{W}$),

$$P \int_0^1 \beta(a, P) da = \frac{t \bar{W}}{\eta \tau (e^{1/\tau} - 1)} \equiv B,$$

a constant that does not depend on P . Aggregate dollar demand has two components. The dollar value of trades on private signals is $(P \int \beta da) s_p = B s_p$, where s_p is the realized sufficient statistic—fixed, since it is built from traders' own (realized) signals. The remaining, supply-accommodating component is pinned down by market clearing: from (11a), $\int \delta(a, P) da = \Theta(P) - h(P) \int \beta(a, P) da$, where $h(P)$ is the price-inference map through which traders learn from the transaction price. Its dollar value is $P \int \delta da = P \Theta(P) - h(P) B = \vartheta - B h(P)$, using

$\bar{\theta} = 0$ and $P\theta(P) = \vartheta$. Adding the two components,

$$D(P) = \underbrace{B s_p}_{\text{trades on own signals}} + \underbrace{\vartheta - B h(P)}_{\text{supply-accommodating}}.$$

At market clearing, $h(P) = s_p$, so $D(P) = \vartheta$: the normalizing constant in (124) coincides with aggregate dollar demand.

From schedule to elasticity. Only the inference term $-B h(P)$ varies with the transaction price. By the log-linear price function (21), the inference map is affine in the log price, $h(P) = (\tau/\tau_p)(\ln P - 1/(2\tau))$, so

$$\frac{\partial D(P)}{\partial P} = -B \frac{\tau}{\tau_p} \frac{1}{P}.$$

Substituting into (124) and using $\tau_p = t/\text{HHI}$ from (23),

$$\varepsilon = -\frac{1}{\vartheta} B \frac{\tau}{\tau_p} = -\frac{1}{\vartheta} \frac{t \bar{W}}{\eta \tau_p (e^{1/\tau} - 1)} = -\frac{1}{\vartheta} \frac{\bar{W}}{\eta} \frac{\text{HHI}}{e^{1/\tau} - 1}. \quad (125)$$

Two features stand out. First, ε is *constant*: both $B\tau/\tau_p$ and the normalizer ϑ are independent of P , so ε does not depend on the price level.

Second, and more importantly, ε depends on the cross-sectional *distribution* of risk tolerance through the Herfindahl index HHI, not merely on its mean. This is a sharp departure from standard large economies. There, a law-of-large-numbers aggregation result (Appendix E) renders an economy with heterogeneous wealth, precision, and preferences observationally equivalent to a homogeneous one with the same *aggregate* risk tolerance; consequently every equilibrium quantity—including the demand elasticity—depends on the cross-sectional distribution of risk tolerance only through its mean. With CRRA preferences, this means that only *average* wealth $\bar{W}$ would matter, not how wealth is distributed. This is precisely the structure of Peress (2004), whose equilibrium price, risk premium, and price informativeness are all expressed through a single aggregate risk-tolerance measure ($\nu = \int t(W_0) dG$ in his notation). In CHILE, by con-

trast, the elasticity (125) depends on HHI: the wealth distribution matters in its own right, not only through its mean. This sensitivity of an aggregate object to the distribution—rather than just the average—is a manifestation of granularity in asset markets (Glebkina, Malamud, and Teguia, 2025).

Finally, the elasticity is tightly related to the liquidity measure $\mathcal{L}$ from the main text. Writing the conditional variance of the gross return $R = V(v)/P$ as $\text{Var}[R \mid P] = \text{Var}[V \mid P]/P^2 = e^{1/\tau} - 1$ and using $\mathcal{L} = (\bar{W}/\eta)$ HHI from (23), (125) becomes

$$\varepsilon = -\frac{1}{\vartheta} \frac{\mathcal{L}}{\text{Var}[R \mid P]},$$

the (risk-adjusted) liquidity scaled by the dollar supply and the conditional return variance.

J Average vs. aggregate quantities in large economies

This section illustrates, in the simplest possible setting, how individual trades and risk premia can vanish as the number of agents grows while their economy-wide aggregates remain finite and nonzero. The setting is a standard competitive economy with n identical, uninformed, risk-averse agents—no private information and no features specific to CHILE. The phenomenon is a generic consequence of splitting a fixed amount of aggregate risk among an increasing number of agents.

Consider an economy with n identical agents, and let $\mu = 1/n$ denote the mass of each agent. Each agent has initial wealth W_0 and preferences $u(\cdot)$ over terminal wealth, where u is increasing, strictly concave, and thrice continuously differentiable. There is a risky asset with payoff $V(v)$ and a continuously differentiable supply $\Theta(P)$, and a risk-free asset with gross return normalized to 1. Agents have no private information: each agent observes only the price P and chooses demand x to maximize $\mathbb{E}[u(W_0 + x(V(v) - P))]$.

By symmetry, all agents hold the same position $x^*(\mu)$ in equilibrium. Market clearing requires that aggregate demand equals supply:

$$n \cdot x^*(\mu) = \Theta(P), \quad \text{i.e.,} \quad x^*(\mu) = \mu \Theta(P).$$

Substituting the market-clearing condition into the first-order condition, the equilibrium price $P(\mu)$ is determined by

$$G(\mu, P) \equiv \mathbb{E}\left[u'(W_0 + \mu \Theta(P) (V(v) - P)) (V(v) - P)\right] = 0. \quad (126)$$

Proposition 13 (Individual trade). *Let $\rho = -u''(W_0)/u'(W_0)$ denote absolute risk aversion at W_0 . As $\mu \rightarrow 0$ (equivalently, $n \rightarrow \infty$):*

(i) *The equilibrium price satisfies*

$$P(\mu) = \mathbb{E}[V(v)] - \mu \rho \Theta(\mathbb{E}[V(v)]) \text{Var}(V(v)) + O(\mu^2). \quad (127)$$

(ii) *Individual demand vanishes: $x^*(\mu) = \mu \Theta(\mathbb{E}[V(v)]) + O(\mu^2) \rightarrow 0$.*

(iii) *Aggregate demand converges to a finite, generically nonzero limit: $n x^*(\mu) = \Theta(\mathbb{E}[V(v)]) + O(\mu) \rightarrow \Theta(\mathbb{E}[V(v)])$.*

The key to understanding why vanishing individual demands are compatible with a finite aggregate is the distinction between *averages* and *aggregates*. The average demand across agents is $\frac{1}{n} \sum_{i=1}^n |x_i^*| = |x^*(\mu)| \rightarrow 0$, which indeed vanishes as the proposition states. The aggregate demand is $\sum_{i=1}^n |x_i^*| = n |x^*(\mu)|$ —the product of a diverging number of agents and a vanishing per-agent demand—and this product converges to a finite limit.

Proposition 14 (Individual risk premium). *Under the conditions of Proposition 13, as $\mu \rightarrow 0$:*

(i) *The per-agent risk premium vanishes:*

$$\mathbb{E}[V(v)] - P(\mu) = \mu \rho \Theta(\mathbb{E}[V(v)]) \text{Var}(V(v)) + O(\mu^2) \rightarrow 0.$$

(ii) *The scaled risk premium converges to a finite, generically nonzero limit:*

$$\frac{1}{\mu} \left(\mathbb{E}[V(v)] - P(\mu) \right) = \rho \Theta(\mathbb{E}[V(v)]) \text{Var}(V(v)) + O(\mu).$$

The economics are transparent: a finite amount of aggregate risk, $\Theta(\mathbb{E}[V(v)]) \text{Var}(V(v))$, is shared among n agents. Each agent bears a $1/n$ share and earns a proportionally small premium. The per-agent premium $\mathbb{E}[V(v)] - P(\mu) = O(\mu)$ vanishes, yet the scaled premium $n (\mathbb{E}[V(v)] - P(\mu))$ —the per-agent premium rescaled by the number of traders—remains finite, exactly as with aggregate demand.

J.1 Proofs

Proof of Proposition 13. We expand $P(\mu)$ around $\mu = 0$ using the implicit function theorem applied to $G(\mu, P) = 0$ defined in (126).

Zeroth order. At $\mu = 0$, the per-agent holding is $x^*(0) = 0$, so terminal wealth equals W_0 with certainty. Equation (126) becomes $u'(W_0) \mathbb{E}[V(v) - P_0] = 0$, which gives $P_0 \equiv P(0) = \mathbb{E}[V(v)]$.

First order. Differentiating $G(\mu, P(\mu)) = 0$ with respect to μ and evaluating at $\mu = 0$:

$$\left. \frac{\partial G}{\partial \mu} \right|_{\mu=0} + \left. \frac{\partial G}{\partial P} \right|_{\mu=0} \cdot P'(0) = 0.$$

For the partial with respect to P : at $\mu = 0$ the only surviving term is $\mathbb{E}[u'(W_0) \cdot (-1)] = -u'(W_0)$, since the chain-rule term through u'' carries a factor of μ that vanishes. For the partial with respect to μ :

$$\left. \frac{\partial G}{\partial \mu} \right|_{\mu=0} = \mathbb{E}[u''(W_0) \Theta(P_0) (V(v) - P_0)^2] = u''(W_0) \Theta(\mathbb{E}[V(v)]) \text{Var}(V(v)).$$

Solving for $P_1 \equiv P'(0)$:

$$P_1 = - \left. \frac{\partial G / \partial \mu}{\partial G / \partial P} \right|_{\mu=0} = \frac{u''(W_0)}{u'(W_0)} \Theta(\mathbb{E}[V(v)]) \text{Var}(V(v)) = -\rho \Theta(\mathbb{E}[V(v)]) \text{Var}(V(v)),$$

which establishes part (i). Parts (ii) and (iii) follow from the market-clearing condition $x^*(\mu) = \mu \Theta(P(\mu))$:

$$x^*(\mu) = \mu \Theta(\mathbb{E}[V(v)] + \mu P_1 + O(\mu^2)) = \mu \Theta(\mathbb{E}[V(v)]) + O(\mu^2),$$

so $x^*(\mu) \rightarrow 0$ and $n x^*(\mu) = \Theta(\mathbb{E}[V(v)]) + O(\mu)$. ■

Proof of Proposition 14. Immediate from the price expansion (127): part (i) is $\mathbb{E}[V(v)] - P(\mu) = -\mu P_1 + O(\mu^2) = \mu \rho \Theta(\mathbb{E}[V(v)]) \text{Var}(V(v)) + O(\mu^2)$, and part (ii) follows by dividing

by μ . ■